

Diffraction by DNA, carbon nanotubes and other helical nanostructures

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Diffraction by DNA, carbon nanotubes and other helical nanostructures

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Received 19 November 2004, in final form 1 March 2005

Published 21 April 2005

Online at stacks.iop.org/RoPP/68/1181

Abstract

This review discusses the diffraction patterns of x-rays or electrons scattered by fibres of helical biological molecules and by carbon nanotubes (CNTs) from the unified point of view of the Fourier–Bessel transform of an atomic helix. This paper is intended for scientists who are *not* professional crystallographers. X-ray fibre diffraction patterns of Pauling’s protein α -helix and of Crick and Pauling’s protein coiled-coil are revisited. This is followed by a non-technical comparison between the historic x-ray diffraction patterns of the A and B conformations of DNA, which were crucial for the discovery of the double helix. The qualitative analysis of the diffraction images is supported by novel optical simulation experiments designed to pinpoint the gross structural informational content of the patterns. The spectacular helical structure of the tobacco mosaic virus determined by Rosalind Franklin and co-workers will then be described as an early example of the great power of x-ray crystallography in determining the structure of a large biomolecular edifice. After these mostly historical and didactic case studies, this paper will consider electron diffraction and transmission electron microscopy of CNTs of great current interest, focusing particularly on recent data obtained for single-wall, double-wall and scrolled nanotubes. Several points of convergence between the interpretations of the diffraction patterns of biological helices and CNTs will be emphasized.

(Some figures in this article are in colour only in the electronic version)

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1. Introduction

In his celebrated opus '*On Growth and Form*' (first edition in 1917; abridged edition: Cambridge University Press, 1961), D'Arcy Thomson (1860–1948) describes the goal of the book as: '*My sole purpose is to correlate with mathematical statement and physical law certain of the simpler outward phenomena of organic growth and structure or form . . .*'. The structure or form are those of animals and plants, the hexagonal bee honeycomb, the spiralling nautilus shell, the spherical radiolaria skeleton, the helical narwhal tusk and many others, and the mathematical physics laws are taken from geometry, rheology, structural mechanics, etc.

The author of this venerable and influential book unfortunately did not live long enough to witness the great revolution that started in the early 1950s with the new molecular understanding of growth and form in biology. This breakthrough was the culmination of half a century of steady advances, of which D'Arcy Thomson was undoubtedly aware, in biochemistry, in genetics and especially in the physical methods for finding the structure of materials at the atomic scale. The revolution signalled the start of a tremendous acceleration for all the life sciences.

This paper will first aim at 'correlating' the most fundamental form of molecular biology, the helical structure of proteins and nucleic acids, with the mathematics of the Fourier transform which constitutes the formal basis of the physics of diffraction used to discover the structure. This, of course, is an old topic belonging to the glorious history of 20th century science. But the story deserves to be told again in language less technical than is usual for a crystallography subject. This presentation will hopefully be accessible to a wider readership of students and professional scientists. The language developed in a biological context will be found quite adequate also for understanding the helical atomic structure of the recently discovered carbon nanotubes (CNTs), which, like that of biomolecules, has been unravelled almost exclusively by diffraction methods.

Barring the universal presence of screw dislocations in all defective crystalline materials, inorganic solids made of mono-atomic or highly symmetrical polyatomic subunits rarely show cylindrical rotation symmetry or helical screw symmetry. One exception is the class of solids that have a layered structure, such as some transition metal dichalcogenides (WS_2 , MoS_2 , . . .). These are materials in which the inter-layer interactions are much weaker than the intra-layer bonds, a situation that permits the bending and closing (or scrolling), at the nanoscopic scale, of the flat sheets of the material into cylindrical surfaces, during or after synthesis. The recently discovered CNT, a completely unexpected molecular form of pure carbon, is one notorious and rather exceptional elemental case of such a cylindrical layered structure synthesized at the level of a supporting catalytic nanoparticle. What are straight rows of atoms in the flat carbon sheet of graphite become helical chains on the cylindrical surface of the nanotube, which, as a result, will generally have an achiral helical symmetry, as will be seen in this paper.

By contrast, helices occur ubiquitously in organic compounds and biological organisms built from them, and at all scales from the molecular level all the way up to the macroscopic sizes of the plants and animals of D'Arcy Thomson's book. Here, helices occur as a matter of necessity in the process of covalently binding small asymmetric molecules together in a linear sequence or as a consequence of the dissymmetric interactions involved in the noncovalent packing of larger chiral subunits making up the final structure. Examples of the former are helical chains of amino acids (polypeptides) or nucleic acids (polynucleotides, DNA, RNA). Instances of the latter are bigger assemblies such as multi-strand ropes of protein fibres (e.g. collagen, silk, keratin), the microtubules of the mitotic spindle, actin filaments and other skeletal cellular fibres, bacterial cilia and flagella, muscle fibres, chromosomes, several helical viruses and many other complex systems of the living world.

A few representative biological helices and CNTs will be considered in this paper from the exclusive point of view of x-ray diffraction, electron diffraction (ED) and electron microscopy. Whenever possible, this review will point out similarities between the ideas and techniques applied to these widely differing helical organic and inorganic structures. In the first part of this paper, we will begin by presenting the elementary ideas of wave scattering in the kinematical approximation and introduce the central concept of layer-lines for the description of diffraction by fibrous biopolymers. Then, the Fourier–Bessel transform of an atomic helix, the universal tool to ‘correlate’ the real space of material helices with the reciprocal space of their diffraction patterns, will be derived. Applications to a few of the most important and most typical biological examples will then be discussed in some detail, the α -helix of proteins, the protein coiled-coil, the DNA double helix and the spectacular helix of the tobacco mosaic virus (TMV). As we shall see, three great scientists have been mostly responsible for the discovery of these beautiful natural structures, Linus Pauling, Rosalind Franklin and Francis Crick, to whose many contributions the science of biology is forever indebted and to whose memory this paper is dedicated.

The second part of this paper is devoted to ED and transmission electron microscopy (TEM) of CNTs, for which we will frequently make reference to the concepts previously developed for biological helices. The Fourier–Bessel transform of an atomic helix can be extended to obtain a closed analytical form of the scattering factor of a complete monolayer nanotube. This tool will then be exploited to analyse very recent diffraction results, particularly for single-walled and double-walled nanotubes, bundles and rings of such nanotubes and nanotube scrolls. Several of the examples chosen are among those which may find applications in nanoelectronics in some distant future.

2. Short round-up of relevant diffraction concepts

2.1. Diffraction of x-rays

Diffraction is the coherent scattering of a *wave* by a material object. The *intensity* of the scattered wave is collected on a recording medium in the form of a ‘*diffraction pattern*’. Optical diffraction was first demonstrated by Thomas Young in the early years of the 19th century. His celebrated two-slit diffraction experiment established the *wave nature of visible light*. Great advances in physical optics followed this seminal discovery throughout the 19th century. Most notable was the birth and development of *atomic spectroscopy* based on the separation of visible wavelengths either by refraction prisms or by *diffraction gratings* generalizing Young’s diffraction set-up. The existence of *invisible radiations*, in the infrared and ultraviolet regions of the spectrum, and their refraction and diffraction were demonstrated even before Maxwell formulated his electromagnetic theory in the 1860s (brief histories of optics giving details of these events are available on several readily accessible websites). The direct confirmation of Maxwell by Heinrich Hertz in 1887 was provided in the even more remote spectral region of ‘Hertzian’ or radio waves [1a].

Then came the discovery of the mysterious x-rays by Röntgen in 1895. Their practical applications to radiography immediately followed the discovery. But their electromagnetic nature and their place in the electromagnetic spectrum were established only about two decades later. In a concentrated period of three years [1b] the following momentous events took place in breathtaking succession: (i) Rutherford discovered the *nuclear model* of the atom in 1911 [2]; (ii) Barkla demonstrated and named the *characteristic x-ray* radiation of the elements in 1912 [3]; (iii) Bohr developed the *quantized model of hydrogen and hydrogen-like ions* in 1913 [4]; (iv) von Laue proved the *wave nature* of x-rays in 1913 via their *diffraction* by an

atomic lattice [5]; (v) the *use of x-ray diffraction in crystallography* was established by the Braggs in 1913 for the determination of the three-dimensional atomic architecture of solids and molecules [6, 7]; (vi) an *absolute scale of x-ray wavelengths* was established by Moseley in 1914 [8].

Moseley made a most remarkable unification of these great advances in physics when he interpreted Barkla's characteristic x-rays in terms of the atomic models just established by Rutherford and Bohr. From the Balmer–Bohr formula applied to inner electronic shells, Moseley found that the characteristic atomic wavelengths are given by $1/\lambda = RZ^2(1/n^2 - 1/m^2)$ where $R \approx 1.1 \times 10^7 \text{ m}^{-1}$ is the Rydberg constant and Z the atomic number. For example, for the prominent K_α line of elements (transition between the L and K shells), the wavelength is approximately given by $1/\lambda = RZ^2(1/1^2 - 1/2^2) = 3RZ^2/4$. For copper ($Z = 29$) $\lambda \approx 0.15 \text{ nm}$ while for molybdenum ($Z = 42$), $\lambda \sim 0.07 \text{ nm}$ [9].

2.2. Diffraction of matter waves

Later in the 20th century, *quantum mechanical matter waves* associated with *electrons, neutrons* and even *whole atoms and molecules* were discovered. First, experiments were conducted to prove de Broglie's conjecture that a material particle of momentum p behaves like a wave of wavelength $\lambda = h/p$ where h is Planck's constant [10]. This was done with slow electrons reflected from surfaces [11] and with fast electrons transmitted through thin films [12]. Then, as with x-rays, the diffraction of matter waves became thoroughly used as a new tool for both bulk and surface crystallography.

A broad spectrum of wavelength values thereby became available to do diffraction experiments, from a few Å (keV x-rays, eV electrons or thermal neutrons) to a small fraction of an Å (hard x-rays, keV electrons). Today, the diffraction of electron waves is used in standard characterization techniques such as low energy electron diffraction (LEED), reflection high energy electron diffraction (RHEED) and high energy electron diffraction (HEED) or ED, the latter in conjunction with TEM. The use of neutron scattering for bulk crystallography and dynamics had to await the availability of thermal beams of high intensity from fission reactors [13] in the middle of the 20th century, while the scattering of thermal neutral atoms (mainly helium [14]) was developed in the later parts of the 20th century, especially for the study of surface structures and dynamics.

2.3. Cross sections

In the kinematical (first Born) approximation, which will be found sufficient in this review, the differential cross section for elastic scattering of a wave from the initial $\mathbf{k}_i$ to the final $\mathbf{k}_f$ wave vectors by a collection of atoms at nuclei positions $\mathbf{r}_j$ is the square of the modulus of a complex amplitude $A(\mathbf{q})$

$$\frac{d\sigma}{d\Omega} = |A(\mathbf{q})|^2 = \left| \sum_j f_j(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}_j} \right|^2, \quad (1)$$

where $\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i$ is the scattering wave vector and where $f_j(\mathbf{q})$ is the atomic form factor, an intrinsic atomic property. Note the use here of 'rationalized' wave vectors $\mathbf{q}$ of solid-state physicists, which incorporate factors of 2π , rather than the traditional (hkl) notation of crystallography. For unpolarized x-rays an extra factor $(1 + \cos^2 2\theta)/2$ must come in the diffracted intensity to take into account the two polarizations, where 2θ is the scattering angle.

The behaviour of $f_j(\mathbf{q})$ versus $\mathbf{q}$ for the elements B, C, N, O, P is shown in figure 1 [15]. For x-rays which are scattered by individual electrons [16], $f_j(\mathbf{q})$ is the Fourier transform

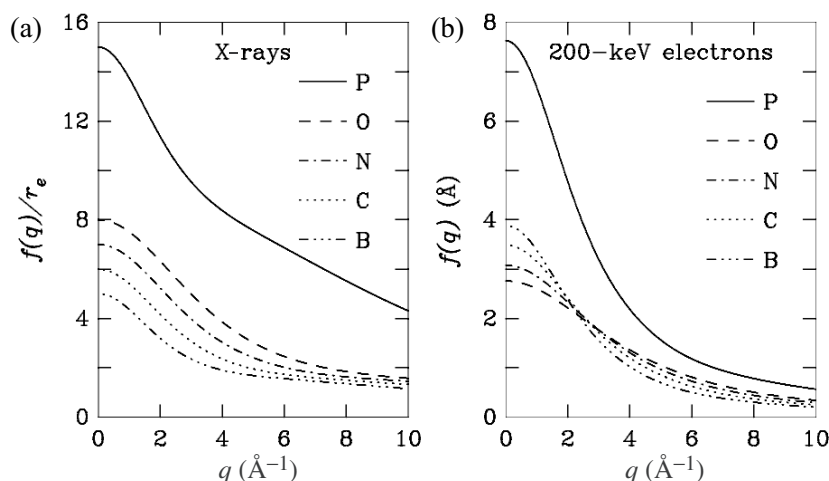


Figure 1. The atomic scattering factors of a few elements for Cu K_α x-rays (a) and for 200 keV electrons (b).

of the atomic electron density and the natural length scale is the classical electron radius $r_e = e^2/4\pi\epsilon_0 mc^2 = 2.8 \times 10^{-15}$ m. For fast electrons, which are scattered by the electrostatic potential of the nucleus screened by the orbital electrons, $f_j(q)$ is the Fourier transform of the screened nucleus Coulomb potential and the length scale is the Bohr radius, 0.53×10^{-10} m. For neutrons $f_j(q)$ is the Fourier transform of the neutron–nucleus interaction potential. Given the short range of the latter and the small size of the nuclei on the scale of the neutron wavelength, $f_j(q)$ is a constant independent of q called *the scattering length* and is also of order 10^{-15} m. These length scales for the different radiations explain why ED, when combined with high-resolution TEM (necessary to locate the sample), works with one single nanoscopic object such as a CNT or a single virus, while x-ray or neutron diffraction requires macroscopic samples.

Since the diffraction amplitude $A(\mathbf{q})$ is related to the Fourier transform of the density of scattering centres, the latter is in turn obtainable from the Fourier transform of the amplitude. Unfortunately, the measurement of the diffracted intensity $I(\mathbf{q})$, proportional to the scattering cross section in equation (1), throws away the phase $\varphi(\mathbf{q})$ of the complex amplitude $A(\mathbf{q}) = \sqrt{I(\mathbf{q})} \exp[i\varphi(\mathbf{q})]$. This precludes deriving the structure simply from a direct inversion of the intensity pattern. So, in the absence of the phases, the painstaking, trial-and-error method initially consisted in making informed guesses about the structure on the basis of the geometry of the pattern and its most prominent features, of chemical information on the molecules, etc and varying the structure until the square of the computed Fourier transform was found to give a satisfactory agreement with the measured diffracted intensity pattern. Later, in the second half of the 20th century, experimental and theoretical methods were introduced to retrieve the phases, at least partially, allowing a direct inversion of the diffraction patterns. Examples of these methods will be briefly discussed in this review.

2.4. Some fundamentals of fibre diffraction

Given the inversion difficulty just mentioned, the structures of simple three-dimensional crystals of *inorganic materials* were first investigated with x-rays [7].

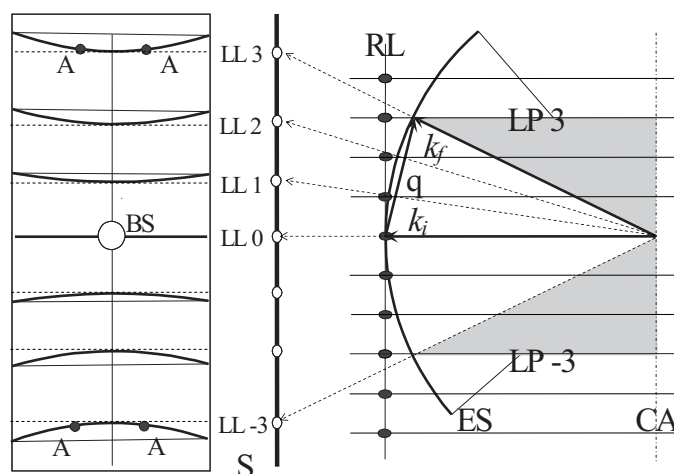


Figure 2. Principle of the geometrical construction of the hyperbolic layer-lines in the x-ray diffraction pattern of a linear array of point scatterers. RL, real lattice; ES, Ewald sphere; CA, cone axis; LP, layer-planes; S, observation screen; BS, beam stop; LL, layer-lines. The shaded regions are cones whose intersections by the screen generate the layer-lines.

However, not very long after the von Laue–Bragg discoveries, x-ray studies were devoted to complex organic polymers, at least whenever the large, unwieldy molecules could be made to condense in a partially ordered phase capable of causing non-isotropic scattering. For example, very long polymer chains—*polysaccharides*, such as cellulose [17], *polypeptides*, such as keratin (of natural hair or wool fibres), [18] and *polynucleotides* such as DNA [19]—could be aligned in fibre form and produce distinct features in their x-ray fibre patterns interpretable in terms of elements of a molecular structure. Other biomolecules having a globular shape such as pepsin [20] (a digestive enzyme) or haemoglobin [21] (the oxygen carrier in the blood), could even be made to condense in three-dimensional crystal lattices, which remain stable when kept in their aqueous mother solution and gave sharp, spotty diffraction patterns. These early, pioneering works [17–21] proved that functional biomolecules of a given species had both a definite atomic composition and a fixed spatial structure, rather than being a random, variable aggregate of atomic or molecular subunits, which would have precluded crystallization of any sort. They also gave hope that the atomic details of that structure could eventually be worked out by the methods of x-ray crystallography so successful for inorganic solids. In this review, we will be concerned only with the relatively simple diffraction by long, filamentous molecules, which are naturally arranged in fibres or which can be drawn into fibres. Diffraction by three-dimensional crystals of large globular proteins is a more complex matter reserved for specialized x-ray crystallographers.

For the interpretation of x-ray diagrams of fibres, an important structural concept was first introduced by Polanyi [17], namely that of *layer-planes* and *layer-lines* in reciprocal space. The concept is most easily explained with a simple model frequently used in the standard teaching of elementary diffraction theory. Let us first consider (figure 2) a *single linear ‘polymer’* consisting of identical ‘atoms’ or point scatterers occupying equidistant positions $z_j = jP$ on the z -axis, where j is an integer and P is the polymer repeat period. The scattering amplitude in equation (1) reduces to

$$A(\mathbf{q}) = f(q) \sum_j e^{iq_z jP} = f(q) \frac{2\pi}{P} \sum_{l=-\infty}^{+\infty} \delta\left(q_z - l \frac{2\pi}{P}\right), \quad (2)$$

where we have made use of the well-known identity (the fundamental relation for diffraction by ordered structures)

$$\sum_{n=-\infty}^{+\infty} e^{2\pi i x n} = \sum_{m=-\infty}^{+\infty} \delta(x - m). \quad (3)$$

The vertical component q_z of the transfer wave vector is, therefore, quantized to integer multiples of $2\pi/P$. In reciprocal space, the horizontal planes of equation $q_z = 2\pi l/P$ are called *layer-planes*.

Now, for *elastic scattering*, the allowed values of the wave vector k_f of the scattered waves must lie on the intersection of the layer-planes with the *Ewald sphere* of radius $|k_f| = |k_i|$. These intersections are latitudes on the sphere and constitute the bases of cones with a common apex at the sphere's centre (figure 2). The intersections, by the plane of the observation screen S , of the extrapolated conical surfaces give the loci of the non-zero diffracted intensity. As conical sections parallel to the cone axis, these intersections are evidently *hyperbolae* which are called *layer-lines* (see the front view of the screen S in figure 2). Along each layer-line the diffracted intensity is governed by the atomic form factor $|f(q)|^2$, which, for a point scatterer, is a monotonically decreasing function of q (see figure 1).

If, instead of a point-like 'atom', the monomer is a thin continuous horizontal line or a narrow slit in an opaque screen, the usual devices used in the class room to demonstrate optical diffraction, then the intensity along the layer-lines, would be reduced to a sharp point and the diffraction pattern would be the familiar set of such points aligned along the '*meridian*', perpendicular to the slits. In the intermediate case, i.e. if the 'atom' is a horizontal *line or slit of finite width*, the intensity along the layer-lines will likewise be extended over a finite width (recall that there is reciprocity between the width in real space and the width in Fourier space). The intensity will also be oscillatory, reflecting the interferences typical of the diffraction by a finite aperture.

More generally, the diffraction pattern of any linear periodic polymer whose repeated monomer may have any three-dimensional shape whatsoever and may contain any number of atoms, will also be organized in hyperbolic layer-lines similar to those of the mono-atomic example discussed above. Suppose indeed that the j th monomer has its atoms at positions $\mathbf{r}_{j\mu} = \mathbf{r}_\mu + jP\mathbf{e}_z$. One immediately sees from equation (1) that the scattering amplitude is obtained by simply replacing $f(q)$ by the monomer form factor $F(\mathbf{q}) = \sum_\mu f_\mu(q) \exp(i\mathbf{q} \cdot \mathbf{r}_\mu)$ in equation (2). The sharp layer-line organization remains. $|F(\mathbf{q})|^2$ governs the intensity distribution along the layer-lines, which is now broken up into a pattern of maxima and minima reflecting the interferences of the waves scattered by the monomer atomic content.

When the x-ray radiation is perpendicular to the polymer, as in figure 2, the pattern of layer-lines is symmetric with respect to the straight, '*equatorial*' layer-line. In some symmetry conditions, the pattern of intensity distribution may also be north-south and east-west symmetric: any feature A on a layer-line will be reproduced identically in the four quadrants. For example, this pattern symmetry will obtain when averaging the intensity over a continuous rotation of a centro-symmetric polymer around its vertical axis.

Consider now a *fibre* containing a large number N of such identical parallel polymers at 'positions' ρ in the fibre (this symbolic ρ might include a rotation around the polymer axis and an axial translation within the repeat period P , i.e. a screw operation). The diffraction amplitude of the fibre will be obtained by multiplying the scattering amplitude $A(\mathbf{q})$ of the ρ th

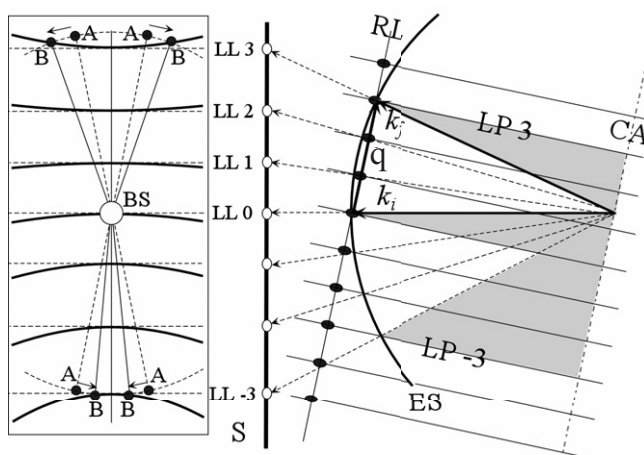


Figure 3. Same as figure 2 for a linear polymer with its top tilted towards the x-ray beam. A feature A which appeared symmetrically in figure 2 has moved towards asymmetric positions B on the Bragg diffraction circle.

polymer by a phase factor $\exp(i\mathbf{q} \cdot \boldsymbol{\rho})$ and summing over all the polymers:

$$\sum = \sum_{\boldsymbol{\rho}} A(\mathbf{q}) e^{i\mathbf{q} \cdot \boldsymbol{\rho}}. \quad (4)$$

If the fibre is a gel (parallel molecules but disordered distribution of $\boldsymbol{\rho}$ s), due to the randomness of the phases $\mathbf{q} \cdot \boldsymbol{\rho}$, the intensity $|\Sigma|^2$ will reduce approximately to $N \langle |A(\mathbf{q})|^2 \rangle$, where the angle brackets indicate averaging over the angular positions of the molecules. In this case, the diffraction pattern of the fibre is representative of the angularly averaged individual polymer pattern and should display streaks along the layer-lines characteristic of a single molecule, whereas, if the fibre is crystalline, the positions $\boldsymbol{\rho}$ are on a two-dimensional lattice and the sum over $\boldsymbol{\rho}$ in equation (4), on account of equation (2), amounts to a sum of delta functions at the nodes of the two-dimensional reciprocal lattice. In this case, the intensity along the layer-lines is broken up into sharp spots and the overall pattern is just that of an ordinary three-dimensional crystal for which the information on the unit cell content is to be found in the relative intensity of all the spots. All intermediate cases are of course possible.

What happens to the diffraction pattern of a polymer when the incident radiation beam is not perpendicular to the polymer axis? Such a geometry is often intentionally used to enhance certain features of the pattern in x-ray diffraction or may be quite unavoidable in electron microscopy and diffraction for which the control of the orientation of the nanoscopic target may be more problematic. Figure 3 illustrates the symmetry breaking that occurs when the top of the polymer leans towards the incoming beam. Due to the finite radius of the Ewald sphere for x-rays on the scale of (inverse) interatomic distances in solids, any north-south symmetry which might have been present is destroyed. In particular, the equatorial layer-line acquires a downward curvature and the meridian separations between the layer-lines are smaller in the north than those in the south. In addition, the fibre inclination makes a peak A move along its modified layer-lines towards asymmetric positions B, as shown in figure 3. All these features are beautifully illustrated in figures 4 and 5, which are real and historically important examples described later in this review.

The effects upon tilting the polymer are somewhat different for the diffraction of high-energy electrons, as will be discussed in the section on CNTs.

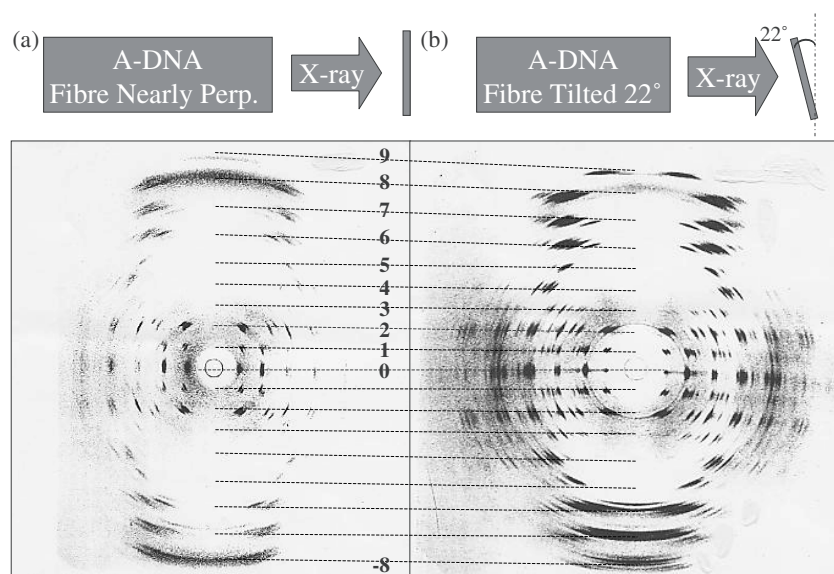


Figure 4. The x-ray diffraction pattern of an A-DNA fibre in near normal incidence (*a*) and at incidence tilted by 22° from the normal (*b*). The layer-lines are numbered 0–9. Note the deformation of the layer-lines and the movement of the diffraction features towards asymmetric positions on the north and south layer-lines.

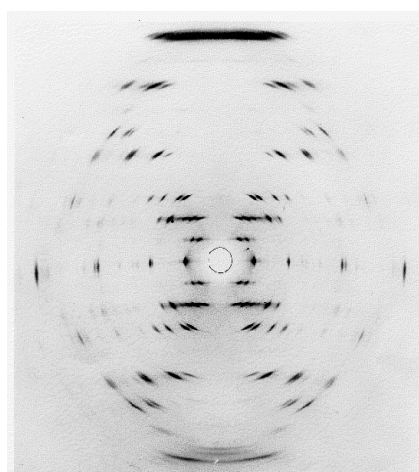


Figure 5. X-ray diffraction pattern of a B-DNA crystalline fibre [56]. Note the north–south asymmetry caused by the tilting of the fibre by 13° away from the normal to the beam.

3. Helices

A helical polymer is one in which the monomer units making up the periodically repeated part of the molecule are aligned along a regular circular helix and are all equivalent under the symmetry operations of the helical screw axis. Four examples considered in this paper are polypeptide chains of amino acids in the α -helix configuration, the sugar-phosphate backbone of nucleic acids in the DNA double helix, helical viruses and CNTs.

When fibrous polymeric materials began to be investigated by x-rays in the 1920s and 1930s [18], their flexible chain molecules were suspected of being folded back and forth in the space occupied by the molecules. The main reason was that a folded structure would help explain the ease with which some of the natural fibres, such as wool or silk, could be mechanically stretched to a large extent by reversible unfolding. *Planar folding* modes were first envisaged [18]. However, in the 1930s and early 1940s *helical folding* was suggested as an alternative for a growing number of polymers including proteins, basically for stereochemical reasons, namely that chiral monomers could naturally be expected to linearly polymerize in helical form (see [24], 323ff and [26], 60ff for the genesis of the idea of helical molecular structures). A specific helical diffraction theory was, therefore, called for quite early to compare the form factor of an atomic helix with that of a planar folded structure in view of helping with the interpretation of such x-ray diffraction results as collected up to that time for fibrous materials. But the speculative nature of the helical idea, the preliminary and rather poorly resolved x-ray fibre patterns and the incomplete structural information on the elemental chemical units, were such that only mild pressure existed for the formulation of such a theory.

All this changed in the late 1940s and early 1950s when helices were, so to speak, more and more in the biological air. Indeed, the demand for a theoretical framework became pressing when quite specific models were proposed by several groups, particularly by Linus Pauling and his collaborators, for possible helical structures of natural filamentous proteins [22], artificial polypeptide chains and even portions of complex globular proteins. The proposed models could by then rely on a mass of x-ray data and other chemical information, much of which was accumulated by Pauling's school, on the spatial structure of single amino acids, short peptides and other simple substances related to proteins. These studies had determined quite accurately the various acceptable bond lengths, bond angles and interaction distances, that is to say the stereochemical properties of the amino acid residues that were likely to be conserved when these units are assembled in polypeptide chains. In addition, the general concept of hydrogen bonding had been quantitatively strengthened and its prominent role in biological structures had been further investigated, especially by Pauling [24]. Yet, it was not Pauling's group that came up with the proper formula for the form factor of a helix. The correct and full theory was first published by Cochran, Crick and Vand (CCV) [23] for the explicit purpose of explaining the diffraction of x-rays by certain synthetic polypeptides. A version of the formalism was independently developed by Alexander Stokes in the context of the research on the structure of nucleic acids but was published only well after 1951–53 [24, 25], the crucial discovery years for the structure of helical biomolecules. The CCV fundamental paper was not only very timely for the intended interpretation of protein fibre diffraction but also turned out, unexpectedly as we shall see, to play an even more important role for the understanding of the x-ray patterns of DNA and hence for the discovery of its double helix structure (the detailed circumstances under which these events took place are admirably told in [26–28]).

3.1. Diffraction by an atomic helix. The CCV formula

The derivation of the CCV formula, which we give here [29], is somewhat more economical than that in the original paper [23] (if the reader finds the present demonstration too curt, a very detailed, step-by-step derivation can be found in [16b]). We note that the plane waves in the scattering amplitude in equation (1) are not particularly adapted to the cylindrical geometry of a helix. Cylindrical waves should be more appropriate. As elementary solutions of the linear wave equation, plane waves and cylindrical waves should be related by a linear relationship. This is the so-called *Jacobi–Anger expansion* of a plane wave in cylindrical waves, which can

be found in any mathematical physics textbook [30]:

$$e^{i\mathbf{q}\cdot\mathbf{r}} = \sum_n e^{in(\psi_q + \pi/2)} J_n(q_\perp r) e^{-in\varphi} e^{iq_z z}, \quad (5)$$

where (r, φ, z) and $(q_\perp, \psi_q, q_z)$ are the cylindrical coordinates of $\mathbf{r}$ and $\mathbf{q}$ around a vertical axis and $J_n(x)$ are the regular cylindrical Bessel functions of integer order n . The mathematically oriented reader will easily derive this 19th century formula by expanding the plane wave in a Fourier series of $\exp(-in\varphi)$, which is what equation (5) is. The Fourier coefficients, calculated in the standard reciprocal way, are found to be the integral representation of the Bessel functions J_n times the phase factors shown in equation (5) [30].

Now, on a continuous (right-handed) helix of radius r and pitch P , the cylindrical coordinates $\mathbf{r}_j = (\rho_j, \varphi_j, z_j)$ of identical atoms sitting at regular axial intervals p_a are given by $\rho_j = r$, $\varphi_j = \varphi_0 + 2\pi(z_j - z_0)/P$ and $z_j = z_0 + jp_a$, relative to an arbitrary atom (r, φ_0, z_0) taken as the origin. Introducing these values in the scattering amplitude in equation (1), where the plane waves have been given their Jacobi–Anger form (5), and using the identity of equation (3) once more, one obtains the celebrated CCV formula:

$$A(\mathbf{q}) = \frac{2\pi}{P} f(q) \sum_{n,m} J_n(q_\perp r) e^{in(\psi_q - \varphi_0 + \pi/2)} e^{iq_z z_0} \delta\left(q_z - n\frac{2\pi}{P} - m\frac{2\pi}{p_a}\right), \quad (6)$$

where $f(q)$ is the single atomic scattering factor. To gain some familiarity with this formula, let us first consider a few particular cases, which we shall need later.

3.2. Continuous helix

The case of a continuous helical wire can be obtained by taking the limit $p_a \rightarrow 0$ in the above formula. Only the terms with $m = 0$ survive and we get

$$A_{\text{cont}}(\mathbf{q}) = \frac{2\pi}{P} f(q) \sum_n J_n(q_\perp r) e^{in(\psi_q - \varphi_0 + \pi/2)} e^{iq_z z_0} \delta\left(q_z - n\frac{2\pi}{P}\right). \quad (7)$$

This result can also be obtained directly from the Jacobi–Anger formula (5) by integration over the linear coordinate of a continuous helix. As expected, we have a pattern of layer-lines corresponding to the layer-planes $q_z = n2\pi/P$. Along the n th layer-line, the intensity, proportional to $f^2(q) J_n^2(q_\perp r)$, oscillates like the square of the Bessel function and attenuates at large scattering angles due to the decrease of both the amplitude of the Bessel oscillations and the atomic structure factor.

The diffraction pattern of a continuous helix is seen from equation (7) to be invariant under rotation around the helical axis and to be north–south and east–west symmetric, the same oscillations of the Bessel functions being reproduced in the four quadrants. Remember now the general behaviour of these functions [31]: when the distance $q_\perp$ to the meridian increases, $J_n(q_\perp r)$ remains small until $q_\perp r \approx n$, where it rises to its first maximum. So, as the layer-line label n increases, the first maximum on the successive layer-lines occurs further and further away from the meridian. It follows that the diffraction pattern should exhibit a prominent St Andrew’s cross made of the first maxima of the (squared) Bessel functions in the four quadrants. An example is shown in figure 6(a) for a continuous phosphorus helix. From these first maxima, the intensity oscillates away from the meridian and, thus, remains confined to the east and west angles of the central cross. So, *there is no intensity in the meridian angles of the central cross*. From the reciprocity between real and reciprocal spaces expressed in the dimensionless argument $q_\perp r$ of the Bessel functions, it is further easy to see that *the arms of the central cross are almost perpendicular to the rising slope of the helix*. From this argument, it follows that the meridian angle α of the cross is related to the helix radius r via the relationship $r = P/2\pi \tan \alpha/2$. In figure 6(a) the angle is about 60° , as for DNA.

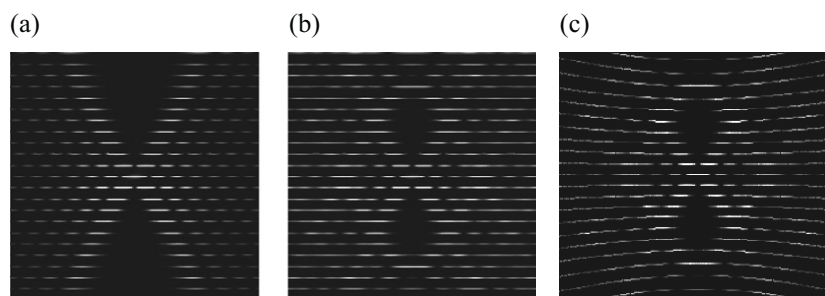


Figure 6. (a) Simulated patterns of x-ray diffraction by a *continuous* phosphorus helix. Note the central St Andrew's cross, the Bessel oscillations towards the east and west of the layer-lines and the absence of intensity in the north and south meridian angles. (b) Simulated diffraction pattern by an *atomic* phosphorus helix; note the diamond pattern and the nearly empty north and south diamonds. (c) The same as in (b), but this pattern includes the effect of the finite radius of the Ewald sphere for x-rays, which give layer-lines their hyperbolic shape.

3.3. Continuous cylinder

One can generate a continuous cylinder either by rotating a continuous helix one complete turn around its axis or by translating it through one period P along the axis. The corresponding diffraction amplitude is obtained from equation (7) by integration either over φ_0 or over z_0 ; only the $n = 0$ term survives, giving

$$A_{\text{cyl}}(q) = 2\pi r \sigma f(q) J_0(q_{\perp} r) \delta(q_z), \quad (8)$$

where σ is the area density of 'smeared' atoms. This expression (squared) also gives the exact intensity of the equatorial layer-line of a continuous helix (equation (7)) and the leading equatorial term for an atomic helix (equation (6)). Several illustrations will be discussed later in this review.

3.4. Atomic helix

Coming back to the full CCV formula of equation (6), we ask what new features appear in the diffraction pattern of a continuous helix (figure 6(a)) when introducing the atomicity? The additional axial periodicity p_a in the real space of the molecule causes a new periodicity in the reciprocal space pattern. This is mathematically produced by the extra summation over the integer m in equation (6).

To understand this, we first note that the pattern is still organized in layer-lines as demanded by the periodicity of the diffracting object and expressed by the delta function in equation (6). If we call P_a the true axial repeat of the helix, regularly spaced layer-planes should appear at $q_z = 2\pi l/P_a$, where l is the integer labelling the layer-planes and layer-lines. Once l is given, the intensity along the corresponding layer-line is governed by equation (6) where the various integer couples (n, m) must be chosen to satisfy the 'selection rule' of the delta function:

$$\frac{l}{P_a} = \frac{n}{P} + \frac{m}{p_a}. \quad (9)$$

Figure 6(b) shows one example of the *calculated* layer-line pattern produced by scattering Cu K α radiation ($\lambda = 0.15$ nm) from a mono-atomic helix with the geometrical parameters of the *phosphorous helix* in a single strand DNA backbone. The parameters are $P_a = P = 10p_a = 3.4$ nm, $r = 1$ nm and the selection rule equation (9) is, therefore, $l = n + 10m$. For example,

on the equatorial layer-line $l = 0$, equation (9) selects the first couple ($n = 0, m = 0$); i.e. J_0 is the leading term of the diffraction amplitude, as in equation (8). The following, much weaker term on the equator layer-line would be ($n = 10, m = -1$), etc. For the next five non-equatorial layer-lines equation (9) will select the leading terms (1, 0), (2, 0), (3, 0), (4, 0), (5, 0) and ($-5, 1$), while from the sixth to tenth layer-lines one will select ($-4, 1$), ($-3, 1$), ($-2, 1$), ($-1, 1$), (0, 1), etc. Keeping in mind that the index n is the order of the Bessel functions in equation (6) (and that $J_{-n} = (-1)^n J_n$), one sees that the St Andrew's cross pattern of a continuous helix tends to be repeated periodically in the meridian direction every ten layer-lines. This creates a diamond pattern in the diffraction picture attenuated at large momentum transfers by the atomic scattering factor $f(q)$. *The meridian diamonds are approximately empty of intensity*, as is clearly demonstrated by the illustration of figure 6(b). This important feature is characteristic (but not exclusive) of a helical distribution of matter in the diffracting object. However, one effect of the finite radius $2\pi/\lambda$ of the Ewald sphere in reciprocal space is that the repetition of the cross pattern every ten layer-lines may not amount to a mere folding back of the arms of the cross of figure 6(a) from the fifth layer-line upwards. For instance, while there is no intensity on the meridian of the $l = 2$ layer-line in figure 6(b), there *is* strong intensity on the meridian of the $l = 8$ layer-line, in spite of the fact that both layer-lines involve J_2 as the leading term. This is due to the fact that, when the layer-line index l increases, the argument $x = q_\perp r$ of the Bessel function increases and eventually reaches non-negligible values on the meridian: the geometrical construction of figure 2 shows that $q_\perp$ has a component in the backward direction, which increases with l according to $q_\perp = 2\pi/\lambda - [(2\pi/\lambda)^2 - l^2(2\pi/P_a)^2]^{1/2}$. For $l = 8$, $x \approx 2.7$ ($J_2 = 0.47$), which is close to the *first maximum* of $J_2(x)$ ($J_2 = 0.486$ at $x = 3.1$). On the other hand, from the same argument, one might expect a non-zero intensity on the meridian of the $l = 9$ layer-line where $x \approx 3.5$, but this happens to be close to the *first zero* of $J_1(x)$ at $x = 3.8$; so, *fortuitously*, there is no or very weak intensity there for the particular parameters of the phosphorus helix illustrated in figure 6(b).

The diffraction pattern is quite sensitive not only to the helix parameters but also to the angular orientation of the atomic helix around its axis [32]. In the numerical simulation of figure 6(b), the intensities have been averaged over the helix angular orientation (as in a gel fibre). Note that figures 6(a) and (b) are direct numerical applications of the CCV formula and do not incorporate the effects of the geometrical projection from the centre of the Ewald sphere, as sketched in figure 2. The layer-lines are straight and equidistant as if the Ewald sphere had a very large radius (this would be appropriate for HEED). In the simulation of figure 6(c), the layer-lines are projected from the centre of the Ewald sphere corresponding to Cu K_α x-rays and indeed they do appear hyperbolic and chirped, as in the real x-ray patterns. A more detailed discussion of the CCV analytical formulae can be found in the original CCV paper [23] and in [32].

The form factor of a multi-atom helical biopolymer will be obtained from equation (6) by adding the amplitudes of the separate mono-atomic helices multiplied by the appropriate phase factor $\exp(iq \cdot r_{0j})$, where r_{0j} is the position of the reference atom in the j th helix [23]. Note that, amongst the St Andrew's crosses associated with each of these helices, *the one with the narrowest meridian angle will correspond to the helix of largest radius, that is the one making the outer part of the polymer*. The meridian diamonds created by this outer helix should be nearly void of intensity. In the process of discovering the structure of DNA, as we shall see, it was this peculiar feature of the x-ray patterns, which revealed that the phosphorus helix of the sugar-phosphate backbone was at the periphery and not inside the molecule.

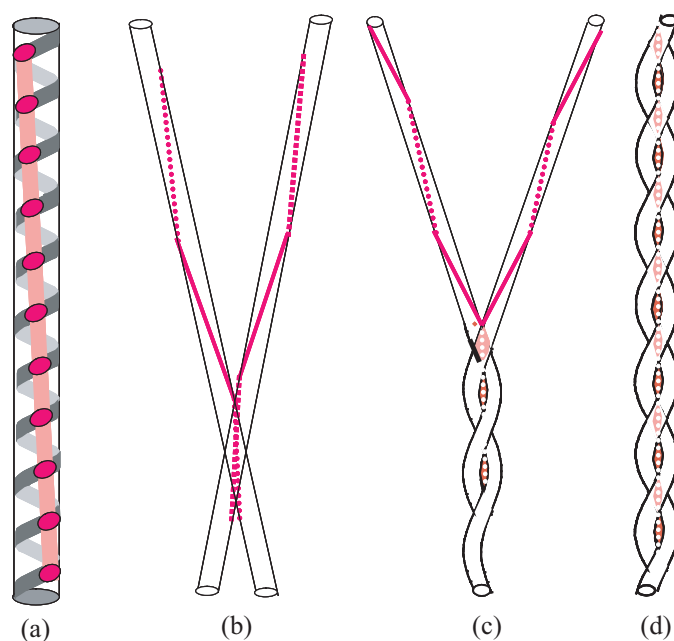


Figure 7. The mechanism of coiling around each other of two polypeptide chains in water. In figure 7(a) the hydrophobic amino acids of a single chain are represented by ovals aligned along a helical stripe. In figure 7(b) to (d) the hydrophobic stripes of the two chains are facing each other along the region of contact of the coiled cell.

3.5. Atomic coiled-coil

Due to steric and free-energy minimizing molecular interactions or/and to specific conditions of biosynthesis, biological helices are frequently coiled around each other. Examples are tropocollagen, keratin and tropomyosin. These are protein polymers consisting of two or more identical polypeptide helical chains wound around each other. The winding is such that the hydrophobic side groups interdigitate along the line of contact between the chains and thereby shield each other from the aqueous environment [33] (see the illustration in figure 7). Crick and Pauling independently suggested the possibility of such a ‘coiled-coil’ arrangement and pointed out that this could modify significantly the x-ray pattern of the individual helices as compared to their straight configuration and hence influence the interpretation of diffraction data from fibrous proteins. The historically important example of keratin will be considered in the next section. One should be careful not to confuse a coiled-coil with, say, the double helix structure of DNA where two helices are also coiling around each other but are doing so by sharing a common straight axis.

An elaborate derivation of the molecular form factor of a coiled-coil was given by Crick [34, 35]. We give here a compact derivation based again on the Jacobi–Anger expansion of equation (5). Referring to figures 8(a) and (b), the coil is described as a ‘minor’ helix of radius r winding at a rapid angular rate ω around a virtual ‘major’ helix of radius R pitch P and slope α , itself winding around the vertical axis at the slower rate Ω . A point $\mathbf{P}$ of the coiled-coil is written in the following parametric form

$$\mathbf{P}(t) = \mathbf{Z}(t) + \mathbf{R}(t) + \mathbf{r}(t) + \mathbf{z}(t), \quad (10)$$

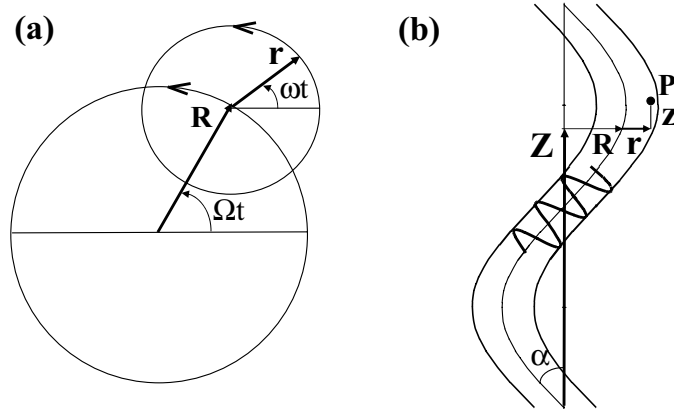


Figure 8. The geometrical description of a coiled-coil: (a) view along the axis; (b) view perpendicular to the axis (see text).

where t is the parameter to which the Z -coordinate of the current point on the axis of the major helix is taken to be proportional:

$$\mathbf{Z}(t) = \left(0, 0, Z = \frac{P\Omega t}{2\pi} \right). \quad (11)$$

From inspection of figure 8, the other three coordinates in equation (10) are clearly *periodic* functions of t . If the slope α is not too large, the position vectors have the simple forms (up to terms of order α^2 ; see [34] for the exact forms):

$$\begin{aligned} \mathbf{R}(t) &= R(\cos \Omega t, \sin \Omega t, 0), \\ \mathbf{r}(t) &= r[\cos(\omega + \Omega)t, \sin(\omega + \Omega)t, 0], \\ \mathbf{z}(t) &= (0, 0, r \sin \alpha \sin \omega t), \end{aligned} \quad (12)$$

where the reference point ($t = 0$) has been conveniently chosen to lie on the x -axis. When t increases by $2\pi/\Omega$, the major helix makes one turn around the z -axis; meanwhile the point $\mathbf{P}$ makes ω/Ω turns around the precessing axis of the minor helix and hence $\omega/\Omega + 1$ turns around the z -axis. Both helices are assumed to be right-handed. If one or the other helix is left-handed, it suffices to change the sign of ω or Ω in the above relations.

Now, inspection of the diffraction amplitude in equation (1) shows that, because of equation (10), the exponential $\exp(i\mathbf{q} \cdot \mathbf{P})$ breaks up into the product of three exponentials each with a periodic exponent, times a fourth exponential $\exp(iq_z Z)$ with a linear exponent. The latter factor gives to the amplitude the form of a *Fourier series* for a mono-atomic coiled-coil with discrete, regularly spaced values of the parameter t , or a *Fourier transform* for a smooth coiled-coil with a continuous t . For each of the three periodic factors, we can use the Jacobi–Anger development of equation (5). That introduces a triple sum over products of three Bessel functions times the product of four exponentials with *exponents linear in t* . Collecting these latter four exponents, the discrete sum over t is then effected by making use once more of the formula in equation (3). As for the straight helix, that selection formula quantizes the possible values of the axial momentum transfer, $q_z = l2\pi/P_{cc}$, where l numbers the layer-lines and where P_{cc} is the *true period* of the coiled-coil. The end result for the amplitude along the

l th layer-line is, therefore,

$$A_l(\mathbf{q}) = f(q) \sum_{n,p,q,m} J_n(q_\perp R) J_p(q_\perp r) J_q(q_z r \sin \alpha) \exp i \left[(n+p) \left(\frac{\pi}{2} + \psi_q \right) + q\pi \right] \\ \times \delta \left(q_z - \frac{2\pi}{P_{cc}} \left[n + \left(\frac{\omega}{\Omega} + 1 \right) p + \frac{\omega}{\Omega} q \right] - m \frac{2\pi}{p_c} \right), \quad (13)$$

where $p_c = p_a \cos \alpha$ is the axial projection of the atomic repeat distance p_a along the minor helix. The δ -function expresses the selection rule for the quartet of integers (n, p, q, m) once the layer-line index l is chosen. The case of a continuous coiled-coil is obtained by setting $p_a = 0$ and thus keeping only the $m = 0$ terms, as in the straight-helix case. The CCV formula itself for a straight helix in equation (6), is a particular case of the general formula equation (13) corresponding to $\alpha = 0$ and $r = 0$, in which case $P_{cc} = P_a$, $p_c = p_a$ and only the terms with $p = q = 0$ survive.

For a single strand polypeptide coiled-coil it is necessary to add the contributions to the total amplitude made by all the various atoms of the reference amino acid residue. That will introduce the different radii of the atoms in the minor helices and extra phase factors in each of the contributions to take account the angular and height offsets of these helices. In a multi-strand rope, it is further necessary to take into account the different axial positions and phases of the single strand coiled-coils making up the rope. Further details are given in [34].

4. Diffraction by helical biopolymers

4.1. Protein α -helix

The α -helix is one frequently occurring mode of primary folding in proteins. It was first proposed in 1951 by Pauling *et al* [22a], although the idea occurred to Pauling several years earlier [26–28]. Novel, planar modes of folding, known as β -sheets, were also advanced at that time by Pauling and Corey [22b] but they will not be discussed here. In globular proteins, α -helical segments and β -sheets follow each other in various proportions and are linearly connected in a continuous path by amino acid sequences with irregular shape.

A representation of the α -helix is shown in figure 9(b) [22a]. The regularly repeating unit or ‘residue’ of the protein chain is an *amide group*—(C_α CONH)—(figure 9(c)) to which one variable side group R (among 20 different kinds possible) is covalently attached via a C_α – C_β carbon linkage. Free rotation around the C_α –C and C'_α –N single bonds is what confers flexibility to the polypeptide chain. Essential for the conception of the α -helix was the fact that the amide group is *planar and rigid*, due to the partial double bond character of the so-called C–N peptide bond (in figure 9(c), atoms C_α , O, H and C'_α are in the plane of the figure). The planarity of the amide considerably restrains the possible *regular* structures of the chain. Pauling found that the rotations around the C_α carbons allow for the construction of only two models of regularly twisted chains in which (i) all amides occupy equivalent positions and (ii) the rate of twist is such that the N–H group of one amide makes a nearly vertical hydrogen bond with the C–O group of another amide placed a few repeat distances above it in the (vertically oriented) helix (figure 9(b)). This multiple *intra*-molecular hydrogen bonding is what stabilizes the helix against collapsing to structures with irregular or other kinds of rotations around the C_α atoms (the β -sheet mode of folding involves *inter*-molecular hydrogen bonds between parallel, extended amino acid chains). Bonding between every third amide appears to be the most frequent helical conformation. The canonical α -helix (figure 9(b)), which came to be generally accepted as representative of helical proteins [37] consists of a right-handed helix of period $P_a = 5.4 \text{ \AA}$ comprising a non-integer number of about 3.6 residues

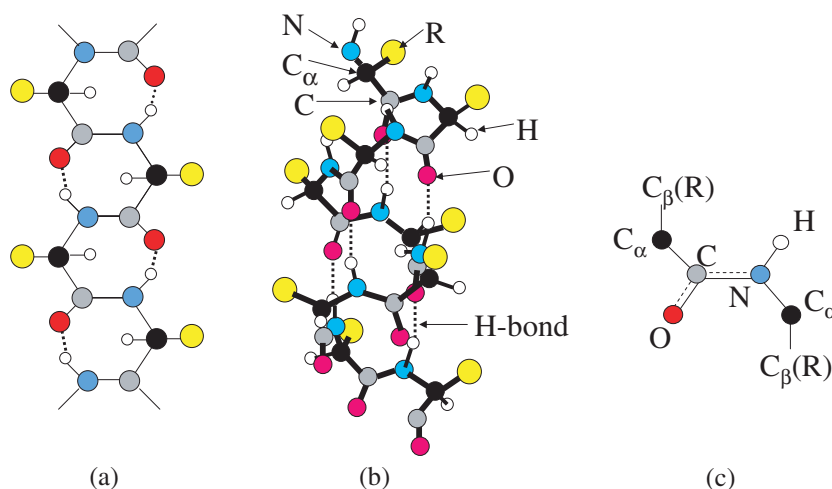


Figure 9. (a) Bamford *et al*'s planar model of a polypeptide chain [36]. (b) Pauling's α -helix model [22a]. (c) The planar amide group of amino acid chains.

per period, i.e. with an axial residue separation of approximately $p_a = 1.5 \text{ \AA}$. The atomic helix has a *true period* of $P_a = 5P = 27 \text{ \AA}$ —that is, 18 amino-acid residues in 5 turns (18/5 helix). The radial extension of the amide helix goes from the inner nitrogen atoms at $1.6 \text{ \AA}$ to the outer C α carbons at $2.3 \text{ \AA}$ from the helix axis, while the first carbon atoms of the side chains, the so-called C β carbons, are at $3.35 \text{ \AA}$ (figure 9(c)). The remainder of the side chains is variable.

The α -helix is compared to the pseudo-planar structure of figure 9(a) published at about the same time by Bamford *et al* [36] for the synthetic polypeptides poly- γ -methyl-L-glutamate (invariable side chain C β H $_2$ -CH $_2$ -COO-CH $_3$) and poly- γ -benzyl-L-glutamate (invariable side chain C β H $_2$ -CH $_2$ -COO-C $_6$ H $_5$). As explained below, the latter structure was quickly abandoned in favour of the α -helix.

4.2. Diffraction by α -helices

In contrast to the case of the DNA structure, the discovery of which, as we will see shortly, depended crucially on x-ray fibre diffraction patterns, the conception of the α -helix just described owed very little to the pre-existing x-ray fibre diffraction data. In the present instance [26–28] Pauling's idea was even developed *in defiance* of some characteristics suggested by the existing x-ray patterns of hair, wool and related fibres. One such pattern is shown in figure 10(a) [18].

The meridian arc in figure 10(a) (marked by an arrow) and in several similarly diffuse patterns of unstretched keratinous fibres was interpreted [18] as arising from an axial repeat of about $5.1 \text{ \AA}$ along the axis of the protein chain called α -keratin by Astbury (to distinguish it from the stretched, β -keratin state; evidently, the existing protein nomenclature of α -helix and β -sheet has its origin with Astbury's work on keratin). On the basis of this and several other crystallographic data, Astbury constructed a *planar folded, ribbon-like model* of α -keratin involving the repetition of hexagonal ring systems of amino acids, somewhat analogous to the known cellulose structure (see [18], p 101 for a sketch of the proposed structure). Contrasting with this interpretation, the theoretical α -helix conceived by Pauling predicted a $5.4 \text{ \AA}$ axial repeat apparently incompatible with Astbury's $5.1 \text{ \AA}$ repeat. In fact, the hard x-ray evidence

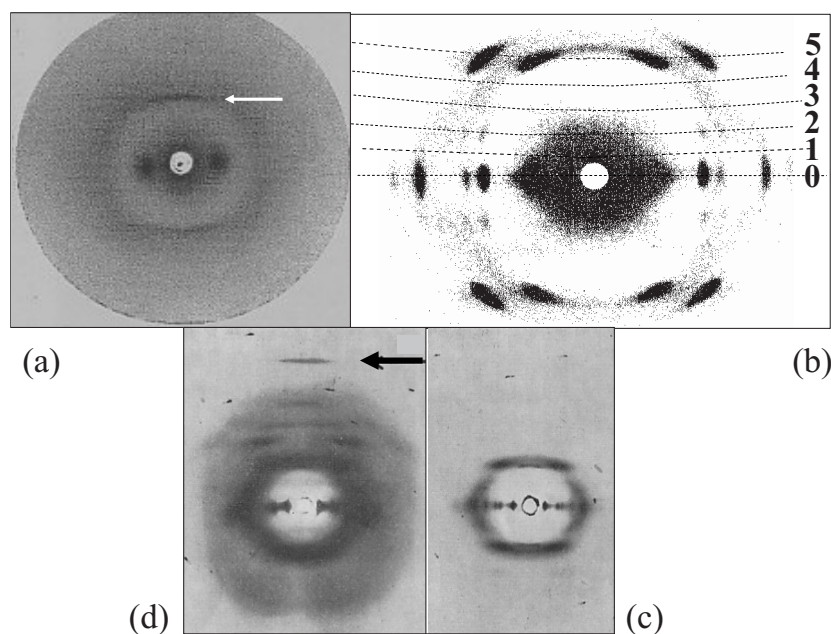


Figure 10. (a) X-ray fibre diffraction pattern of wool keratin [18]; the white arrow indicates the position of the principal 5.1 Å meridian reflection. (b) Diffraction pattern of an artificial poly-gamma-methyl-glutamate crystal fibre [36] with the first few layer-lines superimposed (---); the strong fifth layer-line occurs at a 5.4 Å spacing. (c) X-ray picture of poly-gamma-benzyl-glutamate crystalline fibre [38]; similar to (b) but reduced about 3 times. (d) Same as in (c) for a fibre tilted by 60° towards the x-ray beam and showing the meridian 1.5 Å reflection (black arrow).

in support of the α -helix came only after (and some was even triggered by) the introduction of the concept, which thereby rapidly gained general acceptance.

The first piece of evidence was provided by a much richer diffraction pattern, shown in figure 10(b), obtained by Bamford *et al* [36], for the above-mentioned synthetic polypeptides. For the interpretation of this pattern, these authors suggested the quasi-planar, ribbon like model shown in figure 9(a). But a reinterpretation was soon proposed by Pauling and Corey [22c] and by Cochran *et al* [23] based on the new 18/5 helically folded structure described previously (figure 9(b)). This constituted the very first successful application of the CCV theory [23], which these latter authors had just developed for the purpose. To understand the general appearance of the diffraction picture, let us recall that, according to the theoretical expression in equation (6), layer-lines to which only high-order Bessel functions contribute will tend to be weak. Indeed, not only does $J_n(q_{\perp}r)$ decrease in amplitude when n increases but this function remains vanishingly small until $q_{\perp}r$ reaches n : if n is large this requires large r , which may fall outside the molecule or large $q_{\perp}$, which may fall outside the recorded diffraction image or outside the range of the atomic scattering factor. In contrast layer-lines to which low-order Bessel functions contribute are expected to be strong [23]. The equatorial layer-line primarily involves J_0 and always shows strong features, whose positions are dictated by $\sum_j f_j J_0(q_{\perp}r_j)$ modulated by the structure factor of the fibre lattice. The peaks on the equatorial line thus give combined information on the radii r_j of the contributing atomic chains and on the lattice structure. Pauling and Corey [22c] give a detailed discussion, which need not be reproduced here, of the intensity distribution expected for a hexagonal packing of α -helices and found excellent agreement with the equatorial data of Bamford *et al* [36]. Consider next

the non-equatorial layer-lines whose expected positions are superimposed on the diffraction pattern in figure 10(b). From the selection rule of equation (9), which here takes the concrete form $l = 5n + 18m$, it appears that the $l = 1$ layer-line selects the first couple ($n = -7, m = 2$) and hence involves J_7 , while the $l = 4$ selects ($n = 8, m = -2$) and involves J_8 . So the first and fourth layer-lines should be very weak or absent, which is just what is observed in the pattern of the synthetic polypeptide in figure 10(b). The second ($l = 2, n = 4, m = -1$) and third ($l = 3, n = -3, m = 1$) layer-lines involve J_4 and J_3 , respectively, and should show relatively weak features at large distance from the meridian, as confirmed by figure 10(b). In contrast, strong peaks, involving J_1 , should occur along the $l = 5$ layer-line ($n = 1, m = 0$), *symmetrically on each side of the meridian axis*. This arises from the coherent contributions of the groups of 3.6 amides repeated every period $P_a = 27/5 \text{ \AA} = 5.4 \text{ \AA}$ of the main chain. This prediction [22c, 23] again agrees closely with the observation of very strong, *off-meridian peaks* on the fifth layer-line in figure 10(b), but it conflicts with the *meridian arc* at $5.1 \text{ \AA}$ of Astbury's data on wools in figure 10(a). The reason for the discrepancy will be considered below.

The second piece of evidence supporting Pauling's theoretical model was provided by a measurement by Perutz [38] of x-ray diffraction at large scattering angles along the meridian, for substances purported to contain the α -helix. Perutz pointed out that a strong x-ray *meridian* reflection should arise from the axial spacing $p_a = 1.5 \text{ \AA}$ between the amino acid residues (the term $n = 0, m = 1$ on the $l = 18$ layer-line). This feature, which had lain far outside the resolution of the existing planar photographs, such as those of figures 10(a) and (b), was indeed detected by Perutz [38], with a cylindrical photographic film (with axis perpendicular to the fibre axis), at the predicted large angle on the meridian for artificial polypeptides, for several natural proteins of the α -keratin type and even for globular haemoglobin (confirming that the latter must incorporate a good deal of α -helices). This is shown in figure 10(c) for poly- γ -benzyl-L-glutamate. In this experiment, the $1.5 \text{ \AA}$ reflection was reinforced by inclining the fibre towards the incident x-ray beam and rocking it back and forth through the expected Bragg direction for that spacing in the crystalline polypeptide fibre (the angle is given by Bragg's law $2p_a \sin \theta = \lambda$, i.e. $\theta = 30^\circ$ from the horizontal plane, for the copper K_α line $\lambda = 1.5 \text{ \AA}$).

The last, clinching piece of evidence for the validity of the general α -helix structure came from the resolution of the discrepancy between the $P = 5.4 \text{ \AA}$ off-meridian reflection expected from the model helix and the meridian reflection observed by Astbury for α -keratin at the smaller spacing of $5.1 \text{ \AA}$. Crick [35] and, independently, Pauling and Corey [39] suggested that the latter data could be explained by assuming that the α -helix in Astbury's fibre, instead of having a straight axis, might be deformed into a coiled-coil as a result of the natural tendency of filamentous proteins to pack by coiling around each other in pairs, triplets or higher order ropes [35], rather than in the strictly parallel arrangement realized in Bamford *et al*'s synthetic polypeptide crystals.

The simple but *incorrect* way to take the coiling into account is to view the deformed helix as being locally straight, with its axis inclined at spherical angles (α, ϕ) from the vertical. The position of the incriminated meridian peak could then be evaluated by averaging the diffracted intensity of the straight but inclined helix over its azimuth ϕ . The apparent axial repeat would be approximately $P \cos \alpha$ for those coil portions leaning forwards or backwards ($\phi = 0$ or π) in the vertical plane parallel to the x-ray beam, while the repeat would be the full P for those portions leaning sideways ($\phi = \pm\pi/2$) in the plane perpendicular to the beam. The precession of the coil axis would thus lead, in the diffraction pattern, to a horizontal meridian streak at the right position of $5.1 \text{ \AA}$, provided the inclination angle α be taken equal to 18° since $5.1 = 5.4 \cos 18^\circ$.

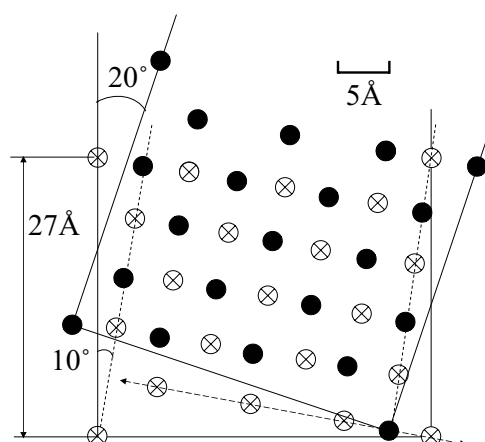


Figure 11. The interdigitated 'net-diagrams' of two identical Pauling α -helices [40] crossing each other at 20° (see text). The crosses indicate the positions of the side chains in the net-diagram of one helix. The double arrowed, broken line indicates the helical chain of chemically linked amino acids. The black dots are those of the net-diagram, turned over, of the second helix.

The *correct* calculation, however, requires *summing the diffracted amplitudes* of the various helix portions rather than their intensities, and this is just what motivated Crick to develop the diffraction theory for a coiled-coil [35] as summarized in section 3.5. As concrete applications of his theory, Crick considered the case of two-strand and three-strand ropes. He first showed that it was indeed energetically and sterically possible for two α -helices to coil around each other in such a way as to interdigit their side groups in a regular fashion, one side group of one helix falling in a hole between side groups of the other. He found that, given the geometrical parameters of Pauling's α -helix, *every seventh side group* of one helix would neatly fall into *every seventh hole* of the other, so that the minor helical axes would each follow a major helix with a pitch angle of about 10° only, much less than the 18° used previously by the inclined helix model.

The ingenious way by which Crick discovered this arrangement is by using the concept of 'net-diagram', which is a planar, conformal representation of the straight cylindrical molecule [35]. One imagines wrapping a piece of paper around the vertical helix, marking the paper where the side groups protrude and unwrapping the paper flat again. One obtains an array of points (figure 11, crosses) slanting at about 10° from the vertical. The slanting is produced by the non-integral number (3.6) of residues per helical period. The points can be linked by sets of straight parallel lines, which can be chosen in a number of directions. The lines of the set, which make the small angle with the horizontal (double arrowed, dashed line in figure 11), represent the flat development of the helix of adjacent, chemically linked side groups. Then, one tries to superpose a second, identical net-diagram, *turned over (back to back) and rotated by 20°* (figure 11, black dots), in such a way that the marks of one fall between the marks of the other in a systematic fashion. The geometry of Pauling's α -helix is such that this can be done in only one way shown in figure 11. This net-diagram construction demonstrates that two straight α -helices can be crossed at an angle of about 20° between them and have their side groups interdigitated in the region of contact, as suggested by figure 7(b). Every seventh side group of one helix will neatly fall into every seventh hole of the other. The number 7 arises because it is about twice the number (3.6) of residues per α -helix turn. Then, a slight twist deformation of the two helices (costing very little energy [35]) will ensure that

they can interlock in this way along their entire length, as illustrated in figures 7(c) and (d). Such was the elegant solution found by Crick [35].

Crick then considered the contribution of one of the strands to the diffraction amplitude. The appropriate geometrical parameters for that strand turn out to be $\omega/\Omega = 36$, $p_a = 1.5 \text{ \AA}$, $p_c = p_a \cos 10^\circ = 1.48 \text{ \AA}$, $P_{cc} = Mp_c = 186 \text{ \AA}$, where $M = 36 \times 7/2 = 126$ is the number of atoms in the true period. These numbers apply to every family of atoms defining one mono-atomic coiled-coil of the strand, for instance the C_α coil or the nitrogen coil, etc (with, of course, different minor coil radii). The radius of the major helix is taken to be about $5.2 \text{ \AA}$. The selection rule in the δ -function of equation (13) then takes the concrete form

$$l = n + 37p + 36q - 126m. \quad (14)$$

We will focus here only on the two major *meridian* features discussed previously, namely the $5.1 \text{ \AA}$ and the $1.5 \text{ \AA}$ reflections (Crick [35] discusses other features). On the meridian ($q_\perp \approx 0$), only the terms with $n = p = 0$ in equation (13) survive due to the vanishing of $J_n(0)$ for $n > 0$. The selection rule (14) then simplifies to

$$l = 36q - 126m, \quad (15)$$

36 and 126 being multiples of 18; l is likewise a multiple of 18. Since the layer-line separation is $1/P_{cc}$ (modulo 2π), it follows that non-vanishing meridian intensities appear only at multiples of $18/P_{cc}$ corresponding to real spacings submultiples of $P_{cc}/18 = 10.35 \text{ \AA}$. The two submultiples at $5.17 \text{ \AA}$ and $1.48 \text{ \AA}$ spacings appear to fit the observed reflections at $5.1 \text{ \AA}$ and $1.5 \text{ \AA}$, respectively, to within one or two per cent, well within the experimental accuracy. These reflections belong to the triplets ($l = 2 \times 18, q = 1, m = 0$) and ($l = 7 \times 18, q = 0, m = 1$), respectively. Examination of the argument of the respective Bessel terms $J_1[2\pi(l/P_{cc})r \sin \alpha] = J_1(0.21r)$ and $J_0[2\pi(l/P_{cc})r \sin \alpha] = J_0(0.74r)$ in the diffraction amplitude then indicates that the 5.1 reflection must be assigned primarily to *the first few carbon atoms of the side chains* for which $r \approx 3.35 \text{ \AA}$, while the 1.5 reflection comes mainly from *the atoms of the main amide chain* for which $1.6 \text{ \AA} < r < 2.3 \text{ \AA}$.

As stated previously the correct treatment of a multi-strand coiled-coil requires adding up the contributions of each strand to the diffraction amplitude, each contribution being multiplied by the appropriate phase factor to take into account the relative position of the strand within the rope. For a two-strand rope in which the strand axial separation is half the coiled-coil period $\Delta z = P_{cc}/2$, the overall phase factor multiplying the single strand amplitude $A_l(q)$ of equation (13) will then be $1 + \exp(2\pi i l \Delta z / P_{cc}) = 1 + \exp(\pi i l) = 2\delta_{l,2s}$, where s is an integer; i.e. only the even layer-lines survive with an intensity *4 times* that of the single strand. For a three-strand rope of axially equidistant coiled-coils, the overall phase factor is $1 + \exp(2\pi i l / 3) + \exp(4\pi i l / 3) = 3\delta_{l,3s}$, and only every third layer-line survives with an intensity *9 times* that of the single strand. The strong meridian features discussed above occurred on the 2×18 and 7×18 layer-lines, which are both even and multiples of 3 layer-lines, so these features are substantially stronger for the rope than for each strand in isolation. In their own version of the coiled-coil idea, Pauling *et al* [37], Pauling and Corey [39] describe several other possible rope structures.

We conclude this discussion of the diffraction by protein α -helices by quoting Crick [35]: ‘The coiled-coil hypothesis explains in a straightforward manner the co-existence of the 5.1 and the $1.5 \text{ \AA}$ reflections on the meridian. . . . It is no longer possible to claim that the α -keratin pattern cannot be explained in terms of the α -helix’. In other words, a new and important theoretical concept in molecular biology had been vindicated after several years of apparent conflict with experimental data and reconciled with these data thanks to the introduction of a further refinement of the concept, namely the higher-order hierarchical organization of a rope structure.

The reader will note that it has been possible to draw quite a number of conclusions on the shape of complex molecular helices just ‘*by inspection*’ of the general structure of the theoretical expressions for the helical diffraction amplitudes in equation (6) and (13), that is without doing any hard machine calculations. In one of the earliest reviews of x-ray protein crystallography, Crick and Kendrew [40] commented: ‘*Armed with the appropriate theory it is often possible to recognize the helical nature of a fibre structure at a glance, and sometimes to specify the main parameters of the helix and its subunits with very little trouble indeed*’. In any case, detailed numerical calculations would have been quite taxing when the theory was developed in the early 1950s but they are clearly required if a quantitative account is to be given of the intensity distribution along the layer-lines of well-resolved diffraction patterns such as that of figure 10(b).

4.3. X-ray diffraction by DNA

The penetrating ‘glance’ of the trained eye talked about by Crick and Kendrew [40] was going to work wonderfully for the discovery of the helical structure of DNA. Indeed, speaking of the role of his CCV theory for the discovery, Crick quipped in a later interview [27] ‘*... it did mean that I had the expertise at my fingertips*’.

But, before we engage in the description of the diffraction by DNA, it might be helpful for the general reader of this review to be reminded of the principal chemical components of the molecule. This is shown schematically in figure 12.

4.3.1. The A-DNA and B-DNA patterns. We shall consider here the two x-ray diffraction photographs of DNA shown in figure 13. These are historically quite unique pictures since they have contributed most directly to precipitate the discovery of the DNA structure half a century ago. Without them the discovery might have been postponed for a number of years. The dramatic events and the people involved in obtaining these photos have been the subjects of popular books [41, 42] as well as of two exciting historical accounts of early molecular biology [27, 28]. One recent opinion on the somewhat controversial role of the principal protagonists at the source of figure 13 can be found in a short commentary by Fuller [43].

The two pictures of figures 13(a) and (b) are usually referred to as the A and B patterns, respectively. A-type patterns were first obtained by Wilkins and Gosling in 1950 but were not initially published in print [28, 43]. Pattern B, the famous ‘*photograph 51*’ as it is often referred to in the popular scientific literature [27, 44], was taken in May 1952 by Franklin and Gosling but was similarly set aside, for nearly a year. It was published only in March 1953 [47] simultaneously with the celebrated paper of Watson and Crick announcing their double helix model of DNA [45] and a related paper by Wilkins *et al* [46], all in the same issue of *Nature*. Pattern A was discussed at length in a detailed report by Franklin and Gosling [48] shortly thereafter.

Above all, pattern B is justly presented as the key crystallographic input, which led to the great double helix discovery. Due to the enormous impact of the latter on the life sciences, photograph 51 is viewed by many as perhaps the most important scientific picture of the 20th century. As such it deserves to be understood as widely as possible. The present authors [49] have developed a didactic presentation of it supported by straightforward optical simulations and requiring very little prior knowledge of x-ray crystallography. This interpretation will not be repeated here in any detail. Instead, because pattern A also played a considerable, if more subtle role in the discovery, we will rather develop here a non-technical comparison between the two patterns and present new optical simulations specific to the A pattern.

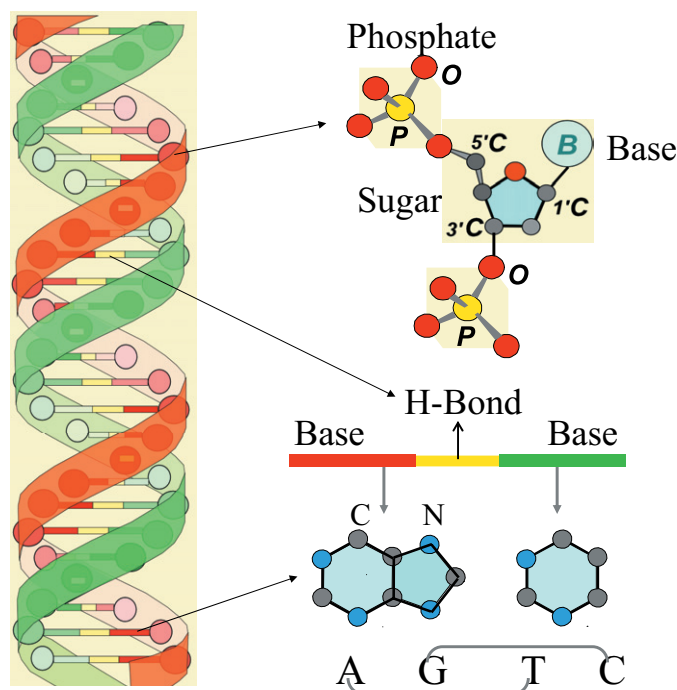


Figure 12. Schematic representation of the B-DNA double helix. There are two right-helix backbones of repeating, identical sugar-phosphate groups, separated by $3/8$ of the helix period. Note the 3'-5' asymmetrical linkage of the sugar pentose to its two phosphate neighbours. The variable base B attached to the 1' carbon can be one of the four groups A, T, G, C whose carbon-nitrogen framework cycles are shown on the lower right. The thin horizontal bars represent the base pairs seen edge-on and linked by hydrogen bonds.

Both pictures in figure 13 are *fibre diagrams of sodium thymonucleate*, meaning the sodium salt of Deoxyribonucleic Acid (Na-DNA) extracted from a thymus gland and purified. The material in the fibre (a few tens of micrometres thick) comprises a macroscopic number of long, roughly parallel, negatively charged DNA strands, Na^+ counter ions and a relatively high quantity of solvating water (up to ten H_2O molecules per nucleotide, the phosphate-sugar-base repeating monomer of DNA). The solvated cations and water molecules are in an amorphous state and cause only diffuse, background scattering.

Gosling and Franklin discovered [46–48] that it was possible to pass *reversibly* from A to B or from B to A by changing the relative water content of the fibre. The drier state giving pattern A was interpreted as that of a *crystalline* arrangement of the DNA molecules, with long range order in the fibre, as evidenced by the ordered reciprocal lattice of relatively sharp spots near the pattern centre. The wetter state giving pattern B with its broader features was called *paracrystalline* and viewed as lacking long range positional as well as rotational order as a result of the disordering effect of the increased solvating sheath of water surrounding the molecules. As explained before, pattern B is that of a gel and is, therefore, representative of the average scattering by a single molecule, while pattern A, at least its centre part, is representative of the crystal lattice, the information on the individual molecular structure in that case being contained in the relative spot intensities.

Both patterns of spots or smears are organized in regularly spaced layer-lines as one would expect from a filamentous molecule having a periodically repeating structural unit in the fibre

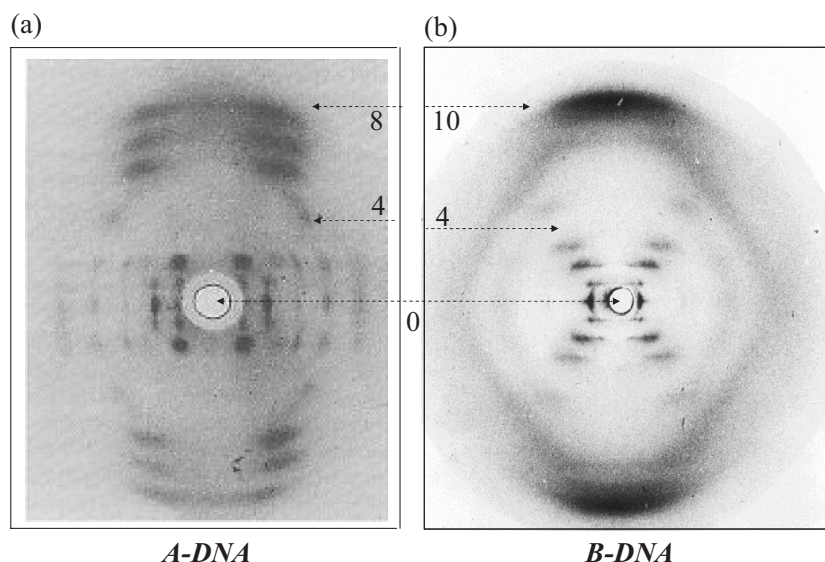


Figure 13. The x-ray fibre diffraction patterns of A-DNA and B-DNA plotted on the same scale. The eighth layer-line of pattern A occurs at about the tenth layer-line of pattern B (see text). The A pattern shows crystalline spots on the first few layer-lines.

direction. The layer-line separation reveals the value of the period. The spectacular reversible change in the type and distribution of spots on going from A to B was taken as reflecting not only the change in the long range spatial arrangement just discussed but also a *reversible change of internal structure* of the DNA molecules themselves. In particular, a large decrease of over 20% in the layer-line spacing occurred on going from A to B (figures 13(a) and (b) are drawn on the same scale), which implied an increase of about as much in the polymer repeat period. The molecules were said to adopt the A-DNA or B-DNA conformations, respectively. This was the beginning of the later observation of multiple polymorphism of DNA double helices [50] under various chemical and physical conditions. The B conformation is believed to be the one adopted by chromosomal DNA (in association with proteins) in the high humidity condition of the living cell.

In previous x-ray studies on DNA fibres, notably those of Astbury [19], the big north and south meridian arcs seen in the B pattern had already been observed (the arcing is due to the deviation from parallelism of the molecules). Astbury's patterns were much more confused than the ones shown here because it turned out that his fibres were in a mixed A and B state. However, they showed the large smears, which he correctly ascribed to a nucleotide repeat of 0.34 nm along a filamentous molecule. He modelled the latter as a sequence of flat 'nucleosides' (planar sugar plus planar base, erroneously represented as lying in the same horizontal plane) stacked on top of each other, like 'a pile of pennies', and covalently linked to the phosphate groups in alternation. Although the flat bases are irregular in their chemical composition and geometrical structure, they scatter coherently because they behave like thin slabs of electrons seen edge-on by the x-ray beam (perpendicular to the fibre), somewhat like a parallel slit grating, to produce the large meridian reflections. Since the B pattern shows about *ten layer-line intervals* separating the centre from the north blob, it follows that *the B-DNA molecule must have an axial repeat of ten nucleotides, i.e. a period of 3.4 nm. That was the first important piece of information coming out of pattern B.*

The most immediately apparent difference between the two patterns of figure 13 is the presence near the centre of a St Andrew's cross in B and its absence in A. As already stated, pattern B reflects the scattering by individual molecules. So the cross must be produced by the long range regularity of the repeating part of the molecule, namely by its sugar-phosphate backbone (see figure 12). In his popular autobiographical novel [41], Watson reported having been dumb struck when seeing photograph 51 for the first time, because for him who had learned the CCV formalism from Crick, that cross screamed 'helix'. *That was the second crucial piece of information given by the B pattern.* Furthermore, from our previous discussion of the CCV theory, that pattern not only meant helix but it revealed at once to Watson and Crick (if only in a confused way since, after all, they did not know the structure yet) the three fundamental dimensions of the phosphate helix: its pitch P_a (3.4 nm, from the layer-line interval), its radius r (1 nm, from the meridian angle of the cross) and its nucleotide repeat p_a (0.34 nm, from the vertical diagonal of the meridian diamonds). The absence of meridian intensity in the diamond and the large radius r indicated that the backbone helix was likely to be at the periphery of the molecule and the bases inside (see the remark at the end of section 3.4), the reverse of unsuccessful models constructed earlier by Watson and Crick [28] and also by Pauling and Corey [51] shortly before the double helix discovery.

Because the A pattern did not show the St Andrew's cross and contained no other obvious clue as to the helical form of the molecule, Franklin took this and other evidence to mean that the A-form molecule might not be helical after all. And since the A-pattern was much more detailed and hence promising for working out the molecular structure, she temporarily put photograph 51 of B-DNA aside and decided to concentrate exclusively on a quantitative interpretation of the A picture. Amongst the most decisive information that she and Gosling produced by analysing A-type patterns [48] was that the A-DNA crystal lattice pertained to the *C-face centred monoclinic space group*, with two molecules per unit cell running along the *c*-axis (that is close to but not quite along the fibre axis), as depicted in figure 14. In effect, the structure was nearly a hexagonal close packing of long rod-like molecules, the deviations from exact hexagonal symmetry arising from interactions between the grooves of the twisted backbones of neighbouring molecules pressed towards each other by the partial removal of water. That C2 monoclinic crystallographic fact, *the third major piece of information* coming out of the diffraction work of Franklin and Gosling, also came to Crick's knowledge prior to publication.

The monoclinic space group implied an absolutely crucial consequence that, apparently, only Crick noticed [27, 52]. He reasoned that because the unit cell *and its content* possess a C2 dyad axis perpendicular to the *a* – *c* face (figure 14), *the DNA molecule itself must also have a two-fold symmetry axis perpendicular to the backbone.* But since the sugar-phosphate backbone of nucleic acids was known to be an *oriented polymer* (due to the asymmetric covalent attachments of each pentose to its two phosphate neighbours; see figure 12) the only possible way to have a dyadic DNA molecule with such polar strands was to assume that it possessed an even number of strands, that is *not one nor three but two counter oriented strands* (figure 14). Four or more strands were incompatible with density measurements.

So a spacious cylindrical cage made of two counter oriented, intertwined sugar-phosphate strands was built by Watson and Crick in a renewed attempt at model building [41]. Inside that cage, Watson tried to install pairs of the four planar bases A, T, G, C perpendicular to the cage axis and to make the bases of each pair interact via hydrogen bonds across the duplex axis (figure 12). In this last attempt, the pentagonal sugar rings had been given the correct vertical (instead of Astbury's horizontal) orientation and the bases the correct keto (instead of enol), tautomeric form of the H-bonds [27, 28]. Exploring the rather limited set of six possible couples, Watson stumbled on the *A–T, G–C pairings* as the only combinations with nearly

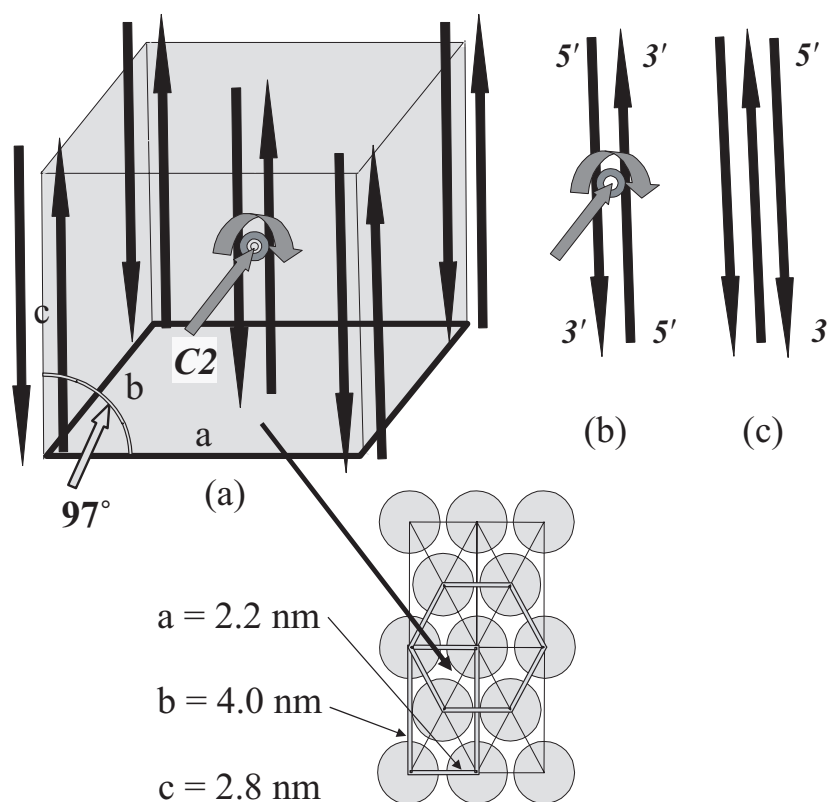


Figure 14. (a) The C-face-centred monoclinic unit cell of an A-DNA crystal fibre. The monoclinic cell, which is close to a hexagonal structure, has the dimensions indicated. (b) The C2 symmetry axis demands that the DNA molecule must itself be dyadic and therefore have two counter-oriented strands. (c) The dyad excludes the possibility of a triple (as well as single) strand molecule.

identical overall sizes, which could be made to fit between the rigid outer frame. That clinching discovery rapidly led to the completion of the beautiful and definitive DNA double helix model (figure 12).

We now complete the qualitative description of the A pattern. Upon drying the fibre, as noted before, the double helix period shrinks by about 20%, changing from 3.4 nm in B-DNA to 2.8 nm in A-DNA. Franklin and Gosling [48] and co-workers [53] further determined that in A-DNA the number of nucleotides per helical period is close to 11 per strand, instead of 10 in B-DNA. Removing water thus contracts the molecules along their axis. But because of nearest neighbour repulsion, this forces the bases to become inclined over the horizontal plane. Later x-ray studies revealed a tilt angle as large as 20° from the horizontal, as shown in figure 15 [53–55]. So most of the bases are no longer seen edge-on by the x-ray beam and hence no longer act as a horizontal slit grating. This explains why the large meridian smears characteristic of the B pattern have disappeared from the A pattern. The effect of the 20° tilt is to produce, instead, the two sets of strong smears on the sixth, seventh and eighth layer-lines on each side of the meridian axis (figure 13(a)). One can understand this by arguing that only a couple of inclined base pairs for every half turn of the helix continue to be seen approximately edge-on by the beam, and thus scatter strongly and coherently. The remaining base pairs are seen at various non-zero angles, partly face on, and cause diffuse scattering. In effect, what the smears represent is *the vestige of a St Andrew's cross* created by the diffraction by the

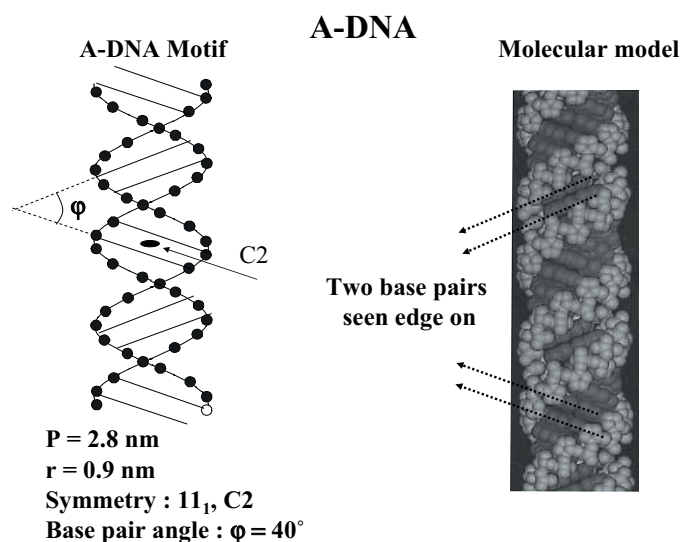


Figure 15. Right: a molecular, space filling model of A-DNA. The backbones are in light grey and the bases in dark. Note that a couple of base pairs define a see-through slit at each half period (broken arrows). Left: the diffraction motif used to simulate the A-DNA diffraction pattern at normal incidence. The backbone is represented by 11 black dots per period. The base pairs are represented as bars inclined at 20° from the vertical. Only the two base pairs seen edge-on in each half-period are included in the model.

couples of edge-on base pairs making up a kind of zigzagging, double slit diffraction grating (see figure 15 and figure 17 in the next section). The reason why one sees only a fraction of the St Andrew's cross in figure 13(a) appears to be the destructive interferences between the waves scattered by the two base pairs of each couple. This will be illustrated below with optical simulations.

Coming back to figure 4 one sees how inclining the crystalline fibre by an angle $\theta = 22^\circ$ from the perpendicular towards the x-ray beam (approximately the tilt angle of the base pairs over the horizontal) affects the A pattern [53]. The north-south symmetry is broken: the equatorial layer-line acquires a downward curvature, the smears of the north arms of the St Andrew's cross move further apart, away from the meridian while the smears of the south arms close in and tend to merge, as predicted by the simple polymer model of figure 3. As noted before, these effects are due to the finite radius of the Ewald sphere for x-ray diffraction by a single molecule. Similar effects were seen when Langridge *et al* [56] measured the x-ray picture of a *crystalline* fibre of a lithium salt of B-DNA when inclining the fibre by about 12° . The beautiful B-type diffraction pattern is shown in figure 5. Compared to the B-pattern of the sodium salt paracrystal in figure 13(b), the pattern in figure 5 shows many more and sharper spots, reflecting a crystalline order. Presumably, the smaller lithium counter ions allow for the B-DNA molecules to come closer and better crystallize. Again, the tilting causes the breaking of the north-south symmetry and the hyperbolic downward curvature of the equatorial layer-line, as explained in figure 3. Also, the symmetric spots of the +8th layer-line on each side of the meridian are widely separated, while the corresponding spots of the -8th layer-line are nearly fused onto the meridian. Although the tilting tends to remove the bases from being seen fully edge-on by the x-ray beam, the big north blob is still present and even enhanced, because the tilt angle was chosen to coincide with the Bragg angle of the flat bases, which, due to the crystalline order, are now all regularly aligned along parallel reflecting

B-DNA

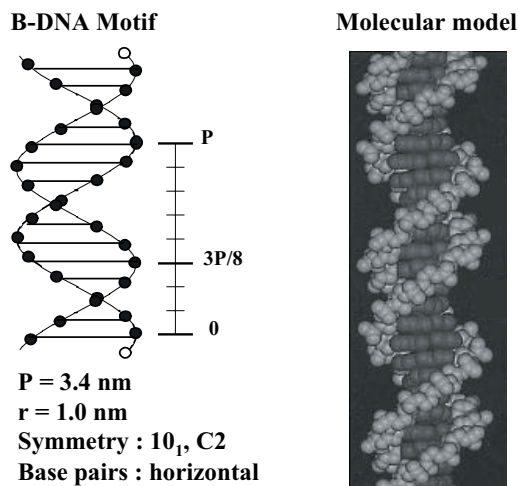


Figure 16. Right: a molecular, space filling model of B-DNA. The backbones are in light grey and the bases in dark. The base pairs are perpendicular to the axis and are all seen edge-on. Left: the diffraction motif used to simulate the B-DNA diffraction pattern at normal incidence. The backbone is represented by 10 black dots per period. The base pairs are represented as horizontal bars.

planes. The south meridian arc has disappeared, as this diffracted direction is not at the Bragg angle.

These complex tilting effects would not be observed in the diffraction of fast electrons for which the Ewald sphere is nearly planar. For this case, the equidistant layer-lines are straight and the primary observable effect of tilting would be a uniform and symmetric increase of their spacing by a factor $1/\cos\theta$ and a corresponding ‘climb’ of the ED spots on the Bragg diffraction circles. This has been observed in ED by CNTs, as will be reviewed later [57].

4.3.2. Simulations by optical diffraction. Diffraction of optical waves by suitable two-dimensional model gratings, the so-called optical transform method, has been used for many decades in crystallography to mimic x-ray diffraction by atomic lattices. The idea of such optical simulations first originated with Bragg himself [58,59]. A collection of optical transforms is given by [60]. Simulations have been performed for the case of DNA [24, 32, 49, 61, 62] as well as, quite recently [63], for ED by CNTs reviewed later. Today, the availability of cheap laser pointers has rendered the method easily accessible to anyone interested.

Taking the geometrical dimensions of A-DNA into account, we have produced, for the purpose of this review, optical diffraction gratings mimicking the principal diffracting elements of this molecular conformation (figure 15). The new gratings are similar to the models successfully used previously to explain the gross diffraction features of the B-DNA conformation [32, 49] (figure 16). The simulations presented here are mere exercises in ‘reverse crystallography’ (reproduce the diffraction pattern of a known molecular structure); they have no pretension at quantitative modelling and are exclusively of didactic value. Indeed, the molecular motifs shown in figures 15 and 16 are two-dimensional projected helices with which one can only hope to qualitatively understand the gross features of the real x-ray images not too far from the pattern centre where the two-dimensional approximation has some validity.

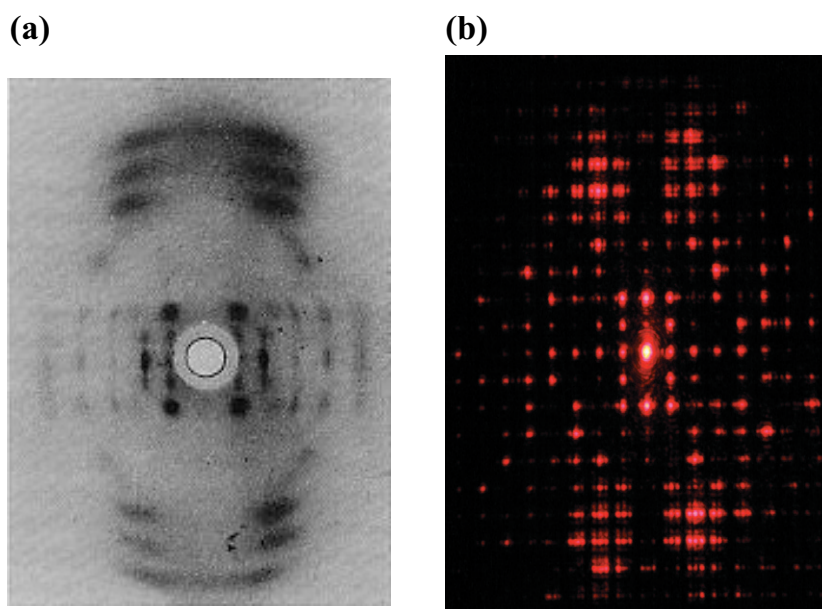


Figure 17. (a) X-ray diffraction pattern of A-DNA (same as in figure 13(a)). (b) Optical diffraction simulation produced by the motif shown on the left of figure 15 (see text).

The optical gratings consist of a large number of such identical motifs densely repeated in parallel on a standard slide (about 5 motifs per mm). The motifs are in two-dimensional crystalline order for the A-DNA simulation while some lateral and longitudinal randomness is introduced for the B-DNA case to simulate the disorder of the paracrystalline phase. The red light from a good laser pointer (4 mW) serves to simulate the x-ray beam and the diffraction pattern is projected on a screen a few metres away. The simulations work in broad day light (the diffraction slides may be ordered from amand.lucas@fundp.ac.be).

The images produced by the two gratings are shown in figures 17 and 18 where they are compared to the real A- and B-patterns, respectively. It is obvious that the simple models do succeed in capturing the gross characteristics of the x-ray pictures. For the simulation of B-DNA in figure 18 the fundamental origin of all the diffraction features has been analysed in detail in [32, 49]. This need not be repeated here. For A-DNA, one observes a two-dimensional reciprocal spot pattern, as expected from the two-dimensional lattice of the simulation grating. Most significant is the fact that the strong x-ray smears are correctly imitated on the 6 to 8 layer-lines in the four quadrants (the lack of east–west symmetry of the simulation pattern is due to the absence of this symmetry in the 11-nucleotide motif in figure 15; only the C2 symmetry is preserved). That these features can be attributed to the edge-on base pairs is further supported by the diffraction demonstration of figure 19. Here, the sugar-phosphate backbones have been removed from the optical model and only the zigzagging base pairs *seen edge-on* have been kept, arranged on a two-dimensional lattice grating. In figure 19(a), only one such base pair per half helix period is considered while in figure 19(b) two adjacent base pairs are represented. The latter motif appears more realistic on careful inspection of the molecular model in figure 15 where a couple of base pairs are seen to define a see-through slit every half period of the model. The corresponding diffraction patterns are both in the form of a spotted St Andrew’s cross with arms perpendicular to the zigzagging directions of the base pairs (40° opening angle). But figure 19(b) clearly indicates that there occurs an alternating intensity

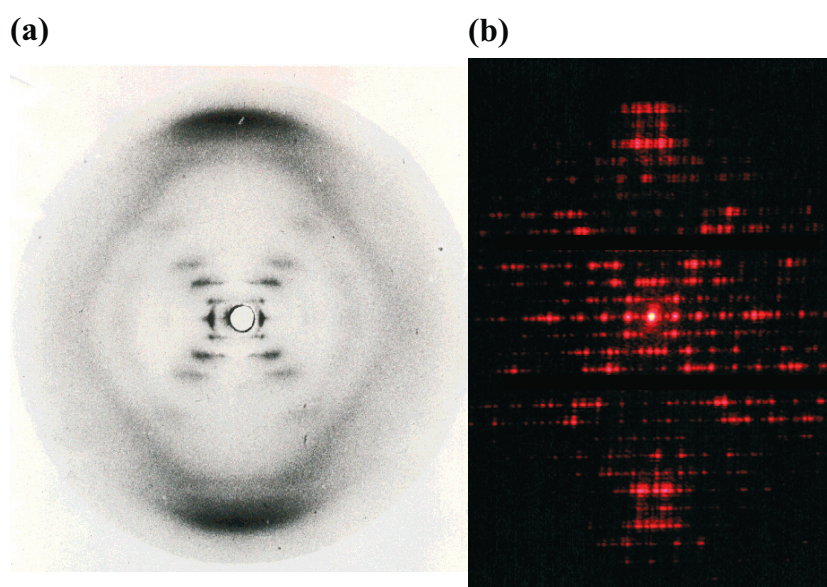


Figure 18. (a) X-ray diffraction pattern of B-DNA (same as in figure 13(b)). (b) Optical diffraction simulation produced by the motif shown on the left of figure 16 (see text).

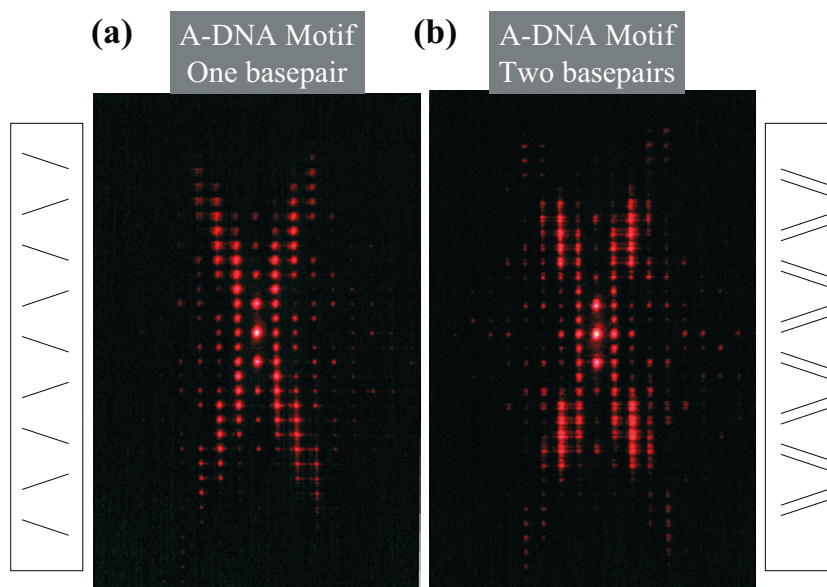


Figure 19. (a) Optical diffraction pattern produced by a two-dimensional lattice of *single* zigzagging lines shown on the left of the pattern. Note the St Andrew's cross of ordered spots. (b) Optical diffraction pattern produced by a two-dimensional lattice of *double* zigzagging lines shown on the right of the pattern. The St Andrew's cross of ordered spots is now similar to that appearing in the simulation of figure 17(b).

weakening and reinforcement of the layer-lines along the vertical direction: weakening from the third to fifth layer-lines and from the 9th to 11th, and reinforcement from the sixth to eighth. This is just what was observed in figure 17 in both the x-ray pattern and in its optical simulation, and this confirms the proposed interpretation in terms of an interference-modulated St Andrew's cross of the two edge-on base pairs.

4.3.3. Is the DNA double helix really plectonemic? The original paper of Watson and Crick [45] achieved three major breakthroughs: (i) it introduced the beautiful double helical structure of DNA sketched in figure 12; (ii) it postulated the complementary A–T, G–C base pairing rules; (iii) it proposed the semi-conservative replication scheme.

As we have explained above, the first item came out of model building supported by x-ray crystallography. The second item was largely the result of serendipity in that process of model building, but it instantly confirmed or was confirmed by the empirical biochemical rules formulated by Erwin Chargaff in the late 1940s [64]: in DNA molecules originating from any organism whatsoever and analysed for their composition in the four bases, the chemical concentrations of adenine and thymine are always the same, and similarly those of guanine and cytosine.

As for the third item, its meaning is that, in replication, the two strands of the double helix must be separated in order for each strand to become a template for copying a new, complementary strand. This process leads to the synthesis of two daughter DNA molecules identical to the parent molecule, each incorporating half of the parental material, hence the name semi-conservative. Although a great revelation for the molecular mechanism of heredity, the scheme proposed by Watson–Crick was at first heavily criticized on the ground that entanglements made it inconceivable that strand separation could be achieved over the entire macroscopic length of a chromosome without introducing multitudinous breaks of the backbones. That the strands are indeed used as templates and somehow transmitted semi-conservatively to the daughters in the replication of bacteria was unquestionably demonstrated by a brilliant experiment of Meselson and Stahl in 1958 [65,66]. Unfortunately, that demonstration left the disentanglement problem fully unresolved as it did not specify anything regarding the spatial structure of the units that were transmitted semi-conservatively. So, it was proposed, at the time that the disentanglement nightmare would not even arise if the DNA double helix were *paranemic* (two anti-parallel strands of separate axes linked together by complementary base pairs) instead of *plectonemic* as in the Watson–Crick model (two coaxial strands winding around each other, a structure initially chosen apparently more for aesthetic reasons than from hard crystallographic, stereochemical or biochemical constraints). The former, ladder-like or side-by-side model would indeed largely eliminate the unwinding difficulty since, in zipper-like fashion, only the relatively weak hydrogen bonds between the bases would have to be broken in replication (perhaps far too easily broken for the long term stability of the genetic material!). Several side-by-side structures were proposed [67] before the discovery of topological enzymes capable of breaking and reforming the strong backbone covalent bonds, thereby assisting the necessary unwinding and unravelling of plectonemic DNA. We refer to [67] and to chapter 11 of [68] for a criticism of the paranemic models and for a description of several more recent experiments, including x-ray diffraction on double-stranded oligonucleotide single crystals [69], all of which have confirmed the validity of the plectonemic structure.

We wish to make a further point here, which we have not seen raised in the DNA structural literature, namely that the original x-ray diffraction patterns of B-DNA show peculiar gross characteristics, which are specific to a plectonemic double helix and appear to be inconsistent with the paranemic arrangement. If we look again carefully at photograph 51, figure 13(b),

we note that *there is a gap in intensity at the nominal position of the fourth layer-line*. This extinction is particularly obvious and spectacular in figure 5. As shown by the model of figure 16 and the optical simulation of figure 18, this cancellation of the fourth layer-line is produced in *plectonemic DNA* by having the two coaxial sugar-phosphate strands axially separated by $3/8$ of the period (or $5P/8$) [32, 49] (although, in the real B-DNA duplex, the two strands are related by a dyad and not by an axial translation, the latter is an approximate symmetry for the nearly spherical electron density of the backbone phosphate tetrahedra, which dominate the diffraction pattern on account of the atomic structure factors shown in figure 1). The extinction was duly noted by Franklin and Gosling, the authors of photograph 51, in their original paper [47] and attributed by them to a likely $3P/8$ separation between the two strands, consistent with the molecular model just built by Watson and Crick [45] (in which the unequal shift of the *counter-oriented* backbones was required for establishing the H-bonds of the base pairs). Now, such a cancellation cannot be achieved with a paranemic arrangement. Indeed, the diffraction amplitude of a double helix (or of any pair of identical periodic polymers for that matter) implies an overall factor of $1 + \exp(\mathbf{q}_\perp \cdot \mathbf{d}) \exp(iq_z z)$, where $\mathbf{r} = (\mathbf{d}, z)$ is the vector distance between the two strands. With $q_z = 2\pi l/P$, this factor vanishes for $l = 4$ when $z = 3P/8$ and provided $d = 0$ (plectonemic case), but it cannot be made to vanish for any *entire* layer-line l (all values of $q_\perp$) if d is non-zero (paranemic case), even if the z axial shift has the favourable value $3P/8$. Thus, that incontrovertible fourth layer-line extinction of the original Photograph 51 of B-DNA constitutes by itself a strong argument against any side-by-side model.

4.4. TMV virus

Small, nanoscopic-size viruses have just two molecular components: a protein coat and a nucleic acid molecule that carries the genetic information. The early application of x-ray diffraction to their study has revealed that such virus particles, like many other biomolecular assemblies, have an invariable and highly ordered internal structure, the result of their mass production by the host cell, which they infect. The coat is built up from a large number of *identical protein subunits* arranged in a highly regular spatial structure, which provides a form of packaging for a single, continuous RNA or DNA molecule responsible for the infectivity of the particles. When that spatial structure is rod-like (as in the tobacco mosaic virus, or TMV) or 'spherical' (more accurately polyhedral, as in icosahedral viruses), the particles can be made to condense in a paracrystal, i.e. an oriented gel (similar to the paracrystal of a DNA fibre), or a three-dimensional crystal (similar to a haemoglobin crystal), respectively, which are very favourable for diffraction studies.

One of the most thoroughly investigated viruses is the helical, ribonucleic TMV virus, the first virus to have been discovered, already at the end of the 19th century. After initial pioneering x-ray studies by Bernal and Fankuchen [70], the decisive progress in its broad morphology was made by the beautiful x-ray work of Rosalind Franklin and her collaborators in the 1950s [71] (in this reference Klug and Caspar give an extensive review of the structural and other works on small viruses up to 1958). The research on TMV later culminated with the electron microscopy and high-resolution x-ray studies of Klug and collaborators in the 1960s and 70s. These works not only determined the structure of the protein subunits down to atomic resolution but also established the complex mode of virus self-assembly from its components [72] (this reference is the written version of Klug's 1982 Nobel Prize Lecture).

It is our intention here to limit the description of the TMV structure to its general features, as already outlined by Franklin *et al* [73]. This will suffice for our purpose, in this paper, which is to bring to the attention of scientists who are not professional crystallographers some

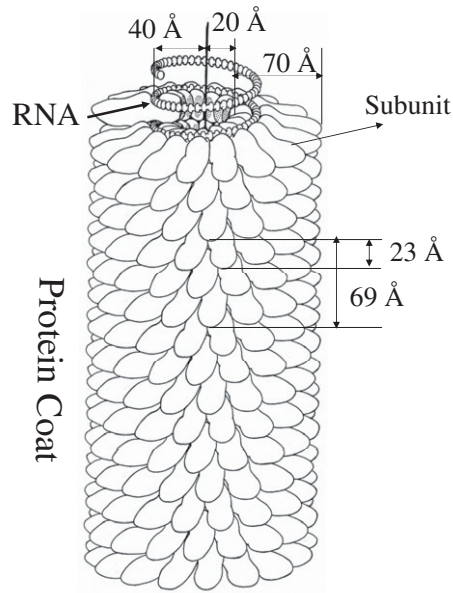


Figure 20. Model and basic dimensions of the TMV virus established by x-ray fibre diffraction [73]. The 40 Å-radius RNA helix is embedded in the helical coat of protein subunits (see text).

of the essential and common elements of the diffraction by several helical nanostructures of widely differing nature. The reader is referred to [71] and [72] for masterly reviews by two of the principal researchers involved in the structural works on viruses.

Figure 20 is a drawing of the outline shape and dimensions of TMV as determined by the last x-ray studies of Franklin *et al* [73]. Such were the novelty and quality of her work that she was invited, shortly before her premature death [42], to construct a room-size model of the virus for public exhibition in the Science Pavilion of the Brussels World Fair in 1958. The rod-like particle contains about 2130 identical protein subunits set along a single, tightly wound circular helix. There are $16 \frac{1}{3}$ subunits per helix turn of 23 Å pitch, so that the structure repeats itself only after 3 turns, i.e. 49 subunits or 69 Å axial repeat. In the notation of the CCV theoretical model of section 3.4, the helix is characterized by the geometrical parameters $P_a = 69 \text{ Å} = 3P = 49p_a$ where p_a is the axial distance between the subunits. The layer-line selection rule must, therefore, be

$$l = 3n + 49m \quad (16)$$

and is common to all the hundreds of mono-atomic helices making up the complete virus coat. The subunits are approximately prolate ellipsoids of revolution of about 23 Å minor axis and 70 Å major axis extending between 20 and 90 Å from the particle axis. This makes the virus look like a thick hollow cylinder with highly corrugated inner and outer surfaces. As shown in figure 20, the subunits have two opposite depressions on their upper and lower sides, so that when they are assembled an interstitial helical groove or tunnel runs between them at about 40 Å distance from the axis. The RNA molecule is embedded there deep inside the coat as a single helix chain threading through the tunnel at the rate of 3 nucleotides per subunit over the whole 300 nm length of the particle (about 10 times the length shown in figure 20).

All this structural information was obtained by analysing x-ray diffraction patterns, such as the beautiful examples reproduced in figures 21(a) and (b) [71, 74, 75]. These extraordinarily rich, fibre-type diffraction diagrams were taken from oriented water gels of TMV particles,

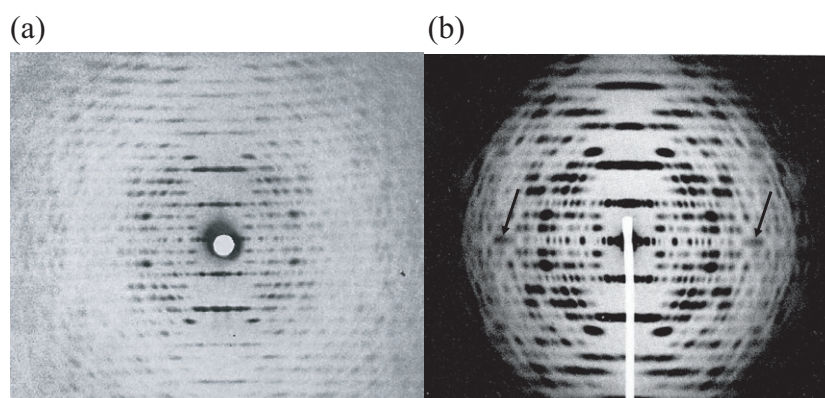


Figure 21. (a) X-ray fibre diffraction picture of native TMV [74]. (b) X-ray fibre diffraction picture of mercury-substituted TMV [75]. Note the stronger streaks marked by arrows, as compared with those of (a).

which are a more or less tight hexagonal packing of TMV rods of random angular orientation around their parallel axes (somewhat similar to the paracrystals of α -helices or DNA molecules discussed previously). The TMV particles that gave figure 21(a) were the native species while those giving figure 21(b) were *mercury-substituted*. This means that in all the protein subunits, a mercury-methyl group (Hg-CH_3) replaces the terminal proton linked to the sulfur atom of the unique cysteine residue (side group $\text{C}_\beta\text{H}_2\text{-S-H}$) in the sequence of 157 amino acids of each subunit (incidentally much of that sequence was later determined to be in the α -helix conformation [72]). The mercury atoms are thus on a helix of their own with the same three-turn repeat as the entire protein subunit helix. Therefore, there must also be 49 Hg atoms equally spaced along three turns of the helix the radius of which turned out to be 56 Å.

The Hg substitution in TMV constituted one of the first applications of the general *heavy atom isomorphous replacement method* in the x-ray crystallography of proteins. This is a powerful technique first applied by Perutz for the x-ray structure determination of large globular proteins, such as haemoglobin, which can be made to crystallize in true three-dimensional crystals. How does it work? (A general introduction to the method can be found in [40].) If the morphology of the molecule is not otherwise disturbed by the ('isomorphous') substitution, the replacement of one or several light atoms of the protein by heavy ones will cause small changes in the diffraction pattern, whose measurement may give access to the phases of the x-rays diffracted by the native molecule. Then, a Fourier inversion of the native diffraction pattern becomes possible and yields the molecular structure. For the present TMV samples, which are in a gel state and not a three-dimensional crystalline state, the method is not fully exploitable. However, the Hg substitution did nevertheless enable the determination of the cylindrically average radial density distribution of the particle from a comparison of the equatorial layer-lines of the native and substituted fibre diagrams (see below) [71].

Coming back to figure 21(a), the hyperbolic shaped layer-lines have a spacing that reflect the $P_a = 69$ Å true axial repeat of the structure. However, they manifestly occur in groups of three, which is readily understood if one assumes that there are three turns of the subunit helix in the true repeat period. There is intensity on or close to the meridian only on multiple-of-three layer-lines corresponding to the spacing P between the successive turns of the helix. The succession of tight helical turns diffracts as a zigzag slit grating to produce the large intensities on either side and close to the meridian. From the CCV layer-line selection rule in

Table 1. First allowed values of the indices n and m in the CCV formula (equation (6)) for the layer-lines $l = 0$ to $l = 13$ seen in figure 20.

l	0	1	2	3	4	5	6	7	8	9	10	11	12	13
n	0 49	-16 33	17 32	1 48	-15 34	18 31	2 47	-14 35	19 30	3 46	-13 36	20 29	4 45	-12 37
m	0 -3	+1 -2	-1 -2	0 -3	+1 -2	-1 -2	0 -3	+1 -2	-1 -2	0 -3	+1 -2	-1 -2	0 -3	+1 -2
J_n	J_0 J_{49}	J_{16} J_{33}	J_{17} J_{32}	J_1 J_{48}	J_{15} J_{34}	J_{18} J_{31}	J_2 J_{47}	J_{14} J_{35}	J_{19} J_{30}	J_3 J_{46}	J_{13} J_{36}	J_{20} J_{29}	J_4 J_{45}	J_{12} J_{37}

equation (16) one can construct table 1, which lists the first allowed couples (n, m) for each of the layer-lines from $l = 0$ to $l = 13$ seen in figure 20, along with the Bessel terms contributing to the diffracted amplitudes in the CCV formula of equation (6). The multiple-of-three layer-lines are listed in bold. The corresponding Bessel functions $J_1, J_2, J_3, J_4, \dots$ indicate that there should be no intensity on the meridian (remember that $J_n(x) = 0$ for $x = 0$ if $n > 0$). A St Andrew's cross should be present, made of the first maxima of J_n^2 on these lines. But the arms of the cross, being perpendicular to the zigzagging subunit helix, are nearly parallel to the meridian since the helix slope is quite small (about 5°). The reason one does not see the cross in figure 21 is the overexposure of the photographs, which was such that, at the resolution of the figure, the first maxima of the low-order Bessel functions J_1 to J_4 on both sides of the meridian effectively overlapped and filled the meridian gaps. Higher resolution photographs with lower exposure did confirm the presence of these gaps [74a].

As was pointed out previously in this paper (see end of section 3.4), it is the outermost helices of the structure, which determine the intensity distribution of the innermost St Andrew's crosses of the diffraction pattern. Based on this fact, Franklin and Klug [74b] were able to analyse the intensity distribution on the meridian region of the third layer-line and thereby demonstrate the deeply grooved and knobbed nature of the outer surface of the TMV particle shown in the model of figure 20.

The pairs of true-repeat layer-lines intermediate between the multiple-of-three ones all involve high-order Bessel functions (see table 1), which explains why their non-zero intensity begins at quite a distance away from the meridian. For instance, the first and second layer-lines involve the dominant terms $J_{16}(q_\perp r)$ and $J_{17}(q_\perp r)$, which begin to rise only for $q_\perp r = 16$ and 17, respectively. Using the metric scale given by the layer-line spacing, values of about $q_\perp \approx 0.20$ and $0.22 \text{ \AA}^{-1}$ can be read for the distance of the first spot to the meridian on these layer-lines. So, the beginning intensity on the first two layer-lines is contributed by atoms sitting at distances $r \geq 16/0.20 = 80 \text{ \AA}$, and $r \geq 17/0.22 = 77 \text{ \AA}$, respectively, i.e. at the outer periphery of the virus particle where the subunits point their tapering ends outwards and where the material density is, therefore, highly contrasted. This is also the case for all the high- l layer-lines which involve high-order Bessel functions.

Is there a clear effect of the heavy Hg atoms in the pattern of figure 21(b) as compared to the native pattern of figure 21(a)? In spite of the fact that each subunit contains over one thousand light atoms for only one single heavy Hg atom, subtle differences exist between the two patterns. Franklin and Holmes [75] have analysed in detail the equatorial layer-lines of the two patterns. The fine oscillatory structure observed along this and other layer-lines results from the oscillations of the relevant Bessel functions. For the equatorial line, $J_0^2(q_\perp r)$ and $J_n^2(q_\perp r)$ are the dominant terms, if n is the number of protein subunits in the true helical repeat of $69 \text{ \AA}$. Franklin and Holmes [75] were able to use the small but measurable differences of the

$J_0^2(q_\perp r)$ oscillations to identify the radius of $r_{\text{Hg}} = 56 \text{ \AA}$ for the Hg helix, as well as the signs (i.e. the phases 0 or π) of the scattering amplitudes at the peaks of J_0^2 . This allowed them to construct the cylindrically average radial density of the virus and, in particular, to locate the radial position of the phosphate backbone of the RNA helix at $40 \text{ \AA}$ from the axis. Then, a broad peak extending from $q_\perp = 0.75 \text{ \AA}^{-1}$ to $q_\perp = 1.0 \text{ \AA}^{-1}$ and centred around $q_\perp = 0.88 \text{ \AA}^{-1}$ (marked by arrows in figure 21(b)), which is visibly stronger in the TMV Hg pattern, was analysed and attributed to the first peak of $J_n^2(q_\perp r_{\text{Hg}})$. They found that from $r_{\text{Hg}} = 56 \text{ \AA}$ and the measured $q_\perp = 0.88 \text{ \AA}^{-1}$, the Bessel function order must be $n = r_{\text{Hg}} q_\perp = 49$, which indeed gave a good fit to the peak position (this corresponds to the $n = 49, m = -3$ couple in the equatorial layer-line selection rule, see table 1). The nearest possible values of n , 46 and 52, did not fit the observed Hg peak position. This result provided strong confirmation that there are indeed 49 protein subunits in three helix turns, as estimated from other parts of the diffraction pattern. The physical origin of the broad equatorial spot at $q_\perp = 0.88 \text{ \AA}^{-1}$ can be explained as follows [71]: when looking down the helix axis, the projections of the 49 Hg atoms over three helix turns are equally spaced at $7.2 \text{ \AA}$ of each other along the helix circumference $2\pi r_{\text{Hg}} = 352 \text{ \AA}$. For the x-ray beam scattered along the equator, the mercury atoms act as a vertical grating of continuous rods separated by a variable spacing. That spacing takes a maximum value of $7.2 \text{ \AA}$ close to the helix axis and chirps towards zero at the exterior parts of the helix. This gives rise to the observed broad streak at about $7.2 \text{ \AA}$ spacing or $q_\perp = 0.88$ in figure 21(b). In general, the spread of the diffraction peaks perpendicular to the helix axis is a manifestation of similar chirping of all apparent distances in this direction. A similar streaking effect will be observed in ED by CNTs in the next section.

The model TMV virus built by Franklin was a *right-handed helix* of protein subunits hosting a single right-handed RNA chain. This correct choice of hand was ascertained only in 1972 (see [120]) from observing the way the TEM micrographs of the virus changes upon tilting the specimen away from the normal to the electron beam. We will come back to this point in the last section of this review, where a similar technique that was applied to CNTs will be discussed (see section 5.4.4).

It will not be possible nor useful to go any further into the many aspects of the analysis of the x-ray data on TMV performed by Franklin (and collaborators) in the last, extraordinarily productive five years of her life. The reader is referred to [71] for an in-depth account and for further references.

4.5. More recent developments in structural biology

Since the heroic beginnings described above in the application of x-ray crystallography to the basic helical molecules of biology, immense progress has been achieved in the last four decades in the determination of the structure of biological macromolecules of ever increasing complexity. While x-ray diffraction alone used to dominate the field, beginning in the late 1950s and 1960s, ED combined with high-resolution electron microscopy progressively assumed more and more importance, as we already mentioned above with the pioneering work of Klug and collaborators on the structure of small viruses [71, 72]. An intermediate review, written in 1978 again by Klug [76], clearly explains the fundamental principles of the new instrumental techniques and the processes of three-dimensional image reconstruction (with the growing involvement of computers) by analysing the data contained in the two-dimensional electron micrographs and diffraction patterns. Among many other distinctive features of what has become known as *electron crystallography*, the most decisive advantage brought to bear by TEM is indeed its capacity of retrieving (optically or computationally) the phases of the diffraction amplitudes from the two-dimensional high-resolution electron images. TEM is now

an indispensable tool in structural biology of highly complex macromolecular assemblies, as reviewed by Downing and Glaeser [77]. Recent spectacular examples in the application of this tool in combination with traditional x-ray crystallography have been the determination, at atomic-resolution, of the helical or hybrid structures of microtubules [78], of the bacterial flagellar filament [79] and of the acetylcholine receptor pore in the cell membrane [80]. Entering into a description of the intricate details of the methods used to obtain these fantastic results is completely outside the scope of this review as well as beyond the competence of its writers.

5. Diffraction by CNTs

5.1. General morphology of CNTs

We now turn to the recently discovered CNTs, which are helical polymers of pure carbon, similar in size to, but far simpler than, the biomolecules discussed previously.

Until about 20 years ago, hexagonal graphite and cubic diamond were the only known crystalline forms of solid carbon. Then, in 1985, the venerable science of carbon was unexpectedly rekindled by the discovery of an entirely new class of pure carbon structures called *fullerenes* [82]. The species first discovered were spheroidal molecules in which the presence of pentagonal rings in the hexagonal graphite network causes the curvature and closure of the graphitic sheets.

Fullerenes were discovered [82] serendipitously in the soot formed when a hot carbon vapour cools off and condenses into molecular clusters in an inert gas atmosphere. The most abundantly produced molecule, C_{60} , comprises 60 carbon atoms, all equivalent, arranged at the vertices of a truncated regular icosahedron. A few years after the initial discoveries a method was developed to produce pure C_{60} and other fullerenes in large amounts [83–85]. It was found that even at room temperature pure C_{60} is a stable molecular crystal [83], predominantly of face-centred-cubic structure, an entirely new form of solid carbon called Fullerite [95].

Beyond C_{60} , the next most abundant fullerene observed in the condensed carbon vapour is C_{70} . This molecule can be conceived as being constructed by cutting C_{60} in two by a plane perpendicular to one five-fold axis and inserting a ring of 10 atoms at the equator. By adding successively n such parallel rings, while maintaining the graphitic three-fold coordination, one can theoretically produce the cigar-shaped molecules shown in figure 22. In the limit of large n , these are particular members of a subfamily of fullerenes called *single-wall nanotubes* (SWNTs). More specifically, the species shown in figure 22(a) is called a ‘*perpendicular*’ SWNT, as by construction it has one family of C–C bonds perpendicular to the cylinder axis. Similarly, starting from C_{60} cut in two this time perpendicular to a three-fold axis and adding equatorial rings of carbon atoms, one obtains ‘*parallel*’ SWNTs having a family of C–C bonds parallel to the cylinder axis (figure 22(b)). Both these types of SWNTs are *achiral molecules*, that is they can be made to coincide with their respective mirror images. Figure 22(c) is one example of a chiral SWNT, which can also be obtained in a similar manner of elongating C_{60} in a direction oblique to the symmetry axes.

Instead, of viewing SWNTs as particular fullerenes elongated in some direction, more generally, any SWNT can be constructed by starting from a flat graphitic sheet or ‘graphene’ of arbitrary width and length and rolling it up into a seamless cylinder whose vertical generator can be parallel, perpendicular to or oblique to a C–C bond. The basic construction is shown in figures 23(a) and (b) where the unambiguous and universally used ‘*wrapping*’ indices (L , M) for a SWNT are made explicit [86]. The rolling up action brings the (L , M) node to the origin (0, 0) by tracing a circumference in the clockwise sense around the vertical axis, so that the

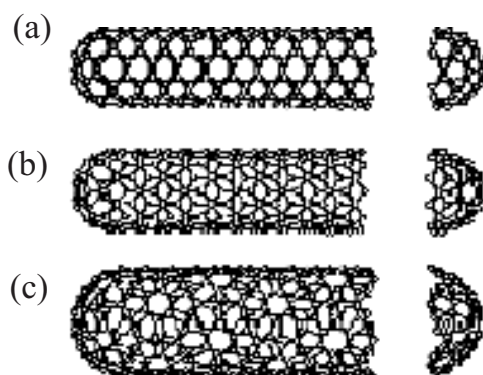


Figure 22. (a) Model of a perpendicular, achiral SWNT obtained by elongating a C_{60} molecule along one of its three-fold rotation axes. (b) Model of a parallel, achiral SWNT obtained by elongating a C_{60} molecule along one of its five-fold rotation axes. (c) Model of a chiral SWNT (courtesy of M Dresselhaus).

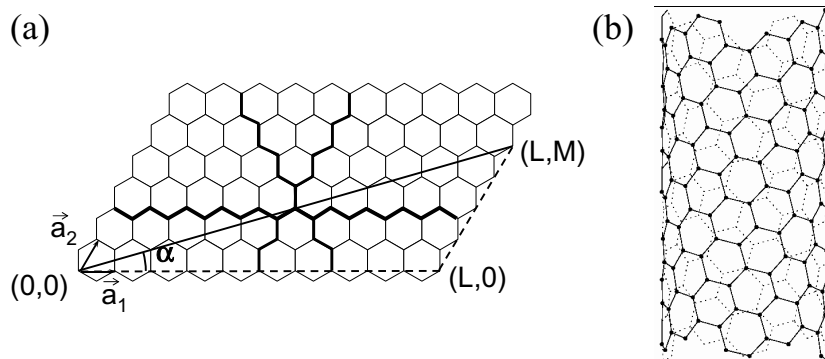


Figure 23. (a) Definition of the (L, M) wrapping indices of a SWNT. The nanotube is generated by rolling up the graphene sheet so as to bring the (L, M) node in coincidence with the $(0, 0)$ node; the front face of the sheet becomes the inner surface of the nanotube. Note the zigzag carbon chains in the three principal directions of the graphene lattice. (b) A $(10, 4)$ chiral SWNT in which the three zigzag chains of (a) become three zigzag helices on the nanotube. Two helices are right-handed and one is left-handed.

front face of the graphene sheet becomes the interior surface of the nanotube. Perpendicular or parallel SWNTs of figures 22(a) and (b) are of the (L, L) or $(L, 0)$ kinds, respectively. In general (L, M) SWNTs with $0 \neq M \neq L$, are chiral objects (figure 23(b)). The diameter of an (L, M) SWNT is given by $2r = (a/\pi)(L^2 + M^2 + LM)^{1/2}$, where $a = 2.46 \text{ \AA}$ is the honeycomb lattice parameter. The conventionally chosen chiral angle defined in figure 23(a) is given by $\alpha = \tan^{-1} M\sqrt{3}/(2L + M)$ and can evidently be limited to the range $|\alpha| < 30^\circ$. Hence, a SWNT is completely specified by giving either its (L, M) or $(2r, \alpha)$ couple. The ends of a SWNT can be open as in figure 23(b) or can be capped by hemi-fullerenes, as in figure 22.

Of course CNTs were discovered by Iijima [87] not as a result of a search for elongated fullerenes. Neither were SWNTs initially observed as isolated species but only as part of *multi-wall nanotubes* (MWNTs) [87]: these structures have a tree-ring organization of several coaxial cylindrical layers of varying chiral angles, separated by the graphitic distance of 0.34 nm . This was first demonstrated [87] using high-resolution TEM and ED. Soon afterwards suitable

experimental conditions were found [88, 89] to obtain abundant quantities of SWNTs, again making this new material abundantly available for fundamental investigations as well as for eventual practical applications. Today, experimental conditions of synthesis can be chosen or, to some extent, tuned to prepare samples containing various proportions of single-, double- or multiple-wall nanotubes.

5.2. Transport properties of SWNTs

The pace of research on CNTs has been accelerated by the discovery of the extraordinary electrical transport properties of SWNTs. Their conductivity is very sensitive to their atomic structure, namely their diameter and chiral angle [86, 90]. It turns out that when the difference $L-M$ between the chiral indices is a multiple of 3, the nanotube is metallic; otherwise there is a semiconducting gap in its band structure. This extreme sensitivity to structure is a consequence of the quantization of the π -electron wavelength by the cyclic boundary condition around the nanotube circumference imposed on the wavefunction. So, a slight change of the couple $(2r, \alpha)$ or (L, M) can result in the SWNT being either metallic or semiconducting. That completes the versatility of pure solid carbon in showing all possible electrical transport behaviour going from a metal to a semi-metal (graphite), to a small gap and on to a large gap semi-conductor (diamond). These transport properties of nanotubes, quite exceptional for an elemental molecular material, could eventually bring about the birth of practical nanoelectronics [90]. Details on the methods of synthesis and on the potential applications of CNT can be found in recent papers [89–92].

5.3. Kinematic form factor of a SWNT

As for practically all large molecules, the atomic structure of CNT has been determined by diffraction methods, especially by ED combined with TEM [87]. In this review, we will be concerned only by the latter methods, leaving aside global methods, such as x-ray or neutron diffraction.

Since electron scattering by atoms is very strong compared to x-rays, the quantitative interpretation of ED patterns of solid targets usually requires a dynamical diffraction or multiple scattering theory. However, a kinematical approach is largely sufficient to analyse the diffraction patterns of an individual CNT, even if it has several layers, because the target is so small that it will scatter electrons only once, if at all. Moreover, it is clear from figures 22 and 23 that SWNTs are helical structures, so their ED patterns can be interpreted within the framework used previously for the biological helices, namely the powerful CCV theory. This circumstance provides a strong unifying principle for the understanding of the structure of these two widely different kinds of materials. A very detailed review of ED and TEM of CNT has already been published in the pages of this journal [81]. Here, we will try to bring out a few of the essential elements and to point out the similarities with the biological helices. This review will also include some very recent experimental results not covered by [81].

The flat graphene sheet rolled into a SWNT is in effect the nanotube equivalent of what Crick and Kendrew [40] called the ‘*net-diagram*’ of a helical protein. As we discussed in section 4.2, this diagram is obtained by wrapping a sheet of paper around the cylindrical molecule, making a mark on the paper at corresponding points in each asymmetric unit of the protein and unwrapping the paper flat. For the present SWNT, the marks can be made at the centre of carbon hexagons. Just as for the net-diagram of an α -helix [40], there is ambiguity in the way the net points can be connected by straight lines. In the α -helix case one of the choices represents the direction of true chemical linkages between units on the helical molecule, while

on the graphitic sheet the atoms are bonded along any one of the trigonal directions of the Bravais lattice, as shown in figure 23. Each choice of direction gives rise to a family of zigzag helical chains on the SWNT [57]. A complete chiral SWNT (L, M) can thus be described as either one of three collections of L helices (we assume $M > 0$ here; when $M < 0$, $L - |M|$ helices suffice). Inspection of figure 23 indicates that with $M > 0$, two of the three helices are right-handed and the third one is left-handed. In such a case, where there is a majority of right-handed helices, the SWNT itself is conventionally considered to be right-handed. Similarly, a left-handed SWNT is made of two left-handed and one right-handed sets of helices. Even the achiral SWNT can be described in this way. One of the three helices degenerates into a horizontal circumference (parallel SWNT) or into a vertical generator (perpendicular SWNT) and the other two have opposite hands. The achiral nanotubes can thus be considered also as a collection of L helices, provided one chooses one of the two sets oblique to the axis. For all cases, the L adjacent zigzag chains are related by a Bravais lattice vector of the net-diagram, which translates into a screw symmetry operation between adjacent helices on the SWNT.

The derivation of the kinematic form factor of a SWNT is then a matter of applying the CCV theory to the helices of the chosen zigzag collection [57]. The broad mathematical steps of the method are as follows. We will assume here that $M > 0$. The general case is given in [93].

(1) For each (n, m) term in the fundamental CCV formula for a mono-atomic helix of equation (6), one first adds the amplitudes $A_1(\mathbf{q})$, with their proper phase factors, of the two primitive, mono-atomic helices, which make up one single zigzag chain:

$$A_2(\mathbf{q}) = A_1(\mathbf{q})[1 + \exp i(q_z \Delta z_1 - n \Delta \varphi_1)], \quad (17)$$

where $(\Delta \varphi_1, \Delta z_1)$ is the screw operation connecting the two primitive helices.

(2) Then, one adds the amplitudes, each with its proper phase factor, of the L zigzag helical chains making up the complete SWNT:

$$A_L(\mathbf{q}) = A_2(\mathbf{q}) \sum_{j=0}^{L-1} e^{i(q_z \Delta z_2 - n \Delta \varphi_2)j}, \quad (18)$$

where $(\Delta \varphi_2, \Delta z_2)$ is the screw operation connecting two adjacent zigzag helices. The right-hand side is a geometrical series which can be summed in closed form.

(3) Now, by using planar geometry on the graphene sheet, the screw operations $(\Delta \varphi_1, \Delta z_1)$ and $(\Delta \varphi_2, \Delta z_2)$ can both be expressed in terms of the (L, M) chiral indices of the SWNT [57, 93]. It turns out that the phase angle in equation (18) is particularly simple: $q_z \Delta z_2 - n \Delta \varphi_2 = (2\pi/L)(n + mM)$. Since $n + mM$ is an integer, the L exponentials in equation (18) are L uniformly spaced points on the complex unit circle. The sum will, therefore, vanish for most couples (n, m) except for those pairs that make this phase angle an integer multiple s of 2π :

$$\sum_{j=0}^{L-1} e^{i(2\pi/L)(n+mM)j} = \frac{e^{2\pi i(n+mM)} - 1}{e^{i(2\pi/L)(n+mM)} - 1} = L \sum_s \delta_{n+mM, sL}. \quad (19)$$

This crucial mathematical step is entirely similar to the one we used for evaluating the amplitude of a multi-strand coiled-coil at the end of section 4.2. Since the (n, m) integers are themselves selected by the layer-line index l via equation (9), the effect of the new selection rule in equation (19) is to eliminate most layer-lines of the isolated zigzag helix and to leave only those layer-lines characteristic of the form factor of the entire SWNT. The layer-lines occur at $q_z = 2\pi l/T$, where T is the true translation period of the SWNT along its axis. Contrary to the case of the biological helices where T may be infinite (e.g. when the ratio p/P between atomic

and helical periods is irrational), for SWNT, T always remains finite. For general values of the (L, M) indices, T is given by $T = C\sqrt{3}/N$, where $C = a(L^2 + M^2 + LM)^{1/2}$ is the nanotube circumference (figure 23) and N is the largest common divisor of $2L + M$ and $L + 2M$. For parallel SWNTs, $T = a\sqrt{3}$ and for perpendicular ones, $T = a$. The somewhat cumbersome details of the complete derivation of the form factor along the l th layer-line have been given in [57, 93] and need not be repeated here. The end result is given by the compact formula

$$F_l(\mathbf{q}) = e^{iq_z z_0} \frac{4\pi C}{\sqrt{3}a^2} \sum_{m,s} J_{sL-mM}(q_\perp r) e^{i(sL-mM)(\psi_k - \varphi_0 + \pi/2)} \times f(q) [1 + e^{2\pi i(s+2m)/3}] \delta_{s(L+2M)+m(2L+M), Nl}. \quad (20)$$

The general structure of this expression and the physical origin of the various factors should be clear from the previous discussion. The last Kronecker δ expresses *the SWNT selection rule* for picking out the integer couples (s, m) , which contribute non-zero amplitudes to the l th layer-line. This equation is valid for $M \geq 0$. When $M < 0$, one has to replace M by $|M|$, L by $L - |M|$ and change the sign of ψ_k and φ_0 .

In boron nitride (BN) SWNTs, one of the two primitive helices of each zigzag pair is replaced by a boron helix and the other by a nitrogen helix. The expression (20) above can then be adapted to this case by replacing the zigzag helix form factor by $f_B(q) + f_N(q) \exp[2\pi i(s + 2m)/3]$ where the appropriate atomic scattering factors f_B and f_N are shown in figure 1.

When dealing with MWNTs or bundles of nanotubes, it will be necessary to add the *amplitude* contributions of all individual SWNT-like layers to obtain the total diffraction *amplitude* of the sample, each term suitably multiplied by a phase factor depending on the linear and angular positions of the layer in question [57, 93]. This, of course, is similar to what had to be done with the CCV amplitudes to obtain the diffraction of x-rays by biological molecules, which are made of a sometimes large collection of mono-atomic helices. A computer code called DIFFRACT has been developed [93] to implement the theoretical form factor in equation (20) and has been used to generate most of the simulations discussed below. The code can be requested from philippe.lambin@fundp.ac.be.

5.4. A few applications

5.4.1. Single-walled nanotubes. Before commenting on the experimental ED of real CNTs, we familiarize the reader with idealized patterns of SWNTs simulated from the theoretical form factor of equation (20). The understanding may also be helped by examining the visible diffraction patterns produced by simple optical gratings mimicking the planar projection of a few CNTs [63].

Figure 24 shows normal incidence patterns of (20, 0), (10, 10) and (19, 2) SWNTs vertically oriented. To understand these pictures, it is useful to remember that the diffraction pattern of a large, flat honeycomb lattice normal to the electron beam, i.e. its two-dimensional reciprocal lattice, is a hexagonal array of sharp spots or $(hk0)$ nodes with hexagon edges parallel to those of the carbon hexagons. In figure 24 one observes that each of these reciprocal spots, while remaining sharp in the vertical direction, has been elongated into a broad streak perpendicular to the nanotube axis. This spot broadening is a consequence of the finite diameter of the diffracting molecule, as we pointed out already for the diffraction by biological helices in figures 5, 13 and 21. The elongation of the original reciprocal nodes occurs in the direction *away* from the meridian and not towards it. This is physically due to the chirping *downwards* of the apparent lateral periodicity of the carbon network towards the left and right sides of the SWNT, as ‘seen’ by the incoming electron waves (the shorter the period, the larger the

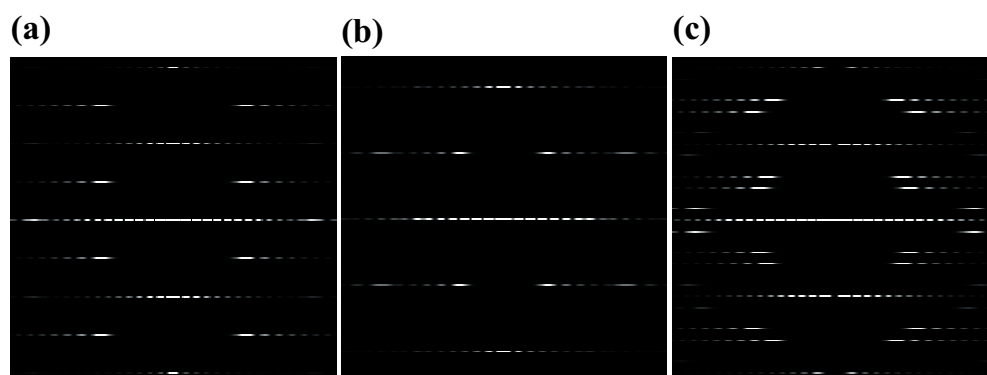


Figure 24. Simulated ED patterns of (a) (20, 0), (b) (10, 10), (c) (9, 2) SWNT. Rotational averaging of intensities has been applied to (a) and (b) to avoid accidental interference cancellations between the top and bottom halves of the nanotube.

scattering angle). The first maximum of the streaks coincides with the spots of the original reciprocal lattice. Finally, the streaks do not fade monotonically away from the meridian but are modulated in intensity. These oscillations of course reflect the ups and downs of the Bessel functions in equation (20). Physically, they are produced by interferences between the electron waves scattered by the left and right halves of the SWNTs. Such oscillations are similar to the optical diffraction oscillations caused by a finite aperture or a finite object, an effect that is well-known to limit the resolution of optical instruments based on far field image reconstructions. The frequency of the oscillations is an inverse measure of the SWNT diameter (see the argument of the Bessel functions in equation (20)).

The streaks now define the equidistant layer-lines of the pattern. Note that for electrons of several 100 keV used in TEM and ED (for 100 keV the de Broglie wavelength is about 0.03 Å, 100 times smaller than the lattice constant a), the Ewald sphere in reciprocal space is very nearly planar, so the layer-lines of the ED patterns of CNTs, are straight and equidistant instead of hyperbolic and chirped as in x-ray diffraction (compare with figure 2).

When comparing the three diffraction patterns of figure 24, one sees that while an achiral SWNT has one single hexagonal set of elongated spots (figures 24(a) and (b)) on each diffraction circle of successive Bragg orders, a chiral SWNT produces two such patterns, one rotated clockwise and the other anticlockwise by the chiral angle. One rotated pattern is produced by the 'front' half of the SWNT, the other by the 'back' half with respect to the incoming beam. The angular separation between the two is, therefore, twice the chiral angle, which can thus be approximately read from the angular positions, measured from the pattern centre, of the first maximum of two equivalent diffraction streaks.

If, by using suitable apertures in a microscopy set-up to reconstruct the nanotube far field image, one selects the diffracted beams of one of the two hexagonal patterns of spots and eliminates the others, one will obtain a picture showing only the front or the back face of the SWNT. Showing only one side of the object, rather than the confused Moiré pattern of superposed front and back faces, may result in more easily interpretable images. Recently, Hashimoto *et al* [94] have used the optical transform of a high-resolution TEM micrograph of a SWNT to filter in this way the optical diffraction pattern and reconstruct the image of one face of the nanotube, thereby determining its (13, 8) chiral indices with a good degree of confidence. A similar method, called *diffraction contrast dark field imaging*, has been exploited by Bernaerts *et al* [95] in the electron microscope itself to localize successive

groups of monochiral layers in MWNT (see below). This principle of aperture filtering and image reconstruction had already been applied long ago and with great success to optical (or computer-generated) diffraction patterns of electron micrographs of helical viruses [76].

The experimental ED patterns of isolated SWNTs [96, 97] have been difficult to measure due to their very weak scattering power and their tendency to be damaged by the energetic electron beam [95]. Bundles of several CNTs with one or a few walls are more resistant to electron beam damage, require less irradiation time and are more easily investigated by high-resolution TEM and ED (see below). However, in recent works, Gao *et al* [98] and Jiang *et al* [99] have obtained high-quality diffraction pictures of completely isolated SWNTs. In Gao *et al*'s work, the SWNT was illuminated by a very narrow, parallel, highly coherent electron beam 50 nm in diameter and the diffracted electrons were detected on an image plate intensifier recording the true, unsaturated intensities. The high quality of the pictures led them to succeed in measuring unambiguously the (L, M) chiral indices of the isolated SWNTs. Figure 25(a) shows the experimental pattern of a small diameter SWNT and compares it to a theoretical simulation in figure 25(b). The analysis of the several intensity oscillations of the equatorial layer-line, given by the dominant $J_0(q_\perp r)$ term in equation (20), allowed the identification of the diameter $2r = 1.4 \text{ nm} \pm 0.02 \text{ nm}$. The chiral angle $\alpha = 16.9^\circ \pm 0.2^\circ$ was measured [98] *via the distances of the sharp layer-lines to the equator* rather than from the ill-defined angular positions of the first maxima on the layer-lines. The necessary geometrical relationships are given in [57, 98]. In addition, this method of determining α is independent of the possible tilt of the nanotube axis away from the perpendicular to the beam [98]. These accurate values of $(2r, \alpha)$ turn out to be consistent only with the (14, 6) wrapping indices [98] for which the expected values are $2r = 1.39 \text{ nm}$ and $\alpha = 17.0^\circ$ when using a honeycomb lattice constant of $a = 2.46 \text{ \AA}$. The nearest couple (15, 6) gives a diameter (1.47 nm) and a chiral angle (16.1°), completely outside the experimental errors. Figures 26(a) and (b) show the observed and computed diffraction patterns of a (35, 20) SWNT ($2r = 3.77 \text{ nm} \pm 0.04 \text{ nm}$, $\alpha = 20.99^\circ \pm 0.2^\circ$). For this much larger tube, the high accuracy of the measurement of the chiral angle was essential for the unambiguous assignment of the wrapping indices [98].

Figure 27(a) is another, high-quality diffraction pattern obtained by Jiang *et al* [99] for an isolated (21, 9) SWNT synthesized by an aerosol method of CNT production. The methods for the quantitative measurements of $(2r, \alpha)$ from the ED pattern were the same as those employed by Gao *et al* [98]. The simulated pattern in figure 27(b), which was obtained with the DIFFRACT computer code [93], is in good agreement with the observed pattern.

Gao *et al* [98] have pointed out two noticeable differences between the measured and simulated patterns in figures 26(a) and (b). First, the distance between the experimental layer-lines are a bit larger than between the simulated ones. Second, the first maxima of the observed non-equatorial layer-lines occur closer to the meridian than the simulated ones. They attributed both effects to a tilt of the SWNT axis by about 13° away from the normal to the electron beam, in accordance with the geometrical mechanism described in [57, 100]. The pattern remains north-south symmetric.

To explain this mechanism we refer to figure 28, which represents a perspective view of the reciprocal space of a vertical SWNT. This construction is obtained as follows. Recall first that the three-dimensional reciprocal space of a flat vertical graphene sheet consists of a hexagonal array of rods normal to the sheet and passing through the $(hk0)$ reciprocal nodes, with parameter $b = 4\pi/\sqrt{3}a = 2.95 \text{ \AA}^{-1}$. By virtue of the description of a nanotube as a rolled up graphene sheet, the reciprocal plane likewise rotates about a vertical line drawn in it parallel to the axis of the nanotube and passing through the (000) node of origin. The reciprocal points $(hk0)$ describe circles in equidistant horizontal planes, which are our familiar layer-planes. The rods, tangent to these circles, sweep the outer regions of the circles. The rods cover the

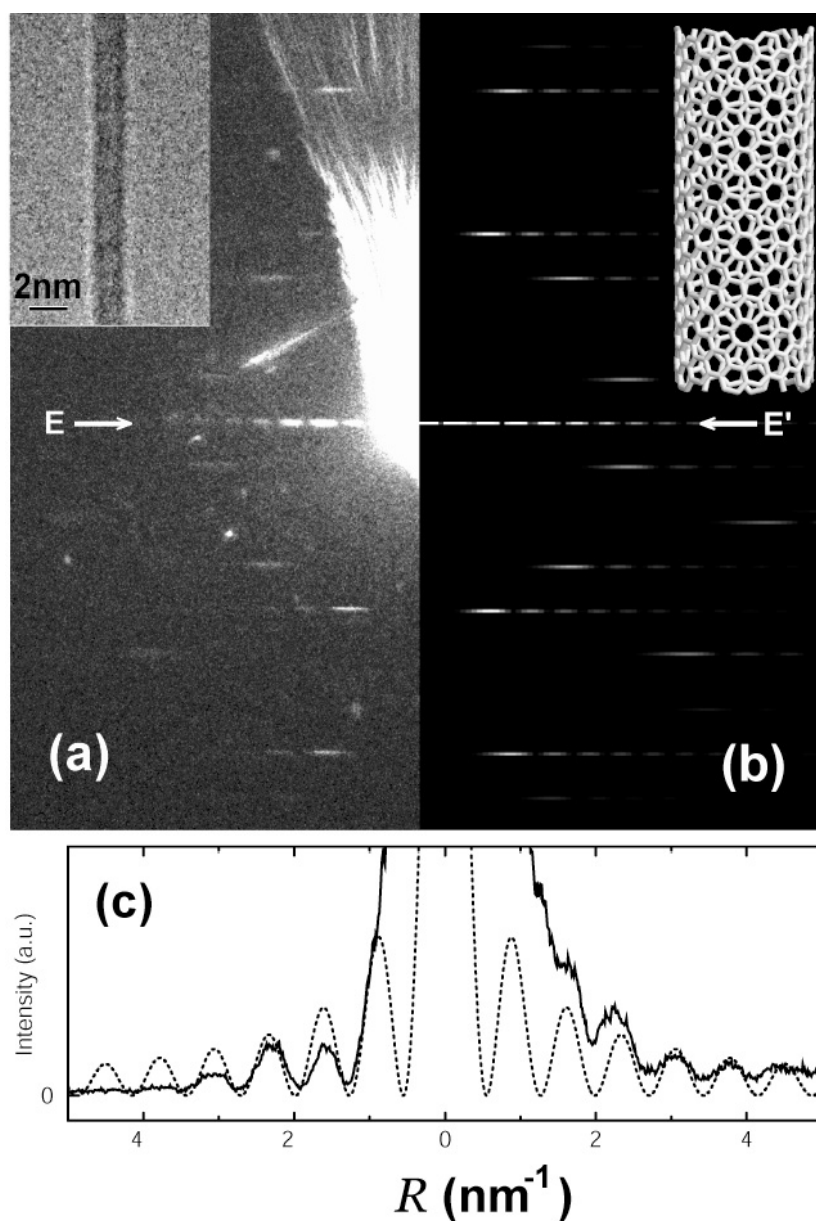


Figure 25. The observed (a) and simulated (b) ED patterns of a (14, 6) SWNT [98]. (c) Compares the measured and calculated distributions of intensity along the equator layer-line. Reproduced with kind permission from [98].

layer-plane more densely close to the circles than far from them. The reciprocal space of a single-walled nanotube, where one expects to find non-zero diffracted intensity, is therefore made of hollow, equidistant, outwardly fading discs with inner radii equal to the distance of the $(hk0)$ point to the rotational axis [100]. In particular, the rod issued from the origin (000) forms a full disc, the equatorial layer-plane. With respect to that equator, the latitudes of the first-order discs, generated by the six first-order reciprocal points, are $\pm(\pi/6 - \alpha)$, $\pm(\pi/6 + \alpha)$ and $\pm(\pi/2 - \alpha)$, where α is the chiral angle of the nanotube. These discs are represented in

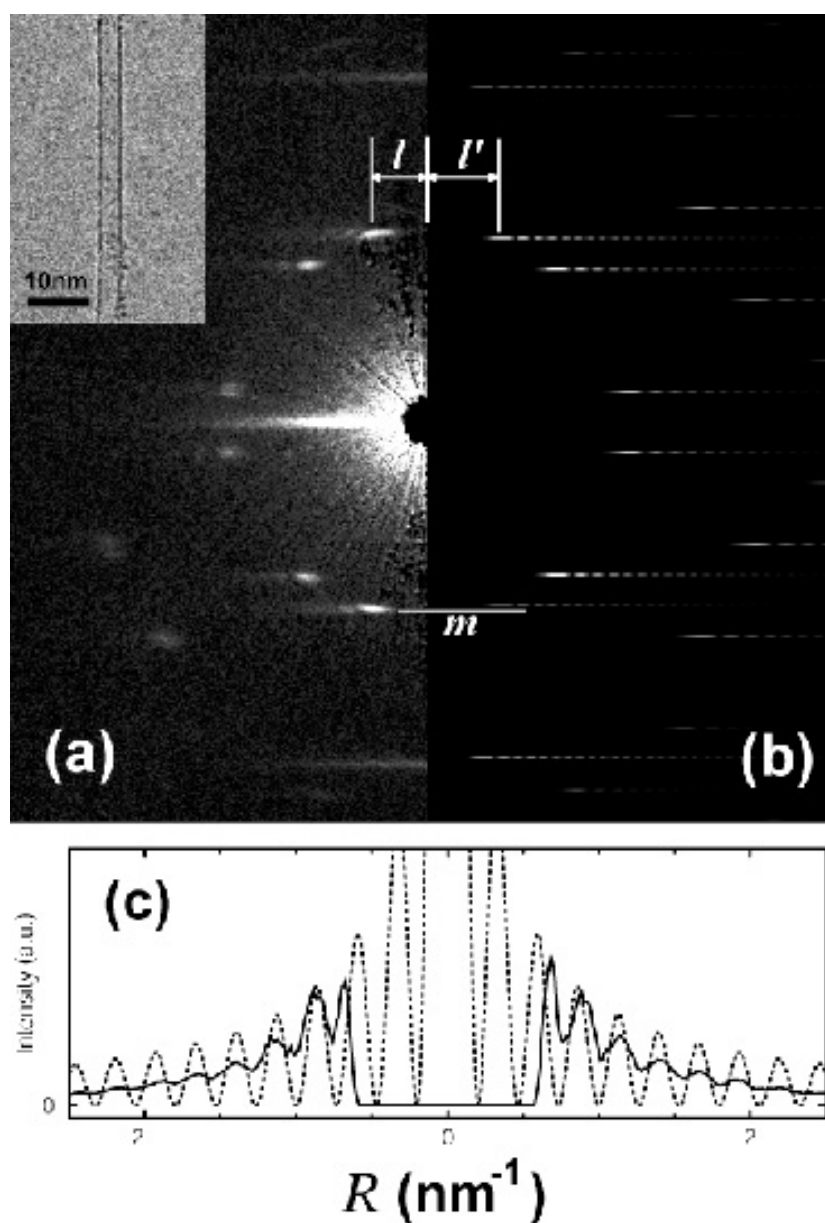


Figure 26. Same as in figure 25 for a bigger (35, 20) SWNT. The inset in (a) shows a high-resolution TEM picture of the nanotube [98]. Note that the streaks of the experimental pattern (left) are closer to the meridian and a little further away from the equator than in the simulated pattern (right), due to the tilt of the SWNT away from the normal to the electron beam. Reproduced with kind permission from [98].

figure 28. The Ewald plane (the plane of the figure) intersects these discs along the outwardly fading and modulated streaks discussed above.

Now, suppose the nanotube is tilted away from the vertical axis so that the layer-planes and their discs in figure 28 are rotated around the equator and become inclined on the horizontal plane. The vertical Ewald plane remains the plane of the figure. Then, the elongated spots

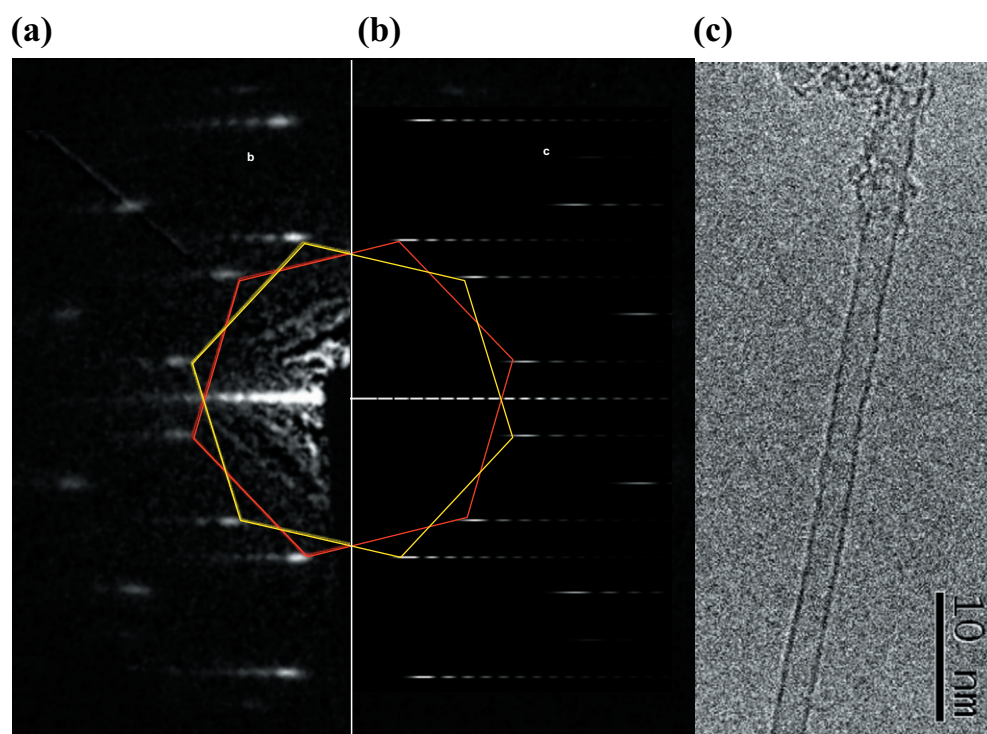


Figure 27. (a) Observed and (b) simulated ED patterns of a large (21, 9) SWNT showing perfect agreement. (c) is a high-resolution TEM picture of the nanotube. Reproduced from [99] with kind permission.

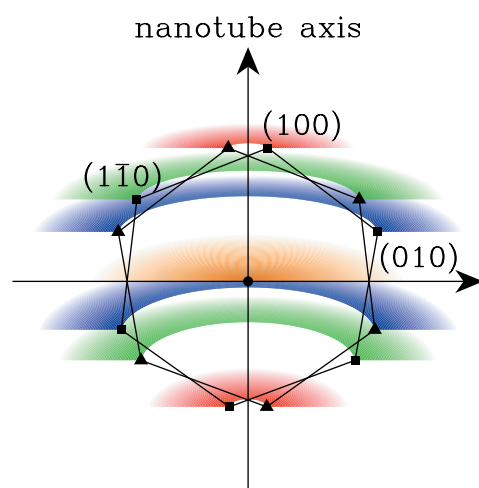


Figure 28. Half the reciprocal space of a vertical, chiral SWNT. The horizontal, fading discs are the loci of non-zero diffracted intensity (see text). The equator disc is full while the non-equatorial ones are hollow, their inner radii being the distances of the $(hk0)$ reciprocal nodes to the meridian axis. Upon tilting the nanotube away from the vertical, the (100) nodes move along the inner edge of the disc, that is upwards and towards the meridian, where they will merge when the tilt angle coincides with the chiral angle.

move symmetrically along the edge of their disc that is progressively *closer to the meridian and a bit further away from the equator*. This is just what happened in the observation reported by Gao *et al* [98] in figure 26. The two upper spots in figure 28 will eventually merge on the meridian when the inner edge of the disc becomes tangent to the Ewald plane. This occurs when the tilt angle coincides with the chiral angle. On increasing the tilt angle further, this merged pair of spots will progressively fade away and disappear. The next pair of spots on the lower layer-lines will successively behave in a similar fashion until the nanotube axis is parallel to the electron beam. ED pictures of MWNTs illustrating these spot movements are reported in [57] and will be discussed further later. Note that the equator layer-line is totally independent of the tilting.

The reader is reminded here of the corresponding spot movements upon tilting in the diffraction patterns of biological helices in figures 3–5. There, the additional effect was the breaking of the north–south symmetry as a result of the finite radius of the Ewald sphere for x-rays.

5.4.2. Isolated double wall nanotubes. In the same paper cited above [99], Jian *et al* have also studied the diffraction of *isolated* double wall nanotubes (DWNTs). Their results are reproduced in figure 29. Depending on the chiral nature of the two concentric nanotube layers, there ought to be at least one and at most four hexagonal sets of spots on each Bragg diffraction circle. The observed pattern (left-hand side of figure 29) shows just three such sets, which are outlined by half hexagons superimposed on the simulated pattern (right-hand side of figure 29). From the accurate geometrical survey of these patterns, the final (L, M) structure of the DWNTs that fits the data best was found to be $(8, 8)@ (20, 5)$ or, in terms of the $(2r, \alpha)$ couple, $(1.09 \text{ nm}, 30^\circ)@ (1.79 \text{ nm}, 10.89^\circ)$. While the moderate-resolution image of the nanotube shown in figure 29 does not resolve the split fringes produced by the projection of their two walls, a single wall is excluded by the presence of three sets of hexagons. A further feature of the diffraction pattern does add support to the assumed double-walled structure. Indeed, the intensity distribution of the equatorial layer-line reveals that the typical Bessel function oscillations are interrupted in a broad range around $q_\perp = 1 \text{ \AA}^{-1}$ (marked by an arrow in figure 29). The origin of this extinction is the nearly complete destructive interference between the amplitudes scattered by the two layers for that particular range of the horizontal scattering vector. Indeed, from equation (20), the equatorial intensity ($l = 0$) is obtained by squaring the total amplitude, which comprises the two main terms

$$A(q) = f(q)[r_1 J_0(q_\perp r_1) + r_2 J_0(q_\perp r_2)], \quad (21)$$

where r_1 and r_2 are the radii of the inner and outer nanotubes, respectively (see also equation (8)). Figure 30 compares the equatorial intensities expected for two such SWNTs in isolation or combined into one DWNT. The two radii, which are those of a sample discussed later in this review, are not very different from those of figure 29. But what is essential for the present discussion is that the radii differ by about the graphitic distance of 0.34 nm. The DWNT in figure 30 is seen to have a broad region of weak intensity around $q_\perp = 1 \text{ \AA}^{-1}$ characteristically produced by that difference of radial distances, as observed in the experimental as well as the simulated patterns of figure 29 (and further patterns discussed later). It is interesting to note here that the present *radial* extinction is rather analogous to the *meridian* extinction of the fourth layer-line in the diffraction pattern of B-DNA since the latter is also produced by the destructive interference between the two *axially separated* helical backbones of the molecule.

It should be clear that in order to determine unambiguously the (L, M) or $(2r, \alpha)$ structure of a CNT, the larger the radius, the higher the accuracy must be with which to survey the geometry of its diffraction pattern. Indeed, the small annular region defined by the experimental

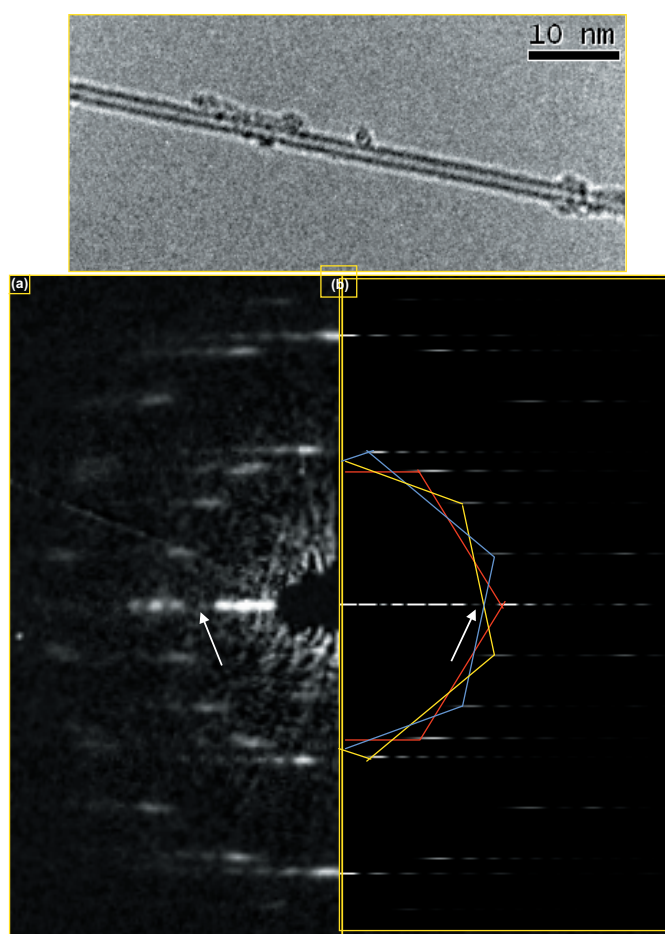


Figure 29. Experimental ED (left) and simulated patterns (right) of a DWNT having one achiral layer and one chiral layer (see text). The TEM picture shown on top lacks the resolution to separate the two layers. The arrows point to a region of the equator layer-line where there is an extinction due to the destructive interference of the scattering by the two walls. Reproduced from [99] with kind permission.

uncertainties Δr , $\Delta\varphi$ around the (r, α) point of the graphene plane contains a number of (L, M) nodes, which grows like $r\Delta r\Delta\varphi$. The most accurate determination so far appears to be that of Kociak *et al* [101] who have succeeded in taking and analysing high-quality ED pictures of isolated DWNTs with an outer diameter as large as 4 nm. One of their TEM and ED pictures is reproduced in figure 31. The high-resolution TEM pictures leave little doubt that the nanotube indeed has a double wall. The most remarkable feature of the experimental ED pattern, also recorded on a proportional imaging plate, is the unusually large number of visible Bessel oscillations both on the equator and on the other layer-lines. This allowed the authors to make a detailed comparison with the theoretical predictions based on equation (20). They could obtain two independent measurements of the chiral indices of the two walls. One set was obtained from fitting the *non-equatorial* layer-lines to each of which the two walls generally contribute independent intensities. Indeed, when the chiral angles of the two walls are sufficiently different, the corresponding layer-lines are separated in latitude, so that the

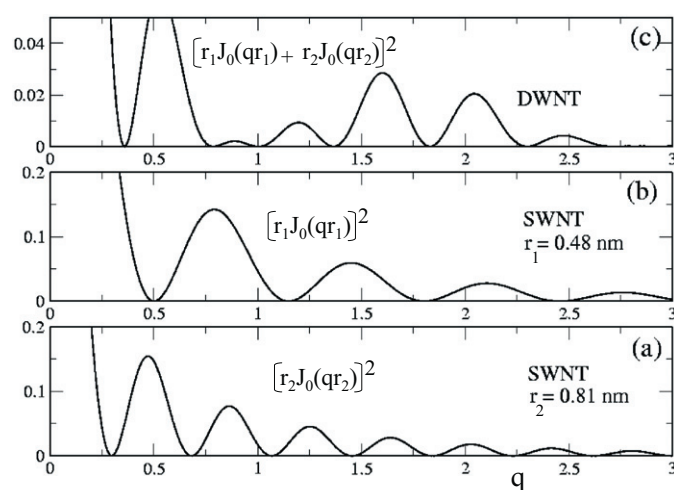


Figure 30. Expected intensities along the equator layer-line for two isolated SWNT (a) and (b) and one compound DWNT (c) with the indicated radii. Note in (c) the suppression of intensity around the transfer wave vector $q = 1 \text{ \AA}^{-1}$.

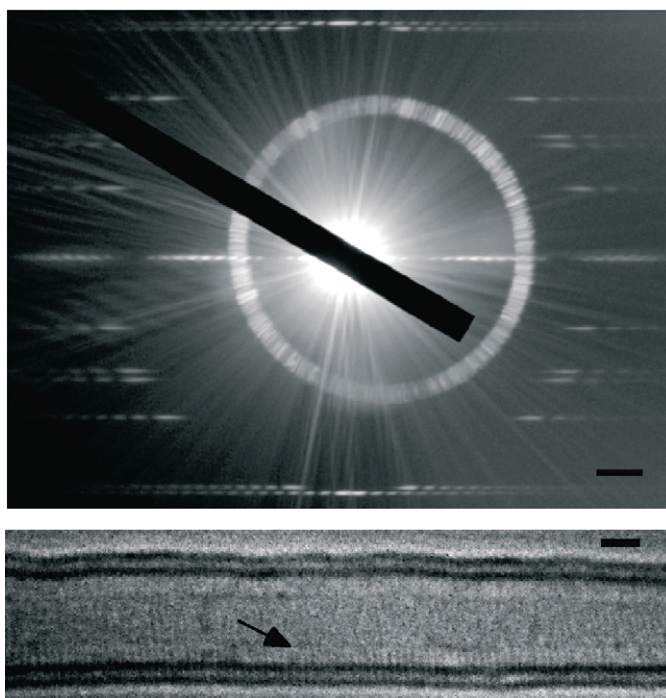


Figure 31. Experimental, high-resolution TEM and ED pictures of a large DWNT recorded on a proportional, unsaturated imaging plate. Note the large number of oscillations in the fine structure of the streaks, which lead to an unambiguous determination of the wrapping indices (see text). The iris-looking feature in the ED pattern is an experimental artefact. Reproduced from [101] with kind permission.

amplitude produced by one wall vanishes along the layer-lines of the other wall. The best set of chiral indices so obtained was found to also account for the oscillations of the *equatorial* layer-line where the two walls do interfere according to equation (21). The end result of the analysis was that the structure (41, 1)@(43, 13) gives the best fit to all the available data.

In a somewhat earlier paper, Kociak *et al* [102] were able not only to measure in this way the chiral indices of an even larger DWNT, (51, 34)@(53, 43) of 6.6 nm outer diameter, but also to characterize, *in the electron microscope*, the conductivity of the nanotube in relation to its measured structure. This combined capacity may be important for future nanoelectronic applications of CNTs. They found that although both walls of this DWNT are semiconductors with non-vanishing gaps (the $L-M$ difference not being a multiple of 3), the nanotube showed an Ohmic conductivity at low bias voltage, suggesting probable doping via the contacts between the nanotube and the electrodes [102].

The extensive analysis of the Bessel oscillations along the equator and other layer-lines of the diffraction patterns of a DWNT should remind the reader of the quite similar technique used in x-ray diffraction of TMV viruses [75] discussed in section 4.4. While here the DWNTs have a material radial distribution, which contributes interfering amplitudes to the equatorial intensities, there it was a comparison between these intensities for the mercury-substituted and the native viruses, which lead to the determination of the mercury helix radius as well as of the radial material distribution of the virus.

As illustrated in the several examples discussed so far and later in this review, the (L, M) wrapping indices of CNTs are most often derived from an analysis of their ED patterns using the helical diffraction formula of equation (20). However, in recent years direct or indirect atomic-resolution *imaging* has sometimes been successfully applied. One direct method, which does not involve diffraction was scanning tunnelling microscopy (STM), a method best applicable to SWNTs, as reviewed by Lambin *et al* [103]. The other, indirect method does involve diffraction and is called imaging nano-area ED (NAED) by Zuo *et al* [104]. Briefly, it consists in Fourier inverting the high-resolution diffraction pattern of a short length of CNT (along with a small support neighbourhood defined by a nanosized electron beam of 50 nm beam waist) by using a numerical algorithmic process of phase retrieval. The atomic-resolution image is iteratively reconstructed by back-and-forth FFT (fast Fourier transform) until self-stabilization of the phases is reached. The theoretical basis of the method is reviewed in [104] to which the reader is referred for references to the founding papers. Using this approach Zuo *et al* managed to reconstruct an atomic-resolution image of a 4 nm outer diameter DWNT and, with the help of the diffraction pattern itself, to ascertain the chiral indices as being (26, 24)@(35, 25). The authors claim that their method could be applicable to atomic-resolution imaging of *single* nonperiodic biological molecules of similar sizes, a feat, which, if realizable [77] could revolutionize once again the field of structural biology.

5.4.3. Bundles and rings of SWNTs and DWNTs. Due to their Van der Waals mutual attraction, CNTs comprising one or a few layers most often occur in bundles of several tens of individual tubes. These rope-like structures are not dissimilar to the biological ropes of α -helices discussed previously. The CNT bundles are most often bent and twisted in the crowded environment of the synthesis or post-synthesis treatments. However, even free standing, straight bundles are spontaneously twisted so that the somewhat flexible CNTs coil around each other, just like polypeptide chain molecules. The deformation energy spent in twisting is recovered from the slight increase in van der Waals energy between the nanotubes along their line of contact. However, the CNTs are stiffer than the more easily deformable polypeptide chains and the interaction between CNTs is not as strong as the hydrophobic force

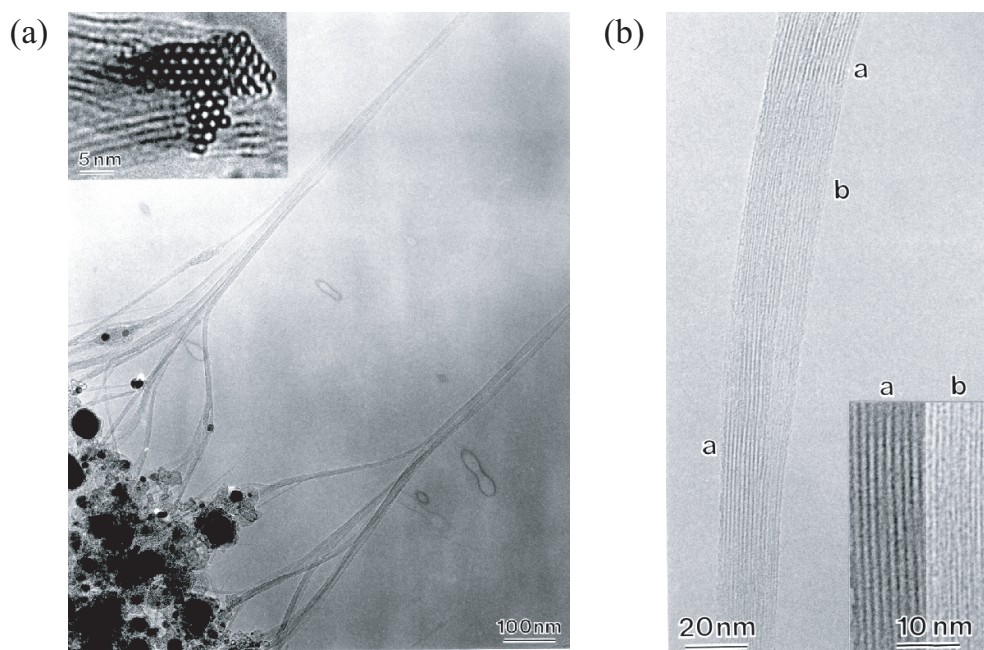


Figure 32. TEM pictures of bundles of SWNT. The inset in (a) shows a polycrystalline bundle in cross section. In (b), the presence or absence of lattice fringes shows that the bundle is twisted by about 1° nm^{-1} . Reproduced from [107] with kind permission.

between biological helices in their normal aqueous environment. So, the rate of twist of CNTs is typically 1° per nm of rope length, which is ten times smaller than for α -helix ropes.

As noted earlier, CNT ropes and MWNTs with many layers (see next section) are resistant to radiation damage and are routinely investigated by high-resolution TEM and ED. There have been quite a number of such studies since the initial discovery by Iijima [87], so we will confine ourselves here to review recent ones [105–107] (these papers refer to many earlier works on bundles).

Bundles of CNTs produced by chemical vapour deposition [105, 106], by electric-arc [107] or by laser ablation [108, 109], appear to be composed of nanotubes of sufficiently uniform sizes to allow them to cluster in quite regular triangular lattices. Sometimes, however, a large bundle [107] will be polycrystalline, which is made of adhering, smaller monocrystalline bundles. The polycrystallinity is revealed by TEM micrographs of bundles that happen to have their axis aligned along the electron beam, such as in figure 32(a) [107].

The analysis of the ED patterns of bundles, especially the intensity distribution along the equatorial layer-line, has to take into account an additional parameter, namely the lattice structure of the bundle. The lattice parameter is given by the nanotube diameter augmented by the nanotube separation, which is generally slightly less than the standard graphitic distance of 0.34 nm on account of the cylindrical shape of the surfaces in contact, which reduces both the attractive and the repulsive interactions as compared to parallel planes. In the diffraction pattern, the lattice regularity causes the breaking up of the layer-line intensity into equidistant Bragg spots whose intensities are modulated by the nanotube form factor. The spot distance depends on the electron beam direction with respect to the crystal lattice. The distribution of radii and of chiral angles of the nanotubes constitute the unknown bundle parameters to be investigated by diffraction. The determination of the relative atomic registry of the individual

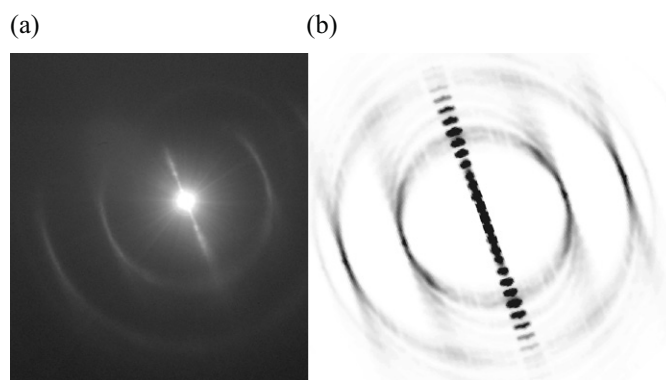


Figure 33. Experimental (a) and simulated (b) ED patterns of a bundle of SWNT such as shown in figure 32. The arcs on the Bragg circles are produced by a random distribution of chiral angles. The divergence of the equator layer-line is due to misalignment of the SWNT in the bundle. Reproduced from [107] with kind permission.

nanotubes with respect to their neighbours in the bundle is beyond the capability of the present techniques.

Henrard *et al* [107] have used TEM and ED to investigate thick bundles of monodispersed but polychiral SWNTs obtained by the arc discharge technique. Figure 32(a) shows the branching or fusing of a multi-bundle whose cross section reveals the different orientations of its several smaller constitutive bundles. Figure 32(b) is a normal incidence micrograph of a twisted bundle in which one can identify regions of successive lattice fringes. The fringes have different periodicities corresponding to (1, 1) and (1, 0) crystal orientations and indicate a twist angle of about 1° nm^{-1} . Figure 33(a) is the ED pattern of a straight SWNT bundle at normal incidence and recorded on a proportional imaging plate. Two major observations are (i) a spotty equatorial layer-line, which opens up outwardly by an angle of about $\pm 5^\circ$ and (ii) quasi-continuous arcs of intensities along Bragg diffraction circles.

Figure 33(b) is a simulated pattern based on the DIFFRACT code implementation of equation (20) [93]. To obtain good agreement with the experimental ED pattern, the authors [107] modelled a bundle of 55 SWNTs with one or two narrow distributions of diameters and a random distribution of 8 different chiral indices (see [107] for details). Moreover, they found it essential to include the effects of ‘*misorientation*’ of the SWNT in the twisted bundle, namely the slow precession of the nanotube axis around the bundle axis. That precession was indeed found to be one major cause for both the observed fanning out of the equatorial layer-line and for the quasi-continuous distribution of intensity along the Bragg diffraction circles. In the simulation, the twist was taken into account by averaging the diffracted intensity of straight, parallel SWNTs over the local precessing directions. One will remember that this procedure of intensity addition is exactly what was first used to account for the x-ray diffraction patterns of polypeptide coiled-coils. In that biological situation where the rate of coiling was rather strong (10° nm^{-1}), the procedure was found to be wanting and needed to be replaced by the exact formula of *amplitude* addition given by Crick in equation (13) (see section 4.2). The question of whether or not the simpler treatment is satisfactory in the present case depends on the coherent length of the electron beam used in the selected area diffraction set-up [107], a property of the experimental system, which unfortunately is rarely known or specified. In principle, for the intensity addition to be valid, the coherent length should be much smaller than the length over which the bundle is twisted by 2π (about 360 nm). Whether the effects of a finite coherent length can be unravelled from those of other instrumental conditions [107] and

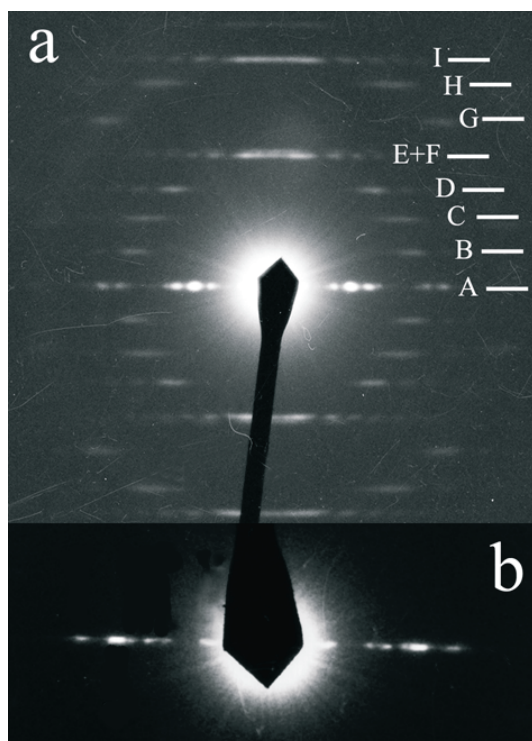


Figure 34. (a) The ED pattern of a bundle of a CVD-synthesized DWNT. All nanotubes are of the same or about the same chirality. The $(11, 0)@(16, 6)$ chiral indices gave the best fit to the distribution of intensity along the equator layer-line A shown enlarged in (b), as well as along all other layer-lines marked B to I in (a). Reproduced from [105] with kind permission.

from those of the great variety of structural parameters in the present SWNT sample remains to be seen.

Colomer *et al* [105, 106, 110] have applied TEM and ED to bundles of DWNTs synthesized by the CVD method. Figure 34 is the diffraction pattern of what is believed to be a small bundle of DWNTs, all having the same or very nearly the same diameter and chiral angle. When comparing the data with the simulations performed with the DIFFRACT computer code [93], the $(11, 0)@(16, 6)$ chiral indices gave the best fit to the distribution of intensity along the equator layer-line A shown enlarged in figure 34(b), as well as along all other layer-lines marked B to I in figure 34(a) [105]. Figure 35(a) shows the ED pattern of another sample of a monochiral nanotube bundle also produced by CVD [106]. The observed ED is best fitted by the simulated patterns of a $(8, 6)@(14, 10)$, as shown in figure 35(b). Figures 36(a) and (b) [110] give the pattern for yet another CVD bundle of DWNTs while figure 36(c) compares the observed equatorial layer-lines to the calculated one based on the chiral assignment $(8, 4)@(14, 7)$. The agreement is perfect, particularly in the weak intensity region observed around $1 \text{ \AA}^{-1}$ (double headed arrows) characteristic of the cancellation of amplitudes for these DWNTs. The double-walled nature of all these nanotubes is, therefore, in little doubt. Figure 37 adds confirmation to this, where one can observe in normal incidence (figure 37(a)) and in cross section (figure 37(b)) a particular single crystal bundle of the same sample but taken in a different field of view (note that, for experimental reasons, it is not often possible to obtain both the TEM micrograph and the ED pattern of the same region

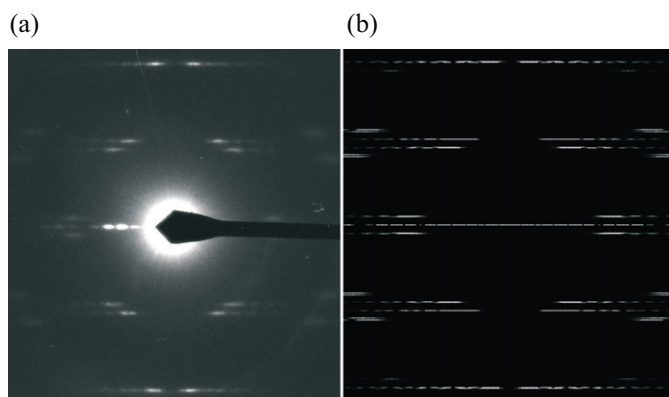


Figure 35. (a) Experimental and (b) simulated ED patterns of another bundle of monochiral DWNT with best fitted indices (8, 6)@(14, 10). Reproduced from [106] with kind permission.

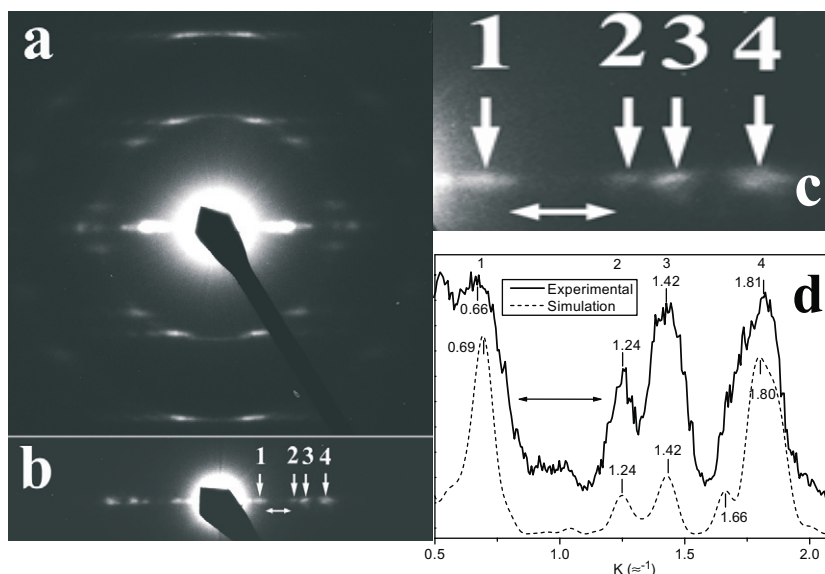


Figure 36. (a) Experimental ED pattern of yet another bundle of monochiral DWNT of indices (8, 4)@(14, 7). The equator layer-line is expanded in (b) and (c). The densitometer trace (—) in (d) is compared to the simulation (---). Reproduced from [110] with kind permission.

of a bundle). Colomer *et al* [110] have discussed some of the growth mechanisms likely to be responsible for such collective coherence as necessarily at work in the synthesis of monodispersed bundles.

Sometimes, during the preparation of samples for TEM-ED observations, closed rings of CNTs may form in low concentrations. Such species were first observed for SWNTs [111]. This spectacular phenomenon was again produced by Colomer *et al* [110] for DWNTs and is illustrated in figure 38. Figure 38(a) shows a perfect circular ring appearing as a closed torus. A seamless toroidal conformation is, however, highly improbable and, as shown by figure 38(b), the ring is in fact an open-ended coil made of about 120 DWNTs overlapping

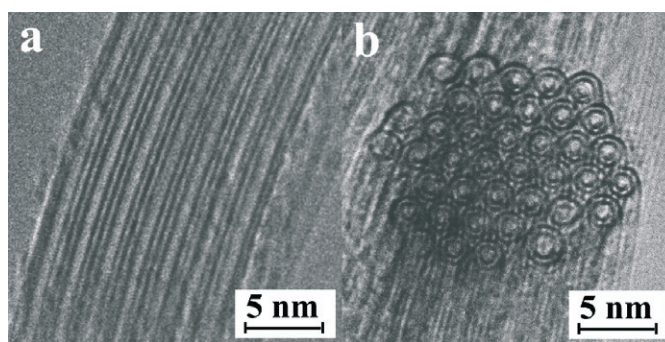


Figure 37. High-resolution TEM pictures of a crystalline bundle of monochiral DWNT similar to the ones studied in figures 34–36. The picture in (a) is taken perpendicular to the bundle axis and shows lattice fringes. (b) is a picture taken along the axis and clearly reveals the perfect hexagonal packing and the two walls of the nanotubes. Reproduced from [110] with kind permission.

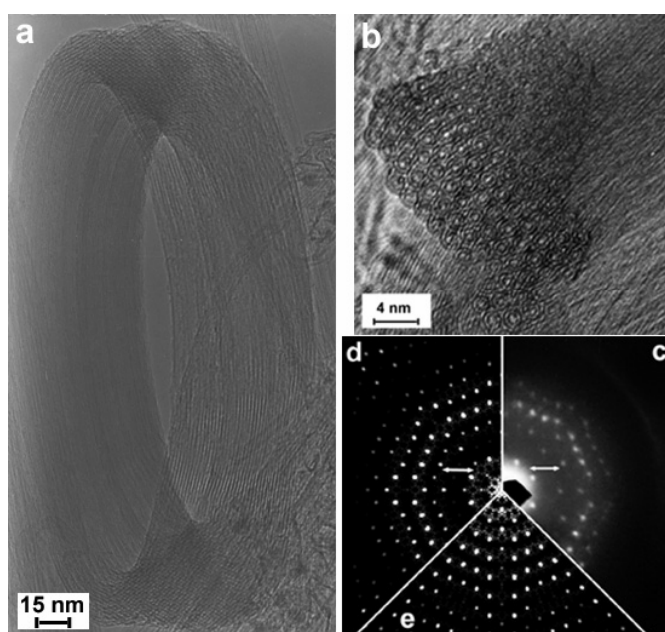


Figure 38. (a) TEM micrograph of a ringed bundle. (b) TEM micrograph of the bundle end seen in cross section. Each nanotube appears as two concentric circles, suggesting that the ring is made of DWNT. (c) Part of the ED pattern of the complete ring shown in (a). The spots are Bragg reflections of the triangular lattice of the twisted bundle of DWNT. The double-headed arrow indicates the characteristic extinction along the equator layer-lines of the twisted DWNT (see text). (d) Theoretical simulation of the ED pattern, assuming a DWNT bundle twisted by $8/6$ of 2π per turn of the ring, and showing the characteristic annular extinction. (e) A theoretical simulation assuming a twisted bundle of SWNT, lacks the annular extinction.

over part of their sticky ends. Some regions of the ring exhibit equidistant lattice fringes, while others do not, due to the variable lattice orientation around the ring circumference. Figure 38(c) shows a fraction of the ED pattern of the whole ring seen along its axis. The complete pattern has eight-fold symmetry and consists of spots aligned along concentric sets of inwardly curved octagons. To understand this, one should first remember, from all the ED patterns previously

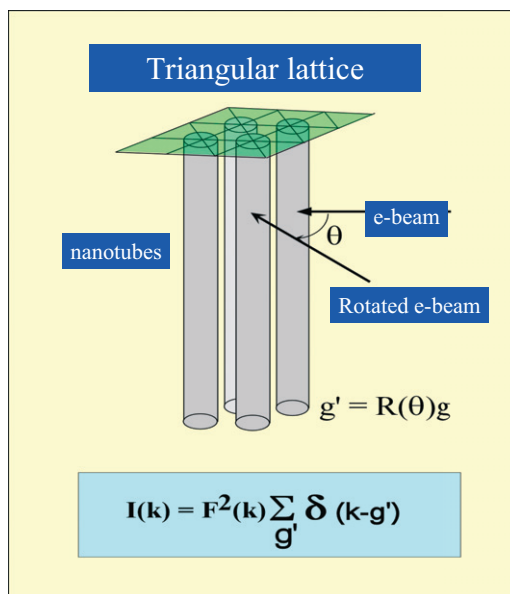


Figure 39. A schematic representation of ED by a triangular lattice of nanotubes. When a nanotube bundle is in the form of a twisted ring as in figure 38(a), the effect of the twisted geometry of the ring can be obtained from rotating the electron beam around a straight bundle. Every time the beam orientation is in a Bragg condition for a lattice reflection, a spot lights up in the local equatorial direction of the ring.

discussed, that the equator layer-line of a straight bundle is always the strongest, so what we are seeing in figure 38(c) is the set of such lines for a bundle whose orientation is continuously rotated around the electron beam. The other layer-lines are not detected. Normally, such a rotation diagram should produce continuous circles traced by the successive Bragg orders along the equator line, but in reality one sees only a discrete set of radial layer-lines. This is due to the fact that the bundled ring is twisted just like some of the straight bundles looked at previously. The twisting is confirmed by TEM micrographs taken along the ring axis, which, like in figure 38(a), also display different lattice fringes, which change continuously around the ring. Because the bundle is twisted, only those sections of the ring where the hexagonal lattice is under Bragg conditions produce diffraction spots along the corresponding equator layer-line.

A semi-quantitative explanation of the appearance of the ED pattern is given in figures 39 and 40. We will again assume that the unknown electron coherence length of the TEM set-up is small enough compared to the ring radius (about 10 nm) so that the various parts of the ring give additive intensities to the ED pattern. On account of the twist, the latter should be equivalent to that obtained by rotating the electron beam around a straight, untwisted bundle, as shown in figure 39. Inspection of this figure shows that if the twist angle was exactly 2π over one turn of the ring, all the (h, k) Bragg spots would produce the complete, six-fold symmetric, two-dimensional reciprocal lattice of the hexagonal crystal of the bundle. Figure 40(a) shows one-sixth of this reciprocal lattice. However, if the twist angle is larger than 2π per turn, then the reciprocal lattice will be explored more than once around one ring circumference. Then, the diffraction pattern of figure 40(a) should be angularly compressed, while maintaining the radial distances of the (h, k) spots to the $(0, 0)$ centre. This is what causes the inward sagging of the polygonal edges in figure 40(b), as in the observed pattern of figure 38(c). From the

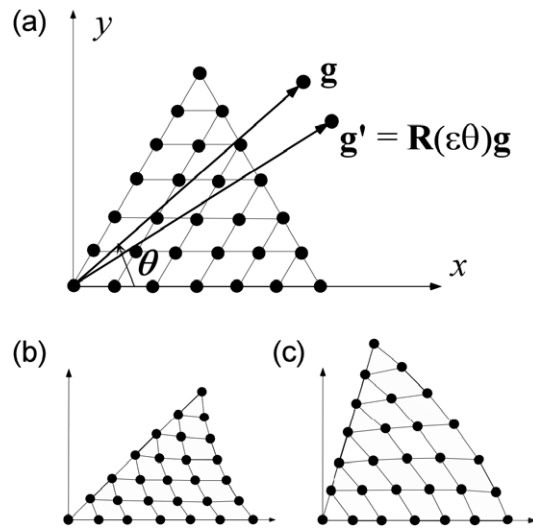


Figure 40. (a) One sixth of the reciprocal lattice of a triangular lattice. The nodes represent the Bragg spots that would be observed along the local equator layer-lines of a twisted ring such as in figure 38(a) if the twist angle, φ , was exactly 2π over one turn of the ring. (b) If the twist angle, φ , is larger than 2π , the reciprocal lattice is angularly compressed by a length-conserving rotation $\mathbf{R}(\varepsilon\theta)$, where θ is the polar angle of the reciprocal node and $\varepsilon = 1 - \varphi/2\pi$. This is the case observed in figure 38(b). (c) If the twist angle, φ , was smaller than 2π , the reciprocal lattice would be dilated.

eight-fold symmetry of the observed pattern, the twist angle is evaluated at $2\pi(8/6)$ per turn of the ring, which amounts to about 0.8° per nm of ring circumference (about 600 nm), a twist comparable to that of the straight bundles. If, on the other hand, the twist angle is smaller than 2π , the model predicts that the pattern of figure 40(a) should be angularly dilated to produce a pattern of polygons with outward bulging edges shown in figure 40(c). This case has not been observed yet.

Finally, figures 38(d) and (e) are computed patterns constructed along the above principles and confirming in a spectacular fashion the correctness of Colomer *et al.*'s interpretation [110, 112]. Figure 38(e) corresponds to a circular, twisted bundle of SWNT, which produces the complete, compressed two-dimensional reciprocal lattice, but not the annular extinction around $1 \text{ \AA}^{-1}$ observed in figure 38(c). Figure 38(d) simulates a DWNT ring and clearly exhibits again that extinction characteristic of all DWNT bundles encountered earlier (see figures 36(b) and (c)).

5.4.4. Multiple-walled nanotubes and scrolls. When the number of walls of a MWNT exceeds two, the determination of the ordered sequence of the (L, M) chiral indices of its constitutive layers becomes quite difficult [113]. Already, for a nanotube with only three layers measured by Iijima [114] and simulated in [115] with the DIFFRACT code, the near equivalence of several possible (L, M) pairs for matching the experimental ED patterns made it hard to assign unambiguously the actual sequence of wrapping indices. The number of solutions grows much larger still for large MWNTs comprising tens of layers, such as those of early studies (see Amelinckx *et al.* [57] and references therein). If, in such MWNTs, the layers had a random distribution of chiral angles, then the ED pattern would look like that of graphite powder, consisting of nearly continuous Bragg circles, and the sequencing would be just impossible on the sole basis of this pattern.

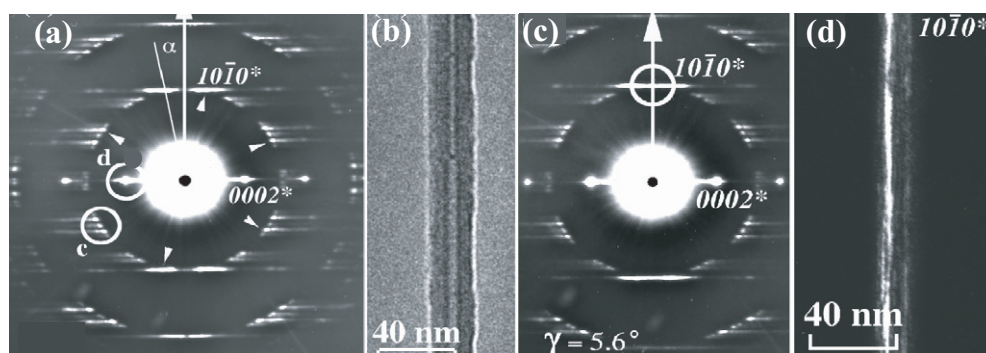


Figure 41. (a) The ED pattern of an MWNT having about 20 layers [95]. One notes the spotty appearance of the Bragg diffraction circles in spite of the large number of layers. (b) Bright-field image of MWNT in (a). (c) The ED pattern of the same MWNT tilted by an angle of 5.6° away from the normal to the electron beam. The tilt causes the (100) streaks to climb over the Bragg circle and merge onto the meridian; the circle there indicates the electrons selected to reconstruct the dark-field image in (d). Note the brightness asymmetry of the image (see text). Reproduced from [95] with kind permission.

The situation is not completely hopeless, however, due to the frequent experimental observations that the ED pattern of a MWNT, far from being continuous, appears quite spotty. In other words, the number of distinct chiral angles must be much smaller than the total number of layers. Figure 41(a) shows one example of the ED pattern of a MWNT having about 20 layers [57, 95]. One first notices the strong, equidistant $(0002n)$ spots aligned along the equator layer-line, which are produced by the left and right nanotube walls acting as diffraction gratings of spacing equal to the graphitic distance of 0.34 nm. Then, the first and second Bragg diffraction circles show a number of discrete spots elongated away from the meridian, as expected from the nanotube curvature. However, in spite of the large number of layers, one can identify only 4 or 5 distinct chiral angles. Several layers of this MWNT must, therefore, have the same or about the same chirality. Bernaerts *et al* [95] were able to accurately measure the angles from the ED patterns, but to go any further in attempting to locate the layers of given chiral angle, they had recourse to another powerful technique of electron microscopy, that of *diffraction contrast, dark field imaging*. In the usual *bright field image*, shown in figure 41(b), all the electrons going through the sample are used to construct the image, which appears, in the bright field of unscattered electrons, as dark shadowy features representing the parts of the specimen that have scattered electrons maximally. As mentioned earlier in this paper (see section 5.4.1) the dark field imaging method consists in using apertures in the microscope to let pass a selected set of diffracted beams with which to reconstruct the image and block all other electrons. Then, the image appears bright in the dark field, which all the discarded electrons fail to reach, as shown in figure 41(d). The spatial positions of the latter image with respect to the bright field image reveal the position of the scattering elements responsible for producing the selected beams. The aperture selection used by Bernaerts *et al* [95] is shown in figure 41(c). The experiment requires tilting the MWNT around the axis perpendicular to both the nanotube and the electron beam in order to bring the desired spots into the aperture. One will recall that such a rotation produces a climbing motion of the $(hk0)$ spots towards the top of the Bragg diffraction circles to which they belong, as explained with figures 28 and 29 in section 5.4.1. Each time the tilt angle passes through one of the measured values of the chiral angle, a pair of symmetric $(hk0)$ streaks merges onto the meridian (figure 41(b)). In figure 41(c) the tilt angle was 5.6° . The corresponding dark field image illuminates the

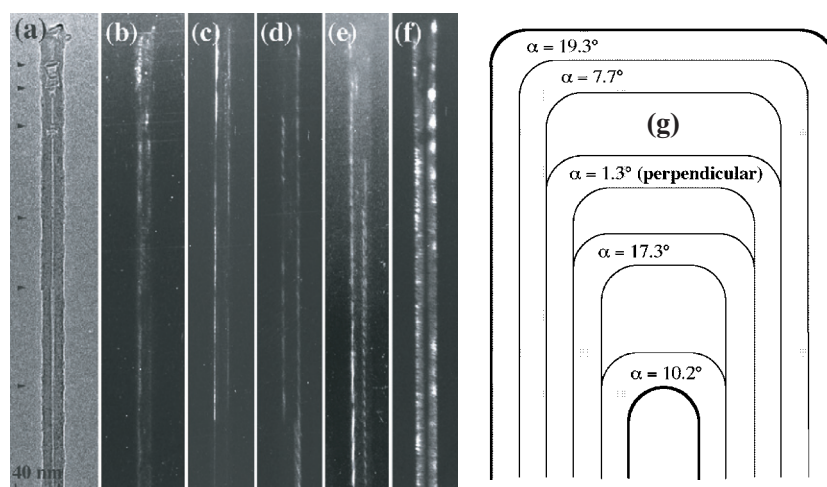


Figure 42. (a) Bright-field image of an MWNT. (b)–(f) Several dark-field images obtained using the technique shown in figure 41 and allowing us to locate the various clusters of layers with the same or about the same chirality. (g) This map gives the locations of isochiral layers deduced from (b)–(f). Reproduced from [95] with kind permission.

radial region of the MWNT where the layers producing these merged streaks are located. As illustrated in figure 41(d), it turns out that the isochiral layers are grouped together in clusters instead of being dispersed throughout the complete width of the MWNTs. Possible reasons for this clustering are discussed below. Moreover, the dark field image of chiral layers appears only on one side of the MWNTs (figure 41(d)). This asymmetry is due to the fact that, with the nanotube tilted at the chosen chiral angle, the atomic helices of the layers that produce the merged diffraction streaks project as cycloid-like curves in the plane normal to the electron beam. The cusp side of the cycloids then act as a vertical diffraction grating of horizontal slits, producing the merged meridian streaks, while the smoothly rounded opposite side of the cycloids scatters in all directions. Hence, the dark field image is formed on the cusp side of the cycloids. Given the sense of tilting, which side is illuminated also determines what handedness the contributing helices have [95]: when the nanotube is tilted with its top leaning towards the incoming beam, a bright left (right) side will indicate right- (left-) hand helices. This technique applied to different ($hk0$) merged beams of the same isochiral layers enables one to determine the handedness of the three families of carbon helices and hence the handedness of the layers themselves.

Figures 42(a)–(f) show the bright field image (figure 42(a)) and five of the dark field images obtained with this powerful technique for a MWNT having two dozen layers. In figure 42(g) a map of the sequence of isochiral clusters is constructed (see [95] for details), including the positions of their termination in the MWNTs. Although this remarkable measurement does specify the sequence of $(2r, \alpha)$ values to within a relative accuracy of a few per cent, the actual sequence of (L, M) wrapping indices of the individual layers remains quite ambiguous for reasons that should be clear from our previous discussions of SWNTs and DWNTs.

Such clusters of adjacent isochiral layers occur frequently in MWNTs. In the defect-free, nested-layer model of MWNTs, clustering might result from a pseudo-epitaxial growth process during which each new layer would locally nucleate on top of the previous layer and would then be committed to grow with the same orientation of the lattice as that of the

substrate cluster. After a few layers grown in this way, an abrupt change of chiral angle would eventually have to take place due to the necessity to maintain the 0.34 nm graphitic interval between the layers. However, irrespective of the method of their synthesis, MWNTs are rarely defect-free. One such defect causes anomalously large lattice fringe spacings, up to twice the graphitic value, in one or the other of the two walls of the nanotube [57]. There are at least three possible explanations for such anomalies. One explanation [116] assumes that in the growth process one layer has formed with an anomalously large radius. The van der Waals attraction then causes the inner cluster of nested layers to adhere eccentrically to one side of the outer cluster, leaving a region of increased spacing on the opposite side. With very little tube deformation, this region of anomalous spacing could meander along the nanotube length and lead to the often observed appearance and disappearance of the fringe anomaly along the tube axis. Another explanation assumes the presence of an edge dislocation parallel to the nanotube axis. Such a dislocation could be associated with an incomplete layer intercalated between two nested layers. Alternatively, the anomaly could be attributed to a *scroll organization* of parts of the MWNT [57], the two ends of the scroll again producing edge dislocations. This latter interpretation has considerable appeal because it naturally explains the isochirality of several adjacent layers since a multiturn graphite scroll obviously conserves its chirality at every turn. According to this view, one or several of the isochiral layers of the MWNT in figure 42 might be scrolls surrounded by a cluster of regularly nested layers or by a larger scroll. With the present state of the TEM and ED techniques, it is impossible to distinguish between all these possibilities, unless the sample could somehow be rotated and looked at in cross section at different locations along the axis.

The coexistence of scrolls with clusters of coaxial, nested tubes in MWNTs has been discussed recently by Lavin *et al* [117]. On the basis of high-resolution TEM images, they suggested that the former structure could transform into the latter by a progression of defects. The independent existence of pure scroll structures resembling MWNTs has been demonstrated unambiguously in recent studies [118, 119]. A method of catalyst-free synthesis to produce CNT-like structures has been described by Viculis *et al* [118], who have provided real space, TEM evidence of a pure nanoscroll configuration. Convincing, reciprocal space, ED evidence has been given by the work of Ruland *et al* [119] for pure scrolled MWNTs synthesized by the auto-catalytic pyrolysis of iron phthalocyanine. Figures 43(a) and (b) show TEM pictures of a large MWNT whose apparently defect-free wall exhibits as many as 45–50 graphitic layers. Figures 43(c)–(e) are ED patterns of three different MWNTs. In spite of the large number of layers, and in contrast to an ED picture, such as in figure 41(a), the patterns are astonishingly simple comprising either one single hexagon of spots on each Bragg circle (figure 43(c)) or at most two such hexagons (figures 43(d) and (e)). In these 3 pictures, the unique chiral angles are measured as $\alpha = 0^\circ$, 9.7° and 25.1° , respectively. By analysing the accurate relationship between the interlayer spacings and the chiral angles in many such samples, the authors convincingly argue that the observed isochirality of the complete MWNT is inconsistent with the nested-layer model while it is readily explained with the nanoscroll structure.

The convergence of ideas and techniques used in diffraction studies of CNTs and of biological helices has been noted several times in this review. One last example is provided by the similarity between the technique just discussed for determining the handedness of isochiral layers in a MWNT [57, 95] and the method used to determine the absolute handedness of the TMV helix from TEM micrographs of the particle [120]. Both techniques involve looking at what happens to the TEM image and ED pattern upon tilting the specimen away from the normal to the electron beam. For the TMV, the measurement was made by Finch [120] who examined several moderate-resolution TEM micrographs of a single TMV particle taken at various tilt angles. As we have seen, tilting the top of a helix away from the vertical

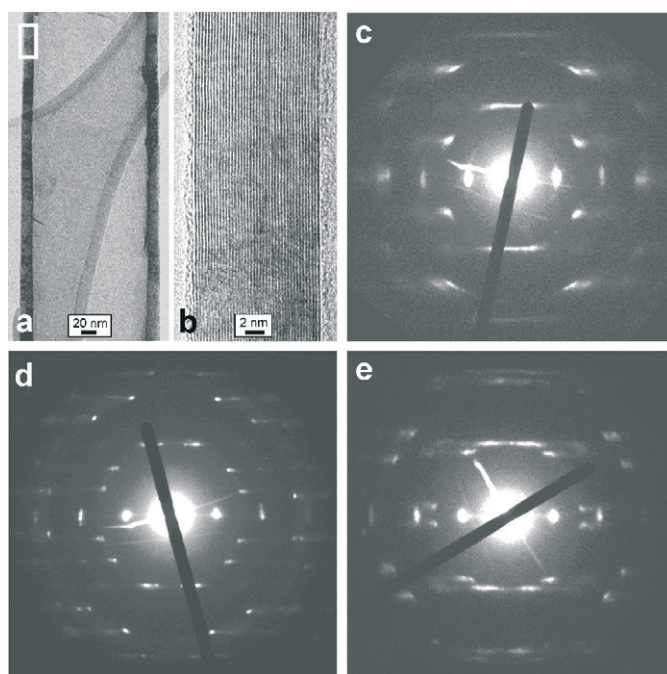


Figure 43. (a) Low-resolution TEM image of a large MWNT. (b) High-resolution TEM image of one wall of the MWNT selected by the white rectangle in (a). (c)–(e) ED patterns of three different MWNT such as the one shown in (a). There are one or at most two hexagonal sets of spots on each Bragg circle, showing the monochirality of all three MWNT, which are therefore most likely to be scrolls (see text). Reproduced from [119] with kind permission.

towards the viewer by an angle equal to the helix slope changes the image from a sine wave to a cycloid-like curve whose cusps appear on the left or right side of the helix, depending on whether the helix is right- or left-handed. Finch found that tilting the top of the TMV particle by about 5° (the slope of the subunit helix) towards the electron beam indeed caused the left side of the particle image to appear crenelated, or like the teeth of a comb, while the right side remained smooth. From this observation, the right-handed nature of the TMV helix was ascertained for the first time, information which could not have been obtained on the sole basis of x-ray diffraction with a macroscopic sample in a gel state. Finch presented additional evidence in reciprocal space by taking the optical Fourier transform of the TEM micrographs of the TMV specimen tilted to the slope angle [120]. On the layer-line of the 23 Å spacing of the basic subunit helix (that is the third layer-line in figure 21(b)), a strong meridian intensity was obtained due to the diffraction by the grating of horizontal slits formed by the cycloid cusps in the TEM micrograph. This is exactly what Bernaerts *et al* [95] exploited by selecting just that meridian streak in the microscope to form the dark field image of the isochiral layers of a MWNT. Following principles similar to those used above, Fan and Bursil [121] have discussed processing techniques of digitized high-resolution TEM bright- and dark-field images to determine the structure of MWNTs comprising a few layers.

5.4.5. CNT peapods. TMVs have a large hollow core filled with water. Similarly, CNTs have a hollow interior, which generally remains empty unless inadvertently or intentionally filled, during or after synthesis, with foreign materials whose surface tension is suitable for

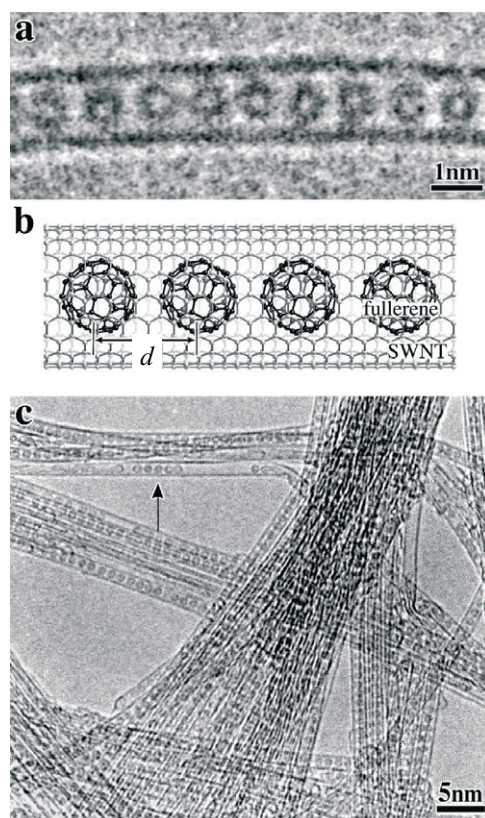


Figure 44. (a) SWNT containing a linear array of equidistant C₆₀ molecules. (b) A computer drawing of the peapod structure in (a). (c) A tangle of peapod bundles. Some of the peapods (black arrow) are not completely filled with C₆₀ molecules. Reproduced from [126] with kind permission.

capillary penetration into the vacant cylindrical space [122]. The smallest expected size of the cavity is clearly determined by the intrinsic minimum radius of a SWNT, which is about 2–3 Å [123, 124], but CNTs typically produced by all methods of synthesis have a cavity substantially larger than that, at least 6–7 Å radius, suitable for encaging even large molecules.

Recently, a very interesting hybrid system of C₆₀ fullerene molecules encaged inside a 1.3 nm diameter SWNT has been observed [125]. The buckyballs are regularly arrayed in a one-dimensional crystal of lattice parameter $d \approx 1$ nm. This pea-pod like structure, illustrated in figure 44, has since been synthesized using several other fullerenes (see [126] and references therein) or even endohedral metallofullerenes, for example, (Gd@C₈₂)_n@SWNT [127]. They are perfect examples of one-dimensional molecular solids whose electronic transport properties may be varied by choosing which fullerene or endo-fullerene is encapsulated into which SWNT.

From the point of view of diffraction taken in this paper, fullerene peapods offer a particularly well suited illustration of the basic principles of fibre diffraction with which we have started in section 2.4. The diffraction of the peapods of figure 44 is shown in figure 45. Let us represent the C₆₀ molecule as a spherical scatterer in which the 60 carbon atoms are smeared uniformly over the spherical surface of radius r . Then, by replacing the discrete summation by a continuous integration over the spherical surface, the isotropic form factor

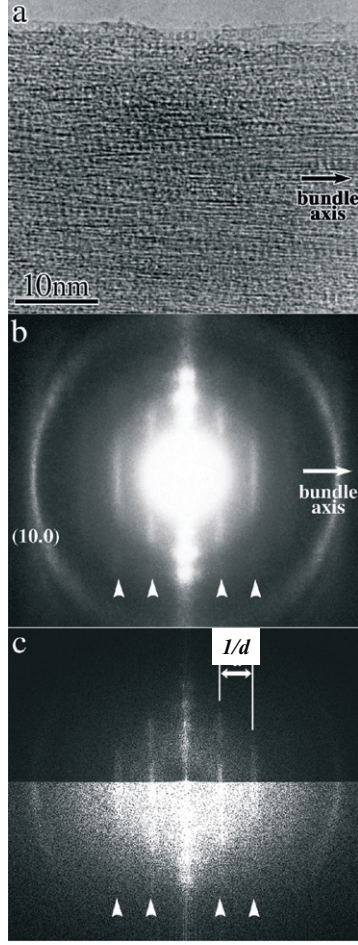


Figure 45. (a) TEM picture of a bundle of approximately parallel nanotube peapods. (b) ED pattern of (a). The white arrow heads indicate the positions of the non-equator layer-lines produced by the C_{60} linear arrays of the peapods. (c) Comparison between the optical transform of the TEM picture in (a) (top) and the ED pattern (middle). Reproduced from [126] with kind permission.

$A_{C_{60}}(q)$ of the molecule is easily found to be

$$A_{C_{60}}(q) = f(q) \sum_j e^{iq \cdot r_j} = 4\pi r^2 \sigma_{C_{60}} f(q) \frac{\sin qr}{qr}, \quad (22)$$

where $f(q)$ is, as before, the carbon scattering factor and $\sigma_{C_{60}} = 60/4\pi r^2$ is the number density of surface atoms. Similarly, the host SWNT can be treated as a continuous hollow cylinder of radius $R = 0.65$ nm, carbon density σ_{cyl} and of form factor A_{cyl} , given by equation (8). Then, the total scattering amplitude of the vertical model peapod of length $L = Nd$ encapsulating N molecules is

$$\begin{aligned} A(q) &= A_{cyl}(q) + A_{C_{60}}(q) \sum_{n=1}^N e^{indq_z} \\ &= f(q) \left[2\pi RL\sigma_{cyl} J_0(q_\perp R) \delta(q_z) + 4\pi r^2 N \sigma_{C_{60}} \frac{\sin qr}{qr} \sum_l \delta_{q_z, l(2\pi/d)} \right], \end{aligned} \quad (23)$$

where $\delta(q_z)$ stands here for the slit function $\sin(q_z L/2)/(q_z L/2)$ and where the Kronecker δ in the second term, which arises from a relation similar to equation (19), quantizes the vertical component of the scattering vector.

So, for relatively small scattering vectors on the scale of $2\pi/d$, the main characteristics of the ED pattern of a single peapod should be

- (i) sharp, equidistant, non-equatorial layer-lines separated by $2\pi/d$ arising only from scattering by the C_{60} molecules and with decaying intensity slowly oscillating like $f^2(q) \sin^2(qr)/(qr)^2$ away from the meridian.
- (ii) an equatorial layer-line where the scattering by both C_{60} and the SWNT interfere with the oscillating intensity $f^2(q)[N_{\text{cyl}}J_0(qR) + N_{C_{60}} \sin(qr)/qr]^2$, where N_{cyl} and $N_{C_{60}}$ are the total number of carbon atoms in the two components of the pea-pod within the length L selected for diffraction.

Note that the atom density $4/a^2\sqrt{3} = 0.38 \text{ \AA}^{-2}$ of the SWNT is about the same as that of the C_{60} molecule $60/4\pi r^2 = 0.39 \text{ \AA}^{-2}$. For a SWNT radius $R = 0.65 \text{ nm}$ and a peapod period $d = 1 \text{ nm}$ the ratio $N_{\text{cyl}}/N_{C_{60}} = 2\pi R d/4\pi r^2 = 2.6$ indicates that the equatorial layer-line is dominated by the faster oscillations of the cylinder Bessel function of the host SWNT.

Most frequently, peapods are bundled, just like empty SWNTs, in imperfect hexagonal lattices of non-uniform period D (about 1.9 nm for $(\text{Gd}@C_{82})_n@SWNT$ [127]). The positions of the fullerene molecules in adjacent peapods, the relative angular orientations of the neighbouring host SWNTs and their chiral indices all appear to be uncorrelated [127]. So we have a typical situation of fibre diffraction. The reciprocal space of the ED pattern of such bundles should involve four principal periodicities, namely $1/D$, $1/R$, $1/d$ and $1/a$, in increasing order. The lattice spacing D and SWNT radius R interfere and dominate the equator layer-line, which should show broad equidistant spots. The uniform intermolecular pea-pod spacing d determines the position of sharp meridian streaks along the non-equatorial layer-lines. The carbon honeycomb lattice spacing a (not represented in the continuous model) should give rise to (100) diffraction arcs associated with the variety of chiral angles and the misorientation of the SWNT in the bundle. All the features observed in the experimental ED pattern shown in figure 45 are therefore accounted for by this simple model.

Acknowledgments

The authors are grateful to J F Colomer, L Henrard, M Gao, H Jiang, M Kociak, W Ruland, A K Schaper and J T Finch for helpful discussions and/or for kindly providing permission to reproduce some of the illustrations of this paper. We acknowledge the very fruitful ongoing collaboration with our microscopist colleagues in Antwerpen, Professors Amelinckx, Van Tendeloo and Van Landuyt. This work has been performed within the inter-university research project PAI-P5/1 'Quantum size effects in reduced dimensionality systems' funded by the Services of the Belgian Science Policy Programming. Part of the work on nanotubes by J F Colomer *et al* cited in this paper was performed under a generous supporting grant by the Belgian Francqui Foundation. We thank the referee of this review for his constructive suggestions.

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